
TROPIC01

ODR_TR01_PSR_004_TRNG_TropicSquare

TROPIC01 True Random Number Generator (TRNG)

Version: 1.0

Abstract

This security report presents the results of a gray-box randomness and reliability evaluation of the true random number generator (TRNG) integrated in the TROPIC01 secure element, performed in-house by Tropic Square. The TRNG met the expected statistical and min-entropy properties at every measured condition, with no degradation across the operating range.

June 3, 2026





1 Summary of the Security Report

1.1 Report Details

- **Evaluation Laboratory:** Tropic Square
- **Evaluation Laboratory Contact:** Tropic Square Security Team, security@tropicsquare.com
- **Evaluation Period:** Measurements captured 06/2025 – 09/2025
- **Type of Evaluation:** In-house evaluation by Tropic Square (using custom samples prepared by Tropic Square).

1.2 Executive Summary

This report documents the evaluation of the true random number generator (TRNG) integrated in the TROPIC01 secure element. The IP vendor claims compliance with AIS31-PTG2 [6] and NIST SP 800-90B [3]. The aim of the evaluation was to verify that both TRNG instances meet the statistical and entropy requirements expected of a cryptographic-grade entropy source across the full stated operating range.

The evaluation was performed by Tropic Square on 30 open samples characterised at five temperatures spanning -40 °C to 125 °C and at three entropy-source supply voltage trim settings, covering both the stated operating range and selected points beyond it.

The collected data were evaluated using the NIST SP 800-22 statistical test suite and the official NIST SP 800-90B entropy assessment test suite. The NIST SP 800-22 test results showed no significant deviations across all tested chips and showed no dependence on environmental conditions. The NIST SP 800-90B assessment produced good min-entropy estimates and the estimates likewise showed no degradation across the measured temperatures and supply voltages.

Overall, the results show that the TROPIC01 TRNG meets the statistical and entropy requirements expected of a cryptographic-grade entropy source and is suitable for use in cryptographic applications across the entire stated operating range.



1.3 Tropic Square Recommendations

Tropic Square regularly publishes FW updates to their products and resolves product weaknesses responsibly, as described in the Coordinated Vulnerability Disclosure Policy (CVD) (<http://tropicsquare.com/security/cvd>).

Version History

Version	Date	Description
1.0	3.6.2026	Initial Report Version



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2 The Evaluation Report

The TROPIC01 true random number generator (TRNG) is a physical true random number generator with:

- analog noise source
- digitization stage
- post-processing (conditioning) stage
- health and online tests

This evaluation was performed in a gray-box manner. The datasets analysed in this report were collected through a debug interface that is not enabled on production devices. The tests do not rely on any internal information about the TRNG IP, with the sole exception of the conditioning component parameters used in the final min-entropy estimate (see Section 2.3).

TROPIC01 contains two identical and independent instances of the TRNG. They are called TRNG0 and TRNG1. The voltage source (VDD) for TRNG0 is provided by the vdd_ana_1 low-dropout regulator (LDO) and VDD for TRNG1 is provided by the vdd_ana_2 LDO.

The IP vendor claims that the TRNG is compliant with AIS31-PTG2 [6] and NIST SP800-90B [3]. The IP vendor claims the following operating conditions (Delivered specification [5]):

Parameter	MIN	TYP	MAX
T junction	-25 °C	25 °C	105°C
Supply voltage	1.212 V	1.35 V	1.485 V

2.1 Test conditions

As per NIST SP800-90B [3], three different TRNG datasets were collected:

- **sequential dataset** - 100 000 000 bits – the digitized noise source, taken before conditioning (raw random numbers (RRN) according to AIS31 [6])
- **conditioned sequential dataset** – 100 000 000 bits – data taken from the output of the conditioning component (internal random numbers (IRN) according to AIS31 [6])



- **restart dataset** – 1000*1000 bits – the TRNG is restarted 1000 times and for each restart the first 1000 bits of raw random numbers are taken

For the sequential and conditioned sequential datasets, a continuous data stream was collected without interruptions. For the restart dataset, each 1000 bit block is continuous.

The above-mentioned datasets were taken for each of the following conditions. In total, 30 chips across wafer corners were used for the evaluation.

- **TRNG instance** TRNG0 and TRNG1 for each chip
- **Temperatures [°C]:** [-40, -25, 25, 105, 125]
- **TRNG core supply voltage:** 'low' ~ 1.212 V, 'default' ~ 1.35 V (applied in production devices) and 'high' ~ 1.485 V

Note

The dataset was also captured outside the operating conditions of the TRNG IP to test its performance for potential randomness degradation outside the specified operating conditions (-40°C and 125°C).

Note that these conditions are detected by the TROPIC01 temperature sensor on production chips.

Additionally, the sequential and conditioned sequential dataset of size 1 GiB (2^{30} bytes) was collected on one chip (chip ID: WID3_CID309) in nominal conditions for both TRNG0 and TRNG1:

1. temperature 25 °C
2. 'default' TRNG core supply voltage



2.2 NIST SP 800-22 statistical test suite

The conditioned sequential dataset was evaluated using the NIST SP 800-22 [4] statistical test suite, specifically the fast-sts implementation from [2]. In addition to the chi-squared test reported in the standard results file, this implementation also provides a Kolmogorov–Smirnov test for uniformity ($P\text{-value(KS)}$). All tests from the suite were run with their default parameters.

2.2.1 Statistical testing on the 1 GiB dataset

First, the 1 GiB conditioned sequential dataset from chip WID3_CID309 taken at nominal environmental conditions was tested. The dataset was split into 1000 sequences, with 8 500 000 bits in each sequence.

The standard results file produced by the test suite for TRNG0 is shown below. It summarizes, for each statistical test, the distribution of per-sequence p-values across the ten bins C1–C10, the uniformity test p-values from the chi-squared ($P\text{-VALUE}$) and Kolmogorov–Smirnov ($P\text{-value(KS)}$) tests, and the proportion of sequences that passed:

RESULTS FOR THE UNIFORMITY OF P-VALUES AND THE PROPORTION OF PASSING SEQUENCES

generator is ...

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	P-VALUE	P-value(KS)	PROPORTION	STATISTICAL TEST
94	80	104	111	92	104	114	84	105	112	0.174728	0.430270	0.9930	Frequency
85	104	102	96	94	110	94	111	103	101	0.775337	0.339659	0.9920	BlockFrequency
89	94	106	114	100	95	81	104	116	101	0.313041	0.700212	0.9950	CumulativeSums
91	95	93	100	94	103	106	93	122	103	0.556460	0.166017	0.9940	CumulativeSums
104	103	91	119	82	96	100	94	87	124	0.069430	0.510129	0.9940	Runs
107	96	116	106	92	108	96	87	110	82	0.279844	0.265407	0.9880	LongestRun
96	102	108	102	98	97	91	96	104	106	0.980883	0.862710	0.9880	Rank
101	101	84	106	83	110	110	89	114	102	0.272977	0.363636	0.9860	FFT
96	98	119	105	101	112	83	96	99	91	0.402962	0.177233	0.9860	NonOverlappingTemplate
101	102	112	123	101	86	100	83	111	81	0.057875	0.069916	0.9910	NonOverlappingTemplate
99	97	97	86	96	102	106	98	110	109	0.870856	0.302478	0.9920	NonOverlappingTemplate
111	100	98	106	94	101	93	103	99	95	0.971006	0.898207	0.9900	NonOverlappingTemplate
107	108	113	102	117	82	83	90	105	93	0.136499	0.010755	0.9930	NonOverlappingTemplate
111	109	86	88	97	90	109	108	105	97	0.504219	0.314687	0.9870	NonOverlappingTemplate
104	108	101	103	91	113	91	97	90	102	0.803720	0.505906	0.9860	NonOverlappingTemplate
93	90	107	110	101	86	97	115	105	96	0.564639	0.631299	0.9950	NonOverlappingTemplate
88	101	115	104	100	115	95	96	98	88	0.534146	0.481107	0.9940	NonOverlappingTemplate
114	94	79	111	98	97	104	91	108	104	0.363593	0.944340	0.9920	NonOverlappingTemplate
103	92	105	103	107	100	113	81	111	85	0.325206	0.318203	0.9880	NonOverlappingTemplate
92	113	109	101	100	80	100	108	112	85	0.244236	0.730493	0.9880	NonOverlappingTemplate
89	104	79	94	124	119	94	97	104	96	0.061260	0.113413	0.9900	NonOverlappingTemplate
104	95	95	99	105	104	106	94	95	103	0.989055	0.990278	0.9920	NonOverlappingTemplate
121	91	92	100	101	90	111	96	103	95	0.476911	0.736839	0.9890	NonOverlappingTemplate



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92	116	100	105	79	106	99	108	107	88	0.289667	0.893021	0.9870	NonOverlappingTemplate
82	112	107	105	103	106	85	95	99	106	0.461612	0.738483	0.9930	NonOverlappingTemplate
104	101	106	116	102	88	103	91	89	100	0.670396	0.247522	0.9920	NonOverlappingTemplate
95	84	90	96	102	105	112	105	116	95	0.459717	0.050729	0.9910	NonOverlappingTemplate
106	96	98	96	115	92	106	100	85	106	0.680755	0.965388	0.9890	NonOverlappingTemplate
104	109	97	102	93	88	110	110	102	85	0.603841	0.839900	0.9920	NonOverlappingTemplate
107	94	100	102	92	104	96	107	101	97	0.982454	0.957965	0.9930	NonOverlappingTemplate
100	93	91	117	88	100	102	100	111	98	0.645448	0.784980	0.9910	NonOverlappingTemplate
109	110	110	97	100	109	91	81	93	100	0.473064	0.117767	0.9860	NonOverlappingTemplate
106	103	100	110	98	94	96	84	96	113	0.697257	0.342388	0.9880	NonOverlappingTemplate
102	90	87	97	113	104	114	108	92	93	0.494392	0.407228	0.9870	NonOverlappingTemplate
102	97	90	94	113	105	104	102	97	96	0.919131	0.690102	0.9920	NonOverlappingTemplate
100	106	119	94	92	87	80	107	97	118	0.106246	0.214401	0.9950	NonOverlappingTemplate
116	100	86	103	98	102	93	89	117	96	0.397688	0.312167	0.9870	NonOverlappingTemplate
102	95	112	112	104	94	80	102	94	105	0.500279	0.227961	0.9880	NonOverlappingTemplate
100	90	112	96	107	93	108	102	95	97	0.867692	0.953777	0.9860	NonOverlappingTemplate
111	99	78	95	102	103	95	107	103	107	0.558502	0.452965	0.9910	NonOverlappingTemplate
98	106	92	105	95	89	117	91	116	91	0.365253	0.963883	0.9870	NonOverlappingTemplate
93	107	91	96	114	77	105	101	103	113	0.259616	0.325562	0.9940	NonOverlappingTemplate
100	103	101	112	100	99	75	107	104	99	0.488534	0.664537	0.9910	NonOverlappingTemplate
103	95	105	94	101	81	102	112	98	109	0.647530	0.424190	0.9910	NonOverlappingTemplate
91	101	95	104	110	95	96	103	116	89	0.689019	0.969098	0.9870	NonOverlappingTemplate
91	113	98	100	89	108	92	103	98	108	0.759756	0.778077	0.9890	NonOverlappingTemplate
112	103	98	98	116	115	103	75	87	93	0.092597	0.011036	0.9930	NonOverlappingTemplate
91	93	107	88	112	101	107	90	103	108	0.647530	0.569335	0.9900	NonOverlappingTemplate
99	86	102	107	82	104	113	92	112	103	0.370262	0.163104	0.9850	NonOverlappingTemplate
91	101	92	122	104	83	84	122	88	113	0.018668	0.582149	0.9930	NonOverlappingTemplate
89	95	100	98	92	93	105	112	106	110	0.771469	0.220933	0.9950	NonOverlappingTemplate
110	116	113	87	91	84	105	91	97	106	0.223648	0.066172	0.9880	NonOverlappingTemplate
85	106	101	100	101	100	80	101	111	115	0.342451	0.098482	0.9870	NonOverlappingTemplate
101	109	111	105	87	108	99	98	100	82	0.544254	0.130682	0.9910	NonOverlappingTemplate
109	103	98	98	88	117	78	110	101	98	0.262249	0.514289	0.9870	NonOverlappingTemplate
100	108	81	108	102	84	120	96	118	83	0.037813	0.743454	0.9930	NonOverlappingTemplate
98	99	89	87	82	117	117	100	98	113	0.133404	0.010736	0.9960	NonOverlappingTemplate
105	101	109	105	103	84	104	94	88	107	0.697257	0.382393	0.9860	NonOverlappingTemplate
104	96	93	102	102	106	83	99	115	100	0.699313	0.829574	0.9850	NonOverlappingTemplate
110	108	93	90	107	90	106	95	110	91	0.632955	0.741598	0.9890	NonOverlappingTemplate
90	116	111	103	103	92	82	100	79	124	0.025193	0.348337	0.9920	NonOverlappingTemplate
99	107	94	92	93	99	120	109	91	96	0.556460	0.813096	0.9870	NonOverlappingTemplate
96	106	80	111	126	102	92	91	108	88	0.065639	0.513338	0.9920	NonOverlappingTemplate
112	110	108	95	95	110	101	90	90	89	0.554420	0.146826	0.9950	NonOverlappingTemplate
94	101	108	113	77	101	107	102	99	98	0.476911	0.867652	0.9940	NonOverlappingTemplate
84	88	94	102	78	109	118	105	110	112	0.067300	0.002138	0.9900	NonOverlappingTemplate
97	88	93	100	86	112	104	90	113	117	0.264901	0.069220	0.9940	NonOverlappingTemplate
95	92	98	93	114	96	102	101	105	104	0.911413	0.377100	0.9950	NonOverlappingTemplate
87	102	125	105	86	83	93	112	107	100	0.078086	0.321987	0.9910	NonOverlappingTemplate
102	100	107	104	99	93	101	95	89	110	0.932333	0.886380	0.9910	NonOverlappingTemplate
103	85	93	109	88	106	100	102	95	119	0.406499	0.466156	0.9900	NonOverlappingTemplate
114	107	101	100	99	94	102	94	93	96	0.919131	0.425376	0.9890	NonOverlappingTemplate
76	94	106	99	96	123	103	103	87	113	0.078086	0.149886	0.9930	NonOverlappingTemplate
102	111	117	95	92	110	87	90	96	100	0.448424	0.246681	0.9950	NonOverlappingTemplate
105	89	110	114	110	91	96	92	101	92	0.566688	0.190611	0.9870	NonOverlappingTemplate
99	86	103	93	101	114	86	93	93	132	0.038818	0.123723	0.9940	NonOverlappingTemplate



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95	96	96	100	92	96	108	118	83	116	0.296834	0.411800	0.9930	NonOverlappingTemplate
105	105	88	114	92	101	108	104	96	87	0.616305	0.749087	0.9860	NonOverlappingTemplate
96	86	101	99	90	110	107	94	120	97	0.429923	0.303362	0.9940	NonOverlappingTemplate
91	101	105	93	96	91	103	116	96	108	0.741918	0.277312	0.9920	NonOverlappingTemplate
111	101	83	88	86	114	118	87	103	109	0.083018	0.053535	0.9910	NonOverlappingTemplate
98	111	101	100	102	87	93	113	93	102	0.769527	0.910245	0.9890	NonOverlappingTemplate
96	99	118	105	100	113	83	96	99	91	0.417219	0.158664	0.9860	NonOverlappingTemplate
98	109	88	108	119	89	100	94	101	94	0.486588	0.500551	0.9900	NonOverlappingTemplate
103	102	94	92	119	110	91	96	99	94	0.628790	0.704720	0.9880	NonOverlappingTemplate
85	104	103	93	92	104	112	93	100	114	0.566688	0.185471	0.9960	NonOverlappingTemplate
107	107	97	101	95	94	108	112	83	96	0.655854	0.625415	0.9910	NonOverlappingTemplate
93	99	110	105	98	92	92	85	112	114	0.463512	0.427253	0.9860	NonOverlappingTemplate
96	96	113	90	103	98	112	98	98	96	0.849708	0.992493	0.9840	NonOverlappingTemplate
106	120	85	109	98	100	98	96	91	97	0.478839	0.369579	0.9880	NonOverlappingTemplate
83	107	98	109	76	106	109	103	117	92	0.097159	0.430956	0.9920	NonOverlappingTemplate
93	95	95	124	85	114	108	84	100	102	0.115387	0.737170	0.9880	NonOverlappingTemplate
91	85	114	105	106	98	112	93	88	108	0.377007	0.383052	0.9910	NonOverlappingTemplate
93	105	88	116	111	106	106	92	98	85	0.383827	0.375388	0.9920	NonOverlappingTemplate
96	116	102	97	105	90	97	84	94	119	0.295391	0.609313	0.9930	NonOverlappingTemplate
106	101	88	98	115	95	96	104	107	90	0.723804	0.928535	0.9920	NonOverlappingTemplate
99	95	103	109	94	104	109	101	101	85	0.854708	0.720176	0.9910	NonOverlappingTemplate
102	84	93	96	90	108	85	120	113	109	0.135720	0.022706	0.9960	NonOverlappingTemplate
103	101	90	111	102	100	99	103	92	99	0.960198	0.982107	0.9900	NonOverlappingTemplate
116	85	116	101	91	108	80	101	100	102	0.168112	0.567281	0.9890	NonOverlappingTemplate
118	105	88	90	88	122	108	93	100	88	0.097159	0.538825	0.9920	NonOverlappingTemplate
95	108	86	88	89	114	100	95	101	124	0.142062	0.085015	0.9910	NonOverlappingTemplate
93	94	103	92	101	114	102	100	110	91	0.798139	0.413694	0.9920	NonOverlappingTemplate
99	103	91	96	100	116	97	84	120	94	0.301194	0.880891	0.9870	NonOverlappingTemplate
101	94	97	95	101	92	103	113	111	93	0.848027	0.534846	0.9940	NonOverlappingTemplate
112	93	87	99	102	106	94	105	97	105	0.836048	0.943402	0.9880	NonOverlappingTemplate
87	119	95	92	95	109	106	105	101	91	0.467322	0.922790	0.9900	NonOverlappingTemplate
83	91	105	93	99	107	111	108	95	108	0.566688	0.099934	0.9910	NonOverlappingTemplate
98	102	105	105	88	108	110	109	88	87	0.574903	0.350381	0.9900	NonOverlappingTemplate
102	96	105	100	98	98	98	99	106	98	0.999482	0.971172	0.9930	NonOverlappingTemplate
77	101	103	86	101	112	103	95	107	115	0.220159	0.025784	0.9930	NonOverlappingTemplate
106	112	83	87	103	88	95	110	107	109	0.314544	0.344287	0.9920	NonOverlappingTemplate
105	94	96	85	105	105	114	108	88	100	0.579021	0.407880	0.9870	NonOverlappingTemplate
110	110	107	95	91	114	106	88	97	82	0.301194	0.106915	0.9950	NonOverlappingTemplate
91	97	101	106	107	88	98	100	101	111	0.878618	0.944970	0.9910	NonOverlappingTemplate
100	130	107	94	105	109	84	93	79	99	0.031012	0.016969	0.9890	NonOverlappingTemplate
85	120	79	97	109	108	92	98	111	101	0.118812	0.486227	0.9920	NonOverlappingTemplate
101	102	114	104	81	107	98	84	94	115	0.244236	0.501370	0.9850	NonOverlappingTemplate
103	108	99	110	103	83	97	103	100	94	0.811080	0.425826	0.9920	NonOverlappingTemplate
107	110	97	79	92	108	101	110	94	102	0.467322	0.757866	0.9890	NonOverlappingTemplate
98	106	99	95	99	100	94	96	113	100	0.968863	0.931811	0.9910	NonOverlappingTemplate
92	105	99	89	120	100	96	100	110	89	0.486588	0.754053	0.9940	NonOverlappingTemplate
105	101	92	91	124	93	109	85	107	93	0.213309	0.973239	0.9870	NonOverlappingTemplate
113	97	74	85	120	115	102	99	104	91	0.034484	0.176989	0.9870	NonOverlappingTemplate
100	102	92	118	83	94	128	108	92	83	0.023705	0.298778	0.9920	NonOverlappingTemplate
93	105	82	120	100	99	113	94	101	93	0.308561	0.600812	0.9860	NonOverlappingTemplate
117	100	88	103	93	93	86	115	95	110	0.285427	0.434368	0.9920	NonOverlappingTemplate
108	104	88	107	91	89	106	104	98	105	0.783019	0.951769	0.9910	NonOverlappingTemplate
94	99	108	102	105	116	87	104	93	92	0.653773	0.439247	0.9870	NonOverlappingTemplate



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96	104	95	102	94	113	92	103	111	90	0.779188	0.986178	0.9870	NonOverlappingTemplate
83	99	111	96	110	100	98	113	93	97	0.576961	0.522446	0.9930	NonOverlappingTemplate
103	111	95	115	81	93	114	94	85	109	0.150340	0.513675	0.9930	NonOverlappingTemplate
99	109	83	92	95	85	111	100	118	108	0.216713	0.079538	0.9960	NonOverlappingTemplate
117	95	97	117	98	104	96	105	98	73	0.120207	0.206706	0.9930	NonOverlappingTemplate
105	101	83	91	101	116	104	83	108	108	0.285427	0.543683	0.9880	NonOverlappingTemplate
91	109	94	112	114	85	104	93	96	102	0.486588	0.561669	0.9890	NonOverlappingTemplate
108	106	84	100	99	100	98	96	105	104	0.899171	0.909125	0.9890	NonOverlappingTemplate
108	95	103	100	103	99	99	88	105	100	0.972382	0.777255	0.9870	NonOverlappingTemplate
113	113	99	92	108	105	97	84	81	108	0.223648	0.205123	0.9860	NonOverlappingTemplate
113	90	95	116	96	94	91	110	99	96	0.534146	0.738147	0.9910	NonOverlappingTemplate
98	104	99	91	88	95	115	119	98	93	0.428095	0.169161	0.9950	NonOverlappingTemplate
105	110	111	94	106	95	99	78	99	103	0.496351	0.150445	0.9910	NonOverlappingTemplate
96	105	103	106	119	81	103	105	97	85	0.292519	0.256833	0.9940	NonOverlappingTemplate
91	86	113	97	102	96	106	94	105	110	0.666245	0.413818	0.9950	NonOverlappingTemplate
95	108	101	96	109	105	89	88	105	104	0.818343	0.893542	0.9900	NonOverlappingTemplate
95	109	89	108	102	97	99	113	99	89	0.743915	0.947738	0.9920	NonOverlappingTemplate
102	102	101	103	99	85	109	101	93	105	0.911413	0.907578	0.9920	NonOverlappingTemplate
96	115	101	114	93	89	91	89	110	102	0.424453	0.352732	0.9880	NonOverlappingTemplate
96	101	101	115	98	98	95	76	112	108	0.304126	0.389904	0.9860	NonOverlappingTemplate
105	80	110	102	100	109	89	100	97	108	0.530120	0.930467	0.9900	NonOverlappingTemplate
105	98	99	93	108	102	99	95	93	108	0.969588	0.946343	0.9890	NonOverlappingTemplate
112	96	96	100	121	94	91	96	100	94	0.548314	0.467654	0.9920	NonOverlappingTemplate
92	94	94	101	98	107	88	101	114	111	0.686955	0.176931	0.9900	NonOverlappingTemplate
99	105	98	99	104	105	88	96	89	117	0.697257	0.855939	0.9970	NonOverlappingTemplate
94	98	108	104	93	105	94	96	113	95	0.883171	0.965034	0.9910	NonOverlappingTemplate
98	111	101	100	102	87	93	113	93	102	0.769527	0.910245	0.9890	NonOverlappingTemplate
94	100	102	100	104	103	102	90	104	101	0.993493	0.952401	0.9890	OverlappingTemplate
91	103	100	97	78	108	99	104	108	112	0.463512	0.071631	0.9910	Universal
97	86	102	116	102	96	87	105	99	110	0.554420	0.377804	0.9900	ApproximateEntropy
86	85	87	81	91	84	78	103	87	88	0.864199	0.777011	0.9862	RandomExcursions
104	76	75	78	102	79	86	95	83	92	0.222534	0.506567	0.9839	RandomExcursions
98	86	74	93	93	80	90	87	90	79	0.771647	0.743234	0.9908	RandomExcursions
78	78	85	111	85	88	84	83	96	82	0.343092	0.456720	0.9943	RandomExcursions
90	84	75	91	87	96	87	92	86	82	0.937744	0.798676	0.9943	RandomExcursions
95	89	83	75	97	97	94	73	80	87	0.504332	0.612077	0.9931	RandomExcursions
84	75	92	75	92	97	89	90	75	101	0.419436	0.250559	0.9862	RandomExcursions
100	87	89	77	80	93	78	90	90	86	0.810681	0.707155	0.9839	RandomExcursions
78	96	93	112	92	71	91	81	75	81	0.081602	0.038782	0.9920	RandomExcursionsVariant
81	101	107	78	104	90	66	69	96	78	0.008045	0.034865	0.9931	RandomExcursionsVariant
85	105	85	108	79	75	85	88	85	75	0.161052	0.027591	0.9920	RandomExcursionsVariant
100	90	96	89	74	85	73	83	85	95	0.511156	0.205850	0.9908	RandomExcursionsVariant
99	92	88	87	80	83	79	76	98	88	0.716708	0.448338	0.9862	RandomExcursionsVariant
95	90	72	91	95	80	86	80	88	93	0.749085	0.857727	0.9874	RandomExcursionsVariant
95	92	75	81	67	83	98	94	101	84	0.203139	0.070944	0.9908	RandomExcursionsVariant
87	95	76	90	80	88	95	78	97	84	0.767179	0.858905	0.9931	RandomExcursionsVariant
84	88	96	86	94	89	81	107	89	56	0.037936	0.074389	0.9954	RandomExcursionsVariant
84	103	86	68	87	98	81	88	79	96	0.297583	0.921984	0.9908	RandomExcursionsVariant
84	84	98	85	99	96	66	80	79	99	0.199426	0.395368	0.9897	RandomExcursionsVariant
90	80	91	89	80	82	100	85	96	77	0.758174	0.956172	0.9874	RandomExcursionsVariant
94	88	77	97	62	83	93	77	95	104	0.069818	0.143267	0.9862	RandomExcursionsVariant
93	88	78	61	79	78	96	104	96	97	0.046772	0.016526	0.9862	RandomExcursionsVariant
93	88	69	65	83	83	96	103	102	88	0.056217	0.008444	0.9897	RandomExcursionsVariant



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90	82	74	80	75	84	81	104	104	96	0.179403	0.015947	0.9885	RandomExcursionsVariant
84	91	79	63	71	90	83	96	107	106	0.014797	0.008867	0.9897	RandomExcursionsVariant
85	85	80	68	82	89	89	96	93	103	0.394957	0.022228	0.9931	RandomExcursionsVariant
99	107	98	104	98	97	103	102	92	100	0.996335	0.844466	0.9920	Serial
104	95	105	97	96	127	86	97	89	104	0.235589	0.492612	0.9940	Serial
101	102	103	101	102	88	95	102	98	108	0.978072	0.942037	0.9840	LinearComplexity

The minimum pass rate for each statistical test with the exception of the
random excursion (variant) test is approximately = 0.980561 for a
sample size = 1000 binary sequences.

The minimum pass rate for the random excursion (variant) test
is approximately 0.979880 for a sample size = 870 binary sequences.

For further guidelines construct a probability table using the MAPLE program
provided in the addendum section of the documentation.



The standard results file produced by the test suite for TRNG1 is shown here:

RESULTS FOR THE UNIFORMITY OF P-VALUES AND THE PROPORTION OF PASSING SEQUENCES

generator is ...

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	P-VALUE	P-value(KS)	PROPORTION	STATISTICAL TEST
92	88	101	112	90	103	91	109	110	104	0.595549	0.091460	0.9920	Frequency
95	105	97	122	102	98	73	92	109	107	0.098330	0.456562	0.9910	BlockFrequency
89	106	99	94	116	90	95	109	99	103	0.672470	0.934834	0.9920	CumulativeSums
83	100	106	104	92	113	100	100	97	105	0.731886	0.400239	0.9930	CumulativeSums
113	103	104	113	91	89	89	98	99	101	0.645448	0.080790	0.9830	Runs
91	91	97	100	102	95	113	103	97	111	0.827279	0.348061	0.9930	LongestRun
98	97	98	106	89	119	103	96	109	85	0.469232	0.852664	0.9920	Rank
106	101	97	104	97	120	99	104	80	92	0.390721	0.219527	0.9930	FFT
93	82	101	112	105	106	99	88	104	110	0.494392	0.341194	0.9920	NonOverlappingTemplate
107	119	70	109	88	106	122	88	85	106	0.003447	0.319004	0.9890	NonOverlappingTemplate
91	87	100	113	90	101	108	102	108	100	0.686955	0.300496	0.9920	NonOverlappingTemplate
81	130	101	96	83	106	93	112	105	93	0.027868	0.789818	0.9880	NonOverlappingTemplate
82	104	103	83	118	102	116	94	101	97	0.177628	0.276727	0.9900	NonOverlappingTemplate
100	88	93	96	86	106	108	94	123	106	0.271619	0.036583	0.9870	NonOverlappingTemplate
76	105	103	113	87	102	107	102	105	100	0.326749	0.111117	0.9870	NonOverlappingTemplate
120	99	88	89	114	101	82	112	108	87	0.074791	0.671897	0.9910	NonOverlappingTemplate
102	83	99	79	111	113	117	97	98	101	0.150340	0.053577	0.9900	NonOverlappingTemplate
104	98	108	93	97	93	116	97	115	79	0.260930	0.701853	0.9830	NonOverlappingTemplate
113	89	96	122	91	112	80	109	94	94	0.073872	0.487398	0.9910	NonOverlappingTemplate
118	113	98	101	87	95	109	90	91	98	0.385543	0.124151	0.9830	NonOverlappingTemplate
91	87	112	102	111	90	102	104	95	106	0.637119	0.245452	0.9920	NonOverlappingTemplate
85	109	99	96	109	95	109	114	86	98	0.431754	0.654069	0.9910	NonOverlappingTemplate
111	105	90	105	99	118	95	101	90	86	0.420827	0.190590	0.9870	NonOverlappingTemplate
84	99	103	104	91	121	107	98	97	96	0.454053	0.490280	0.9890	NonOverlappingTemplate
101	83	87	104	108	125	92	98	101	101	0.194813	0.172605	0.9880	NonOverlappingTemplate
95	99	99	94	101	105	97	115	99	96	0.946308	0.747618	0.9900	NonOverlappingTemplate
111	111	82	87	104	90	99	122	85	109	0.058612	0.593174	0.9880	NonOverlappingTemplate
100	108	100	111	96	112	103	96	92	82	0.576961	0.204734	0.9870	NonOverlappingTemplate
106	111	93	95	114	92	95	97	98	99	0.807412	0.483610	0.9890	NonOverlappingTemplate
119	99	78	109	109	116	93	96	95	86	0.078086	0.230503	0.9900	NonOverlappingTemplate
121	87	96	88	116	106	100	98	103	85	0.162606	0.218539	0.9850	NonOverlappingTemplate
102	92	100	100	107	113	92	98	103	93	0.903338	0.845406	0.9930	NonOverlappingTemplate
104	93	98	104	86	104	94	109	98	110	0.818343	0.546697	0.9910	NonOverlappingTemplate
98	108	113	96	101	98	102	101	105	78	0.562591	0.395921	0.9970	NonOverlappingTemplate
110	96	94	86	108	100	81	120	94	111	0.149495	0.352801	0.9920	NonOverlappingTemplate
107	100	103	89	115	117	81	106	103	79	0.080519	0.239084	0.9910	NonOverlappingTemplate
83	118	89	89	97	113	106	95	103	107	0.241741	0.343992	0.9920	NonOverlappingTemplate
82	112	100	96	112	105	89	87	112	105	0.267573	0.801619	0.9950	NonOverlappingTemplate
113	97	103	82	114	99	108	95	107	82	0.230755	0.741620	0.9920	NonOverlappingTemplate
114	99	95	97	92	107	93	94	114	95	0.689019	0.857341	0.9900	NonOverlappingTemplate
110	94	110	97	98	113	91	102	96	89	0.699313	0.460724	0.9900	NonOverlappingTemplate
106	113	119	97	90	106	99	95	81	94	0.253122	0.030155	0.9900	NonOverlappingTemplate
102	100	106	78	96	109	109	108	100	92	0.504219	0.671106	0.9940	NonOverlappingTemplate
86	103	101	97	123	104	96	87	106	97	0.358641	0.658078	0.9900	NonOverlappingTemplate



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106	98	125	97	92	95	82	93	119	93	0.079051	0.135569	0.9870	NonOverlappingTemplate
104	98	113	103	98	86	91	109	109	89	0.572847	0.671470	0.9890	NonOverlappingTemplate
107	105	96	102	107	88	93	103	96	103	0.930026	0.734222	0.9820	NonOverlappingTemplate
117	106	105	97	93	88	104	98	99	93	0.717714	0.172835	0.9850	NonOverlappingTemplate
110	91	100	88	108	101	108	112	93	89	0.566688	0.607325	0.9920	NonOverlappingTemplate
105	96	100	108	103	95	102	83	110	98	0.801865	0.753580	0.9900	NonOverlappingTemplate
94	99	112	96	104	92	109	90	88	116	0.476911	0.852566	0.9940	NonOverlappingTemplate
90	100	95	109	105	93	104	95	110	99	0.896345	0.717464	0.9920	NonOverlappingTemplate
89	106	94	107	98	100	108	94	105	99	0.928857	0.879407	0.9930	NonOverlappingTemplate
87	101	104	92	87	101	113	104	91	120	0.285427	0.177847	0.9900	NonOverlappingTemplate
96	101	100	98	115	97	80	96	113	104	0.478839	0.722521	0.9890	NonOverlappingTemplate
109	90	91	99	110	99	94	93	100	115	0.664168	0.454105	0.9910	NonOverlappingTemplate
102	115	92	86	101	108	105	94	99	98	0.719747	0.758397	0.9910	NonOverlappingTemplate
97	98	103	107	95	96	102	112	98	92	0.952152	0.775217	0.9870	NonOverlappingTemplate
109	107	96	116	100	94	99	100	84	95	0.616305	0.195412	0.9820	NonOverlappingTemplate
99	97	117	91	98	109	114	88	97	90	0.424453	0.459025	0.9890	NonOverlappingTemplate
104	82	105	93	103	121	108	101	93	90	0.291091	0.755249	0.9910	NonOverlappingTemplate
108	110	90	118	90	96	96	84	112	96	0.251837	0.453709	0.9890	NonOverlappingTemplate
105	110	94	104	92	117	82	95	97	104	0.433590	0.708731	0.9850	NonOverlappingTemplate
94	98	88	97	97	107	107	94	117	101	0.713641	0.291912	0.9900	NonOverlappingTemplate
91	99	89	115	85	93	105	95	121	107	0.190654	0.185121	0.9910	NonOverlappingTemplate
107	87	85	110	91	106	123	106	105	80	0.057146	0.327748	0.9900	NonOverlappingTemplate
92	102	103	95	89	102	109	96	106	106	0.914025	0.608201	0.9890	NonOverlappingTemplate
96	111	94	101	108	93	96	82	111	108	0.522100	0.744529	0.9940	NonOverlappingTemplate
117	111	76	103	91	96	107	96	101	102	0.235589	0.344897	0.9920	NonOverlappingTemplate
120	106	89	102	98	92	90	106	113	84	0.219006	0.295816	0.9910	NonOverlappingTemplate
114	96	95	114	87	118	92	87	109	88	0.128132	0.545577	0.9900	NonOverlappingTemplate
105	97	106	96	93	121	103	103	84	92	0.424453	0.282811	0.9900	NonOverlappingTemplate
107	92	105	104	102	91	102	101	94	102	0.970302	0.957790	0.9890	NonOverlappingTemplate
119	98	108	102	96	82	87	88	125	95	0.043368	0.100440	0.9830	NonOverlappingTemplate
88	100	93	117	97	97	96	81	116	115	0.138069	0.190871	0.9920	NonOverlappingTemplate
115	125	100	94	89	106	95	94	90	92	0.177628	0.040760	0.9890	NonOverlappingTemplate
90	120	87	98	98	116	115	107	88	81	0.046870	0.188448	0.9870	NonOverlappingTemplate
129	84	101	113	89	84	95	117	85	103	0.009136	0.285675	0.9920	NonOverlappingTemplate
102	101	97	89	86	99	99	110	111	106	0.749884	0.203330	0.9900	NonOverlappingTemplate
92	104	92	100	102	107	107	104	103	89	0.916599	0.726732	0.9910	NonOverlappingTemplate
100	100	84	100	115	98	108	100	92	103	0.717714	0.422458	0.9900	NonOverlappingTemplate
95	103	104	108	86	84	116	91	118	95	0.185555	0.451894	0.9880	NonOverlappingTemplate
93	82	101	112	105	105	100	88	104	110	0.506194	0.341194	0.9920	NonOverlappingTemplate
91	98	104	99	97	92	98	93	119	109	0.668321	0.359483	0.9920	NonOverlappingTemplate
102	93	104	127	92	103	94	96	94	95	0.363593	0.469386	0.9910	NonOverlappingTemplate
96	96	97	117	97	94	107	89	98	109	0.709558	0.719688	0.9910	NonOverlappingTemplate
107	93	104	116	89	96	95	84	114	102	0.360287	0.635073	0.9930	NonOverlappingTemplate
85	106	102	99	99	79	94	126	103	107	0.097159	0.074713	0.9910	NonOverlappingTemplate
87	113	97	110	106	110	98	88	91	100	0.522100	0.305590	0.9860	NonOverlappingTemplate
101	103	111	104	93	90	112	103	101	82	0.560545	0.568788	0.9870	NonOverlappingTemplate
93	107	105	115	80	103	94	107	101	95	0.467322	0.595428	0.9890	NonOverlappingTemplate
87	94	110	103	100	103	97	96	109	101	0.890582	0.404488	0.9870	NonOverlappingTemplate
97	116	101	105	88	97	112	74	100	110	0.135720	0.685082	0.9890	NonOverlappingTemplate
108	110	100	97	97	86	100	103	95	104	0.892036	0.625643	0.9920	NonOverlappingTemplate
81	99	110	89	104	101	96	106	101	113	0.512137	0.287836	0.9930	NonOverlappingTemplate
92	109	100	88	100	107	108	103	106	87	0.723804	0.889387	0.9940	NonOverlappingTemplate
106	90	99	90	85	111	111	112	94	102	0.448424	0.137735	0.9950	NonOverlappingTemplate



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93	115	106	92	98	97	99	86	109	105	0.647530	0.677623	0.9910	NonOverlappingTemplate
79	110	98	98	95	96	102	106	124	92	0.176657	0.152684	0.9940	NonOverlappingTemplate
96	99	101	119	99	92	87	103	98	106	0.676615	0.512257	0.9930	NonOverlappingTemplate
88	95	101	95	114	128	104	88	94	93	0.115387	0.290806	0.9920	NonOverlappingTemplate
94	92	115	104	104	97	105	102	97	90	0.830808	0.724530	0.9820	NonOverlappingTemplate
96	99	107	99	97	97	90	93	107	115	0.827279	0.643204	0.9840	NonOverlappingTemplate
96	116	97	86	91	108	103	107	92	104	0.574903	0.532191	0.9940	NonOverlappingTemplate
87	101	99	93	102	95	111	99	103	110	0.851383	0.407445	0.9900	NonOverlappingTemplate
109	95	86	100	101	103	91	112	113	90	0.528111	0.866664	0.9880	NonOverlappingTemplate
87	83	95	89	103	111	113	111	95	113	0.203351	0.009258	0.9900	NonOverlappingTemplate
113	77	117	95	97	87	116	116	98	84	0.020408	0.747605	0.9860	NonOverlappingTemplate
102	92	122	116	120	83	82	100	88	95	0.018540	0.007231	0.9920	NonOverlappingTemplate
104	115	94	88	93	116	100	87	115	88	0.179584	0.511597	0.9860	NonOverlappingTemplate
94	101	122	117	95	101	84	104	95	87	0.161703	0.068966	0.9880	NonOverlappingTemplate
91	98	107	107	102	89	93	97	110	106	0.832561	0.590311	0.9970	NonOverlappingTemplate
106	123	84	111	89	97	90	107	104	89	0.138069	0.312741	0.9910	NonOverlappingTemplate
111	100	96	106	97	106	81	100	102	101	0.755819	0.375684	0.9930	NonOverlappingTemplate
93	98	117	107	99	95	79	114	101	97	0.301194	0.891955	0.9910	NonOverlappingTemplate
102	98	99	96	99	101	103	106	104	92	0.996996	0.906337	0.9920	NonOverlappingTemplate
100	86	104	99	104	106	92	109	88	112	0.639202	0.771393	0.9870	NonOverlappingTemplate
100	94	88	88	109	127	104	88	103	99	0.160805	0.207968	0.9930	NonOverlappingTemplate
111	108	104	98	93	78	108	95	96	109	0.415422	0.238227	0.9880	NonOverlappingTemplate
90	95	95	97	92	114	120	102	83	112	0.183547	0.087955	0.9910	NonOverlappingTemplate
98	107	99	108	88	92	95	101	110	102	0.870856	0.742966	0.9890	NonOverlappingTemplate
97	102	117	99	98	96	102	102	87	100	0.834308	0.694735	0.9890	NonOverlappingTemplate
101	107	106	99	89	97	102	97	97	105	0.976878	0.766528	0.9850	NonOverlappingTemplate
85	97	101	124	129	86	86	93	91	108	0.007750	0.070139	0.9950	NonOverlappingTemplate
95	130	107	105	83	90	103	91	86	110	0.038313	0.013435	0.9910	NonOverlappingTemplate
101	103	95	113	103	107	99	92	86	101	0.812905	0.239174	0.9880	NonOverlappingTemplate
80	112	104	118	120	88	87	106	98	87	0.034484	0.107265	0.9910	NonOverlappingTemplate
100	98	117	90	102	105	86	88	109	105	0.467322	0.764926	0.9860	NonOverlappingTemplate
82	93	98	111	99	101	90	111	110	105	0.488534	0.261368	0.9910	NonOverlappingTemplate
93	98	96	121	87	104	95	104	97	105	0.564639	0.929965	0.9940	NonOverlappingTemplate
105	113	106	108	94	105	100	103	76	90	0.319084	0.062283	0.9860	NonOverlappingTemplate
88	88	89	106	101	97	125	106	101	99	0.263572	0.099741	0.9900	NonOverlappingTemplate
116	108	101	87	96	99	101	92	93	107	0.668321	0.278312	0.9870	NonOverlappingTemplate
104	110	117	102	100	96	90	78	101	102	0.339271	0.065112	0.9870	NonOverlappingTemplate
92	111	88	103	99	104	104	95	91	113	0.693142	0.863713	0.9960	NonOverlappingTemplate
105	111	86	105	103	86	105	103	90	106	0.593478	0.755025	0.9900	NonOverlappingTemplate
98	94	94	96	93	121	99	92	102	111	0.562591	0.274902	0.9910	NonOverlappingTemplate
122	117	111	91	95	92	85	96	98	93	0.138069	0.007448	0.9860	NonOverlappingTemplate
96	102	102	90	92	94	120	94	109	101	0.593478	0.263315	0.9910	NonOverlappingTemplate
113	106	91	109	106	88	102	92	98	95	0.695200	0.245199	0.9910	NonOverlappingTemplate
100	97	105	86	120	109	79	102	98	104	0.227180	0.804494	0.9910	NonOverlappingTemplate
102	104	111	99	91	100	102	103	96	92	0.957612	0.500504	0.9950	NonOverlappingTemplate
99	97	113	101	98	107	100	102	98	85	0.862883	0.238921	0.9940	NonOverlappingTemplate
119	88	98	100	93	95	106	100	99	102	0.715679	0.719405	0.9830	NonOverlappingTemplate
105	108	117	89	100	73	84	109	105	110	0.050305	0.092635	0.9860	NonOverlappingTemplate
90	105	101	94	95	86	96	102	119	112	0.429923	0.108634	0.9920	NonOverlappingTemplate
113	112	96	112	98	99	97	99	94	80	0.415422	0.035752	0.9910	NonOverlappingTemplate
85	104	90	102	122	104	98	80	114	101	0.106877	0.449751	0.9960	NonOverlappingTemplate
104	105	107	94	121	84	107	103	84	91	0.203351	0.196391	0.9880	NonOverlappingTemplate
101	104	99	97	98	97	116	99	96	93	0.934599	0.888525	0.9920	NonOverlappingTemplate



TROPIC01 ODR_TR01_PSR_004_TRNG_TropicSquare TROPIC01 True Random Number Generator (TRNG)

91	107	95	103	96	114	96	100	97	101	0.910091	0.948941	0.9900	NonOverlappingTemplate
95	102	98	97	99	102	101	103	106	97	0.999390	0.934670	0.9940	NonOverlappingTemplate
95	107	84	128	95	88	100	111	95	97	0.109435	0.916459	0.9940	NonOverlappingTemplate
109	88	89	92	97	108	117	104	101	95	0.520102	0.359986	0.9860	NonOverlappingTemplate
89	121	87	107	98	95	84	89	118	112	0.056426	0.095651	0.9890	NonOverlappingTemplate
95	103	103	109	87	83	116	91	118	95	0.177628	0.424989	0.9880	NonOverlappingTemplate
99	108	99	103	97	106	110	89	87	102	0.821937	0.494310	0.9900	OverlappingTemplate
91	94	117	113	109	106	97	90	97	86	0.345650	0.136417	0.9800 *	Universal
114	106	98	107	88	102	93	107	82	103	0.471146	0.293338	0.9880	ApproximateEntropy
94	75	76	67	86	85	99	109	83	78	0.055154	0.192835	0.9894	RandomExcursions
100	79	85	94	76	91	75	79	83	90	0.601644	0.398804	0.9859	RandomExcursions
84	74	80	86	87	99	95	84	91	72	0.565241	0.651262	0.9941	RandomExcursions
95	87	79	99	83	96	82	77	61	93	0.129180	0.154654	0.9836	RandomExcursions
90	74	86	86	85	85	93	89	87	77	0.942075	0.994133	0.9930	RandomExcursions
95	83	82	101	91	100	75	75	72	78	0.195369	0.017136	0.9894	RandomExcursions
99	80	63	84	85	94	88	96	85	78	0.251536	0.750252	0.9918	RandomExcursions
83	82	80	89	88	87	93	86	92	72	0.903500	0.900850	0.9918	RandomExcursions
67	98	84	71	84	99	102	73	90	84	0.070402	0.295137	0.9930	RandomExcursionsVariant
72	98	65	87	83	99	89	95	83	81	0.177514	0.221543	0.9918	RandomExcursionsVariant
90	71	93	68	85	98	91	88	89	79	0.360093	0.403261	0.9906	RandomExcursionsVariant
90	76	90	65	89	87	82	84	96	93	0.458159	0.366148	0.9918	RandomExcursionsVariant
94	77	75	82	81	103	85	89	88	78	0.548434	0.473348	0.9859	RandomExcursionsVariant
87	84	89	78	85	76	99	73	81	100	0.473855	0.518656	0.9859	RandomExcursionsVariant
80	80	86	87	75	79	97	91	90	87	0.859157	0.285610	0.9883	RandomExcursionsVariant
86	82	76	83	83	89	81	101	85	86	0.870485	0.507956	0.9918	RandomExcursionsVariant
73	100	79	100	83	67	91	90	77	92	0.148752	0.727757	0.9953	RandomExcursionsVariant
81	76	92	92	98	87	93	69	79	85	0.482935	0.297564	0.9953	RandomExcursionsVariant
83	93	91	92	79	80	94	87	65	88	0.501320	0.568996	0.9871	RandomExcursionsVariant
88	92	99	68	89	90	81	89	79	77	0.492091	0.304869	0.9906	RandomExcursionsVariant
89	83	81	92	78	91	82	85	82	89	0.983453	0.923930	0.9906	RandomExcursionsVariant
89	92	76	76	67	83	92	114	81	82	0.044304	0.221091	0.9883	RandomExcursionsVariant
90	77	83	75	71	83	97	98	85	93	0.431862	0.049473	0.9906	RandomExcursionsVariant
81	63	86	91	69	87	95	97	90	93	0.149743	0.072218	0.9894	RandomExcursionsVariant
81	77	74	82	69	85	105	83	90	106	0.073017	0.006834	0.9847	RandomExcursionsVariant
86	76	64	81	90	88	100	95	92	80	0.247047	0.060341	0.9883	RandomExcursionsVariant
96	91	107	93	99	94	106	98	118	98	0.739918	0.302127	0.9930	Serial
91	100	85	101	106	108	93	113	95	108	0.622546	0.518575	0.9880	Serial
89	88	103	102	105	106	103	112	98	94	0.805569	0.508265	0.9940	LinearComplexity

The minimum pass rate for each statistical test with the exception of the random excursion (variant) test is approximately = 0.980561 for a sample size = 1000 binary sequences.

The minimum pass rate for the random excursion (variant) test is approximately 0.979774 for a sample size = 852 binary sequences.

For further guidelines construct a probability table using the MAPLE program provided in the addendum section of the documentation.



The results show that all statistical tests passed both the proportion and uniformity criteria, with the exception of the Universal test for TRNG1.

The Universal test failed for TRNG1 with a passing proportion of 0.9800, marginally below the minimum required pass rate of 0.980561 (one test sub-threshold). This is expected and isolated failures of this magnitude are consistent with false positives, which are expected at the 1% significance level given the number of tests performed.



2.2.2 Statistical testing at all environmental conditions

The same test suite was then applied to a conditioned sequential dataset of 100 000 000 bits, collected from all 30 chips across the full range of stated environmental conditions. The datasets from each chip were split into 100 sequences, with 1 000 000 bits in each sequence.

Due to time constraints of capturing the dataset across all environmental conditions on the 30 chips, the dataset size is limited (more than 100 sequences is ideal for the statistical testing [4]). NIST SP 800-22 derives the proportion test threshold from a normal approximation to the binomial distribution at a target significance level of $\alpha = 0.01$; for $n = 100$ sequences this approximation yields a threshold of 97 passing sequences out of 100.

For a sample size of $n = 100$, the normal approximation is not precise, and the exact type 1 error of this threshold deviates noticeably from the nominal value. Modelling the per-sequence outcome as $X \sim \text{Binomial}(n, p)$, where X is the number of passing sequences out of $n = 100$ and $p = 1 - \alpha = 0.99$ is the single-sequence pass probability under the null hypothesis, the exact probability of falsely rejecting at the 97/100 threshold is

$$P(X \leq 96) \approx 1.84\%.$$

The effective significance level of the proportion test is therefore 1.84 % and not the intended 1 %.

The results are presented as a table per environmental condition, in which each row corresponds to a chip at a given trim setting (*high*, *default* or *low*) and each column to one statistical test from the suite. Each cell shows the proportion of sequences that failed the corresponding test; the cell is shaded green if the proportion test passed and red if it failed. The overall percentage of failed tests across the table is given in its title and can be compared directly with the theoretical expected fail rate of 1.84 % derived above.

Tests marked with an asterisk (“*”) aggregate multiple variants of the same test into a single column for readability; the value shown is the worst (highest) fail ratio across all merged variants. This explains the apparently elevated fail ratio (frequent red fail cells) of the NonOverlappingTemplate column. It aggregates 148 variants, so it is expected that at least one variant fails as a false positive even under the null hypothesis. It is merely an artifact of merging the many variants into a single column for readability.

Results for TRNG0 at –25 C, 25 C and 105 C are shown below. Tables for the remaining



temperatures, supply voltages and wafer corners are provided in Appendix 3.1.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -25.0 (°C), TRNG0, WID3
Fail ratio 2.18%, ~ 1.84% expected

WID3_CID246, default	0.99	0.98	0.99	0.99	0.99	0.97	1.0	0.97	0.98	0.99	0.98	1.0	0.98	1.0	0.9434	0.9434	0.96
WID3_CID246, high	1.0	0.99	1.0	1.0	1.0	1.0	1.0	0.99	0.98	0.99	0.98	0.96	1.0	0.99	0.9815	0.9815	0.96
WID3_CID246, low	0.99	0.98	1.0	0.99	0.99	1.0	1.0	0.99	1.0	1.0	0.99	1.0	0.99	1.0	0.9848	0.9697	0.95
WID3_CID253, default	1.0	1.0	1.0	1.0	0.97	1.0	0.98	0.99	0.98	0.98	1.0	0.99	0.99	1.0	0.9841	0.9841	0.95
WID3_CID253, high	0.99	1.0	0.98	0.99	0.99	0.99	1.0	1.0	0.99	0.99	1.0	0.99	0.99	1.0	0.9841	0.9524	0.96
WID3_CID253, low	0.98	0.99	0.98	0.99	1.0	0.99	1.0	1.0	1.0	0.98	1.0	1.0	0.99	0.98	0.9672	0.9508	0.96
WID3_CID254, default	1.0	1.0	1.0	1.0	0.99	1.0	0.98	0.99	1.0	0.99	0.98	1.0	0.99	1.0	0.9701	0.9701	0.95
WID3_CID254, high	1.0	0.97	1.0	1.0	0.99	0.98	0.97	0.98	0.98	0.98	1.0	1.0	1.0	1.0	0.9841	0.9683	0.97
WID3_CID254, low	0.99	1.0	0.98	0.99	0.98	1.0	1.0	0.98	0.99	0.97	1.0	1.0	0.99	1.0	0.9692	0.9692	0.94
WID3_CID255, default	1.0	1.0	1.0	1.0	0.99	0.99	1.0	0.99	1.0	0.99	0.97	0.99	0.98	1.0	0.9589	0.9726	0.95
WID3_CID255, high	1.0	1.0	0.99	1.0	0.99	1.0	0.99	0.97	1.0	0.98	0.99	0.99	1.0	1.0	0.9692	0.9385	0.96
WID3_CID255, low	1.0	0.99	0.99	1.0	0.99	1.0	1.0	0.99	1.0	0.99	1.0	0.99	0.99	0.99	0.9661	0.9831	0.96
WID3_CID256, default	0.99	0.99	0.99	0.99	0.99	0.99	0.99	1.0	1.0	1.0	0.98	0.99	0.97	1.0	0.9821	0.9821	0.95
WID3_CID256, high	0.99	0.96	0.99	1.0	0.97	0.96	0.98	0.97	0.98	0.98	0.98	0.99	1.0	0.96	0.9821	0.9821	0.96
WID3_CID256, low	0.98	0.99	0.97	0.98	0.99	0.98	1.0	0.99	0.99	0.98	1.0	0.98	0.99	0.99	0.9815	0.9815	0.96
WID3_CID258, default	1.0	1.0	1.0	1.0	0.99	0.98	0.98	0.97	0.99	0.99	1.0	1.0	1.0	0.98	0.9836	0.9672	0.96
WID3_CID258, high	0.99	0.99	0.99	0.99	0.98	1.0	0.97	1.0	0.99	0.99	0.98	1.0	1.0	0.99	0.9672	0.9672	0.96
WID3_CID258, low	0.99	0.99	0.99	0.99	1.0	1.0	0.98	0.99	0.97	0.99	1.0	0.99	0.99	0.99	0.9649	0.9649	0.95
WID3_CID259, default	1.0	0.99	1.0	1.0	1.0	1.0	0.99	1.0	0.99	1.0	1.0	0.97	0.97	0.96	0.9683	0.9683	0.95
WID3_CID259, high	0.98	0.98	0.99	0.99	0.99	1.0	0.99	0.99	0.99	0.97	1.0	0.99	0.98	0.99	0.9434	0.9623	0.95
WID3_CID259, low	0.99	0.98	0.99	0.98	0.98	1.0	0.96	1.0	0.99	0.98	1.0	0.98	1.0	0.99	0.9841	0.9841	0.96
WID3_CID260, default	1.0	0.99	1.0	0.99	0.98	0.98	0.98	0.99	1.0	1.0	0.99	1.0	0.98	0.99	0.9649	0.9825	0.95
WID3_CID260, high	1.0	0.98	1.0	1.0	1.0	1.0	0.97	0.98	1.0	0.99	1.0	0.97	0.96	0.98	0.9688	0.9688	0.94
WID3_CID260, low	0.99	1.0	0.98	0.99	1.0	1.0	0.99	0.99	0.99	0.99	0.98	0.99	0.97	0.98	0.9839	0.9516	0.95
WID3_CID261, default	0.98	1.0	0.99	0.98	0.99	0.98	0.99	1.0	0.99	0.99	0.99	0.98	0.97	0.99	0.9672	0.9836	0.95
WID3_CID261, high	0.99	1.0	0.99	0.99	0.99	0.99	1.0	0.99	0.97	1.0	0.98	0.99	1.0	0.99	0.9559	0.9706	0.95
WID3_CID261, low	0.99	1.0	0.99	0.99	0.99	0.99	0.98	1.0	0.98	1.0	0.99	1.0	1.0	1.0	0.9833	0.95	0.96
WID3_CID316, default	0.99	0.99	0.99	0.99	1.0	1.0	0.98	0.97	0.99	0.98	1.0	0.99	1.0	1.0	0.9833	0.9667	0.94
WID3_CID316, high	0.98	1.0	0.98	0.97	0.99	0.99	0.99	1.0	0.99	0.99	1.0	0.96	0.99	0.98	0.9649	0.9649	0.95
WID3_CID316, low	1.0	0.98	1.0	1.0	1.0	0.97	0.99	0.97	0.98	0.99	0.98	1.0	1.0	1.0	0.9706	0.9853	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 1: TRNG0 -25 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 25.0 (°C), TRNG0, WID3
Fail ratio 1.87%, ~ 1.84% expected

WID3_CID246, default	1.0	0.98	0.99	1.0	1.0	1.0	1.0	1.0	0.98	0.99	0.97	0.99	1.0	0.98	0.9848	0.9848	0.95
WID3_CID246, high	0.99	0.99	0.99	1.0	1.0	0.97	1.0	0.99	0.98	1.0	0.99	0.99	0.99	1.0	0.9636	0.9636	0.96
WID3_CID246, low	0.99	0.99	0.99	0.99	0.98	0.99	0.99	0.99	0.99	0.99	1.0	1.0	1.0	1.0	0.9672	0.9508	0.95
WID3_CID253, default	0.98	1.0	0.99	0.98	1.0	0.98	1.0	0.99	0.96	0.99	0.99	0.99	0.98	0.99	0.9677	0.9516	0.95
WID3_CID253, high	0.98	0.99	0.98	0.98	1.0	0.98	0.98	0.99	1.0	1.0	0.99	0.99	0.99	0.99	0.9365	0.9841	0.96
WID3_CID253, low	0.97	0.99	0.97	0.98	0.98	0.95	0.99	1.0	1.0	0.97	0.98	0.98	0.98	1.0	0.963	0.963	0.95
WID3_CID254, default	0.98	1.0	0.98	0.98	1.0	0.97	0.97	0.98	0.98	0.98	1.0	0.99	1.0	0.98	0.9538	0.9692	0.93
WID3_CID254, high	1.0	0.99	1.0	1.0	1.0	0.99	0.98	0.98	0.98	0.98	0.99	1.0	1.0	0.99	0.9683	0.9683	0.95
WID3_CID254, low	0.98	0.99	0.98	0.98	0.99	0.99	1.0	1.0	0.98	0.98	1.0	1.0	1.0	1.0	0.9825	0.9649	0.96
WID3_CID255, default	1.0	0.99	1.0	1.0	1.0	1.0	0.95	0.99	0.99	1.0	0.98	1.0	0.98	1.0	0.9818	0.9455	0.96
WID3_CID255, high	0.99	0.99	0.99	0.99	1.0	0.99	0.99	1.0	0.99	0.99	0.99	0.99	0.99	1.0	0.9857	0.9571	0.96
WID3_CID255, low	0.99	0.98	0.99	1.0	0.98	0.99	0.99	1.0	0.99	0.99	1.0	0.99	0.99	0.99	0.9492	0.9661	0.96
WID3_CID256, default	0.97	1.0	0.97	0.97	1.0	0.98	0.98	1.0	0.99	0.96	1.0	0.98	0.99	0.98	0.9726	0.9726	0.96
WID3_CID256, high	0.97	0.98	0.97	0.97	1.0	1.0	0.98	0.97	0.98	0.99	0.97	1.0	1.0	1.0	1.0	0.9811	0.96
WID3_CID256, low	0.99	0.97	0.99	0.99	1.0	1.0	1.0	1.0	0.98	0.97	1.0	1.0	0.99	0.99	0.9841	0.9841	0.96
WID3_CID258, default	1.0	0.99	1.0	1.0	1.0	0.99	1.0	0.99	0.98	1.0	0.99	0.98	0.98	1.0	0.9672	0.9836	0.95
WID3_CID258, high	0.99	0.99	0.99	0.99	1.0	0.97	0.98	0.98	1.0	0.99	1.0	1.0	1.0	0.96	0.9697	0.9394	0.95
WID3_CID258, low	1.0	0.99	1.0	0.98	0.99	0.99	1.0	0.98	0.99	1.0	0.99	1.0	1.0	0.97	0.9818	0.9636	0.95
WID3_CID259, default	0.99	1.0	0.98	0.99	0.98	1.0	1.0	0.99	0.99	0.98	1.0	0.99	0.99	1.0	0.9714	0.9857	0.96
WID3_CID259, high	0.99	1.0	0.99	0.99	0.99	1.0	1.0	0.98	0.98	0.98	1.0	1.0	1.0	0.98	0.9375	0.9531	0.96
WID3_CID259, low	0.99	0.99	0.98	0.98	0.99	0.99	0.99	0.98	1.0	0.98	0.99	0.98	0.98	0.99	0.9846	0.9846	0.96
WID3_CID260, default	1.0	0.99	1.0	0.99	1.0	0.99	0.98	0.99	0.98	0.97	1.0	0.99	0.98	0.97	0.9844	0.9844	0.96
WID3_CID260, high	0.99	0.99	0.98	0.99	1.0	1.0	0.99	0.99	1.0	0.97	1.0	1.0	0.99	1.0	0.9808	0.9615	0.96
WID3_CID260, low	0.98	0.99	0.99	0.98	1.0	0.99	1.0	0.97	0.99	1.0	1.0	1.0	0.99	1.0	0.9688	0.9844	0.96
WID3_CID261, default	0.98	1.0	0.99	0.98	0.98	0.98	1.0	0.98	0.99	1.0	0.99	1.0	0.99	0.97	0.9677	0.9839	0.96
WID3_CID261, high	1.0	0.99	0.99	1.0	0.97	0.98	0.99	1.0	0.98	0.97	0.99	1.0	0.98	0.98	0.9649	0.9649	0.95
WID3_CID261, low	0.99	0.98	0.99	0.99	1.0	1.0	0.98	0.98	1.0	0.99	0.98	1.0	0.99	0.98	0.9531	0.9688	0.96
WID3_CID316, default	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.97	0.99	0.99	0.99	0.99	1.0	0.95	0.9846	0.9846	0.96
WID3_CID316, high	0.99	0.99	0.99	0.99	1.0	0.99	1.0	0.99	1.0	1.0	0.99	0.99	0.97	0.99	0.9855	0.9855	0.97
WID3_CID316, low	1.0	0.99	1.0	1.0	0.98	0.99	0.98	1.0	0.98	0.98	0.98	1.0	1.0	0.99	0.9701	0.9851	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 2: TRNG0 25 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 105.0 (°C), TRNG0, WID3
Fail ratio 1.89%, ~ 1.84% expected

WID3_CID246, default	0.98	0.99	0.98	0.98	0.96	0.98	0.99	0.99	0.99	1.0	1.0	0.98	1.0	1.0	0.9828	0.9655	0.96
WID3_CID246, high	1.0	1.0	1.0	1.0	1.0	0.99	0.96	0.99	0.97	0.98	0.99	0.97	0.99	0.97	0.9692	0.9692	0.94
WID3_CID246, low	0.97	0.99	0.97	0.97	0.98	0.99	0.99	0.98	0.99	1.0	0.99	1.0	0.99	1.0	0.971	0.971	0.96
WID3_CID253, default	0.99	1.0	0.99	0.99	0.99	0.98	1.0	0.97	0.99	0.98	0.99	1.0	0.99	1.0	0.9706	0.9706	0.95
WID3_CID253, high	0.99	1.0	0.99	0.99	1.0	0.98	1.0	0.99	0.99	0.97	0.99	0.99	1.0	1.0	0.9692	0.9846	0.96
WID3_CID253, low	0.97	0.99	0.98	0.97	0.98	0.99	1.0	0.98	1.0	1.0	0.99	1.0	1.0	0.99	0.9697	0.9848	0.95
WID3_CID254, default	0.98	1.0	1.0	0.99	0.99	0.99	1.0	0.98	1.0	1.0	0.99	0.99	0.98	0.99	0.9697	0.9848	0.96
WID3_CID254, high	1.0	1.0	1.0	1.0	0.97	0.98	0.99	0.99	0.99	1.0	0.99	0.94	0.98	1.0	0.9851	0.9701	0.95
WID3_CID254, low	1.0	0.99	1.0	0.99	0.99	0.99	0.98	0.99	0.99	1.0	0.99	1.0	0.98	0.99	0.9538	0.9692	0.95
WID3_CID255, default	0.99	0.99	1.0	0.99	0.99	1.0	1.0	1.0	0.97	1.0	0.99	0.99	0.98	0.99	0.9538	0.9692	0.95
WID3_CID255, high	0.99	0.99	0.99	0.98	1.0	1.0	0.99	0.98	0.98	0.99	1.0	1.0	1.0	0.97	0.9683	0.9683	0.95
WID3_CID255, low	0.98	0.99	0.98	0.99	0.99	1.0	1.0	0.99	0.98	0.99	0.99	0.98	1.0	0.98	0.9545	0.9394	0.96
WID3_CID256, default	0.99	0.99	0.99	0.99	0.97	1.0	0.99	0.97	0.99	0.98	0.99	0.98	0.99	0.99	0.9688	0.9219	0.96
WID3_CID256, high	1.0	0.98	0.99	1.0	0.99	1.0	0.99	0.98	0.99	1.0	1.0	0.98	1.0	0.99	0.9403	0.9701	0.96
WID3_CID256, low	0.99	1.0	0.99	0.99	0.99	0.99	0.97	0.99	1.0	1.0	0.99	0.98	1.0	0.99	0.9697	0.9697	0.96
WID3_CID258, default	0.99	0.98	0.99	1.0	0.99	0.98	1.0	0.97	0.97	0.98	0.97	1.0	1.0	0.98	0.9855	0.9855	0.96
WID3_CID258, high	0.99	1.0	0.99	0.99	1.0	1.0	1.0	0.98	1.0	0.99	1.0	0.99	1.0	0.99	0.9636	0.9818	0.94
WID3_CID258, low	1.0	0.98	1.0	0.99	1.0	0.99	0.98	0.98	1.0	0.98	1.0	0.98	0.99	1.0	0.9792	0.9583	0.94
WID3_CID259, default	0.99	1.0	0.99	0.99	0.98	0.96	1.0	1.0	0.99	1.0	0.99	0.99	0.99	1.0	0.963	0.9815	0.97
WID3_CID259, high	0.97	0.97	0.97	0.97	0.98	1.0	0.99	1.0	0.98	0.98	0.99	0.99	0.99	0.99	0.9697	0.9848	0.96
WID3_CID259, low	1.0	1.0	1.0	1.0	0.99	0.99	1.0	0.98	0.99	0.97	0.98	0.99	1.0	0.99	0.9683	0.9683	0.95
WID3_CID260, default	1.0	1.0	1.0	1.0	0.99	0.98	0.99	0.97	0.98	0.99	0.98	0.97	1.0	1.0	1.0	0.9636	0.95
WID3_CID260, high	1.0	0.99	0.99	1.0	0.99	0.99	0.98	0.99	0.98	0.95	0.98	0.98	0.99	1.0	0.9706	0.9559	0.96
WID3_CID260, low	0.98	1.0	0.98	0.98	0.99	0.97	1.0	0.99	1.0	0.99	1.0	0.99	0.99	0.99	0.9508	0.9836	0.95
WID3_CID261, default	0.98	0.99	0.97	0.98	1.0	0.99	1.0	0.99	0.98	0.99	0.99	1.0	1.0	0.99	0.9828	0.931	0.93
WID3_CID261, high	1.0	0.99	1.0	1.0	0.99	0.98	0.98	0.99	0.98	0.97	1.0	0.98	1.0	0.99	0.9701	0.9701	0.96
WID3_CID261, low	1.0	0.98	0.99	1.0	0.97	1.0	0.98	0.99	0.95	0.99	1.0	0.99	0.98	1.0	0.9552	0.9552	0.96
WID3_CID316, default	0.99	0.99	0.99	1.0	0.98	1.0	0.98	0.99	0.99	1.0	0.99	1.0	1.0	0.97	0.9531	0.9531	0.95
WID3_CID316, high	1.0	0.99	1.0	1.0	1.0	0.99	0.99	0.98	0.97	0.98	0.99	0.99	0.99	1.0	0.9677	0.9355	0.95
WID3_CID316, low	1.0	1.0	1.0	1.0	0.98	0.99	1.0	0.99	0.99	0.98	0.98	1.0	0.98	0.97	0.9545	0.9697	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 3: TRNG0 105 °C WID3 statistical test results.



Across all 30 chips, both TRNGs, all three trim settings and the full range of stated temperatures and trim voltages, the proportion test fail rates are close to the theoretical 1.84 % expected fail rate derived above. No dependence of the fail rate on temperature, supply voltage or trim setting is observed.

Uniformity test results are not presented but no failures were observed across all conditions and tested chips.



2.3 NIST SP 800-90B min entropy assessment

The NIST SP 800-90B [3] min entropy assessment is computed by the official Entropy-Assessment implementation [7].

The assessment is done using the non-IID track of NIST SP 800-90B. Since a non-vetted conditioning component is used, the entropy estimate is updated using the NIST SP 800-90B section 3.1.5.2 [3].

2.3.1 Entropy assessment on the 1 GiB dataset

First, the 1 GiB dataset from WID3_CID309 taken at nominal environmental conditions was tested. As the restart dataset is forced to be only 1000*1000 bits, it is kept to this size also in this case.

Table 1: Entropy Values for Different TRNGs

TRNG	RRN min-entropy	IRN min-entropy	Restart min-entropy	Final min-entropy
TRNG0	0.945048	0.932087	0.854776	0.932087
TRNG1	0.897038	0.947256	0.832196	0.947256

The final min-entropy estimate for both TRNGs is in line with expectations of the NIST SP 800-90B tests [8].

The Restart min-entropy is noticeably lower than the RRN and IRN estimates for both TRNGs. Two factors might contribute to this:

- The restart test captures the entropy available immediately after the analog entropy source is powered on, before it has had time to fully accumulate entropy. To prevent this from affecting cryptographic operations, TROPIC01 starts the analog entropy source early in its boot sequence and guarantees that at least ~ 200 ms elapse before any output is used for cryptographic purposes, by which point the source has reached its steady-state entropy level reflected in the RRN and IRN estimates.
- The dataset size used for the restart test is fixed at $1\,000 \times 1\,000$ bits, the minimum required by NIST SP 800-90B. We have observed that on the same TRNG and under the same conditions, the entropy estimator returns systematically lower values for 1 000 000-bit datasets than for the larger (~ 1 GiB) sequential datasets used for the RRN and IRN estimates. Part of the apparent gap between the Restart and RRN/IRN estimates might therefore be an artifact of the estimator's behaviour at the smaller sample size.



2.3.2 Entropy assessment at all environmental conditions

The same entropy assessment was repeated across all stated environmental conditions, using the 100 000 000-bit sequential and conditioned sequential datasets together with the $1\,000 \times 1\,000$ -bit restart dataset. The goal was to assess if the environmental conditions have an impact on the min-entropy estimation, which would indicate a degradation of the TRNG's performance at these conditions.

The results are summarised in the following plot, where each point shows the average min-entropy estimate over all chips at the given temperature and trim setting, and the shaded area indicates the min-max range across chips. The plot below is for wafer WID3. The corresponding plots for WID8 and WID11 are provided in Appendix 3.2.

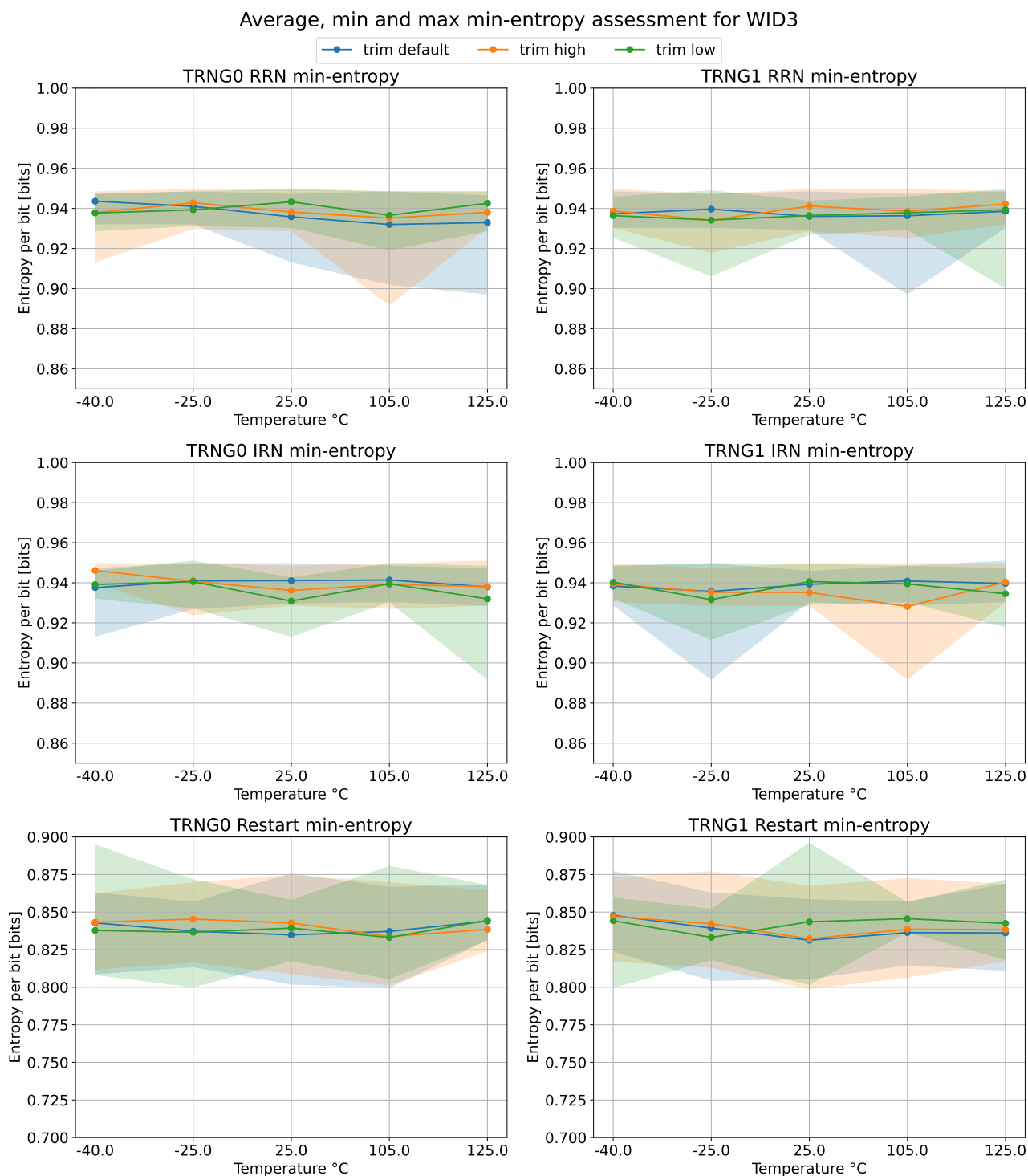


Figure 4: Combined entropy assessment for WID3.



We can see that the min-entropy estimations do not decrease with changing environmental conditions (temperature and voltage) and no dependence of the entropy assessment on these conditions is observed.

The results for individual chips from WID3 for both TRNGs can be seen here. For WID8 and WID11 see 3.2.

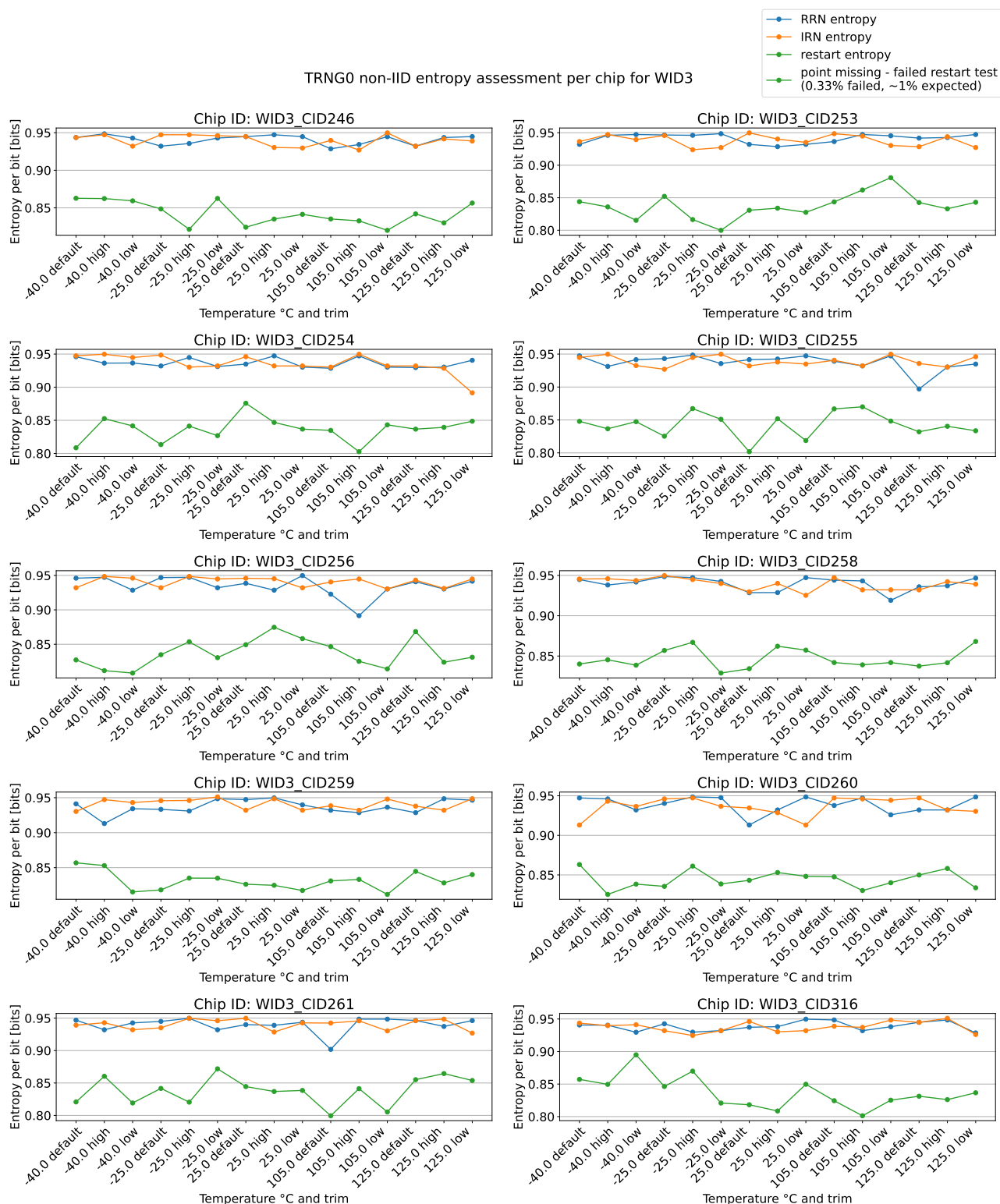


Figure 5: TRNG0 entropy assessment at all conditions for WID3.



Figure 6: TRNG1 entropy assessment at all conditions for WID3.



Across all measured environmental conditions, the min-entropy estimates are in line with expectations from the NIST SP 800-90B tests. The restart dataset consistently yields lower values than the sequential and conditioned sequential datasets, for the reasons already discussed in the previous Subsection 2.3.1. The number of restart-dataset health test failures is also in line with expectations, remaining close to the 1 % statistical significance level.

Most importantly, the min-entropy estimates show no degradation at the measured temperatures and trim voltages, demonstrating that the TRNG is stable across the entire stated operating range.

2.4 Conclusion

Both TRNG instances performed as expected across the full set of measured conditions, covering all 30 chips, three wafers, the stated temperature range and all entropy source voltages.

The NIST SP 800-90B entropy assessment produced good min-entropy estimates on the sequential, conditioned sequential and restart datasets, with no degradation observed across the measured temperatures or entropy-source supply voltages.

The NIST SP 800-22 statistical test suite found no significant evidence against the expected statistical properties of the data, and the test outcomes likewise show no dependence on environmental conditions or chips.

Taken together, these results show that the TROPIC01 TRNG meets the statistical and entropy requirements expected of a cryptographic-grade entropy source and is suitable for use in cryptographic applications across the entire stated operating range.



3 Appendix

3.1 NIST SP 800-22 statistical test results

NIST Test Suite Summary for -40.0 (°C), TRNG0, WID3
Fail ratio 2.41%, ~ 1.84% expected

WID3_CID246, default	0.99	0.98	1.0	0.98	0.99	0.99	0.99	0.97	1.0	0.99	1.0	1.0	1.0	0.99	0.9545	0.9394	0.95
WID3_CID246, high	0.98	0.98	1.0	0.98	1.0	0.98	0.98	0.97	0.98	0.98	0.98	0.99	0.98	0.9531	0.9688	0.94	
WID3_CID246, low	0.99	0.97	0.99	0.99	1.0	1.0	0.99	0.99	0.97	0.99	0.98	1.0	0.99	0.99	0.9649	0.9825	0.94
WID3_CID253, default	0.98	0.99	0.98	0.98	0.99	0.99	1.0	0.99	0.98	0.99	0.98	0.99	0.97	0.9839	0.9516	0.94	
WID3_CID253, high	0.99	0.99	0.99	0.97	1.0	0.99	0.99	1.0	0.97	1.0	0.99	1.0	1.0	0.9643	0.9643	0.96	
WID3_CID253, low	1.0	1.0	1.0	1.0	0.98	0.99	0.98	1.0	1.0	1.0	0.97	1.0	1.0	0.97	0.9623	0.9811	0.95
WID3_CID254, default	0.99	0.99	0.99	0.99	0.97	1.0	0.97	0.97	1.0	1.0	1.0	0.99	0.99	1.0	0.9571	0.9714	0.96
WID3_CID254, high	1.0	1.0	1.0	1.0	1.0	0.99	0.99	0.99	1.0	1.0	0.98	0.99	0.98	0.99	0.9831	0.9492	0.96
WID3_CID254, low	0.98	0.99	1.0	0.99	1.0	0.99	0.99	1.0	0.98	0.99	0.99	0.97	0.98	1.0	0.9677	0.9677	0.95
WID3_CID255, default	1.0	0.99	1.0	1.0	0.99	0.99	1.0	0.97	0.96	1.0	1.0	1.0	0.99	0.99	0.9706	0.9853	0.96
WID3_CID255, high	0.98	0.95	0.98	0.97	0.98	1.0	1.0	0.97	1.0	0.99	1.0	0.98	0.99	0.99	0.9615	0.9808	0.96
WID3_CID255, low	0.99	0.98	1.0	1.0	1.0	0.99	1.0	0.99	0.99	0.99	0.97	1.0	0.99	0.99	0.9841	0.9683	0.94
WID3_CID256, default	0.99	0.99	0.99	0.99	1.0	0.98	0.98	1.0	1.0	0.99	0.99	1.0	0.99	0.99	0.9815	0.9815	0.94
WID3_CID256, high	0.99	0.97	0.98	1.0	1.0	0.99	0.99	0.97	0.98	0.99	1.0	1.0	0.99	1.0	0.9677	0.9839	0.95
WID3_CID256, low	0.97	1.0	0.97	0.98	0.99	1.0	1.0	0.98	0.99	0.96	0.99	0.98	0.98	0.99	0.9559	0.9853	0.96
WID3_CID258, default	1.0	0.99	1.0	1.0	0.99	1.0	0.98	0.96	0.98	0.99	0.99	1.0	1.0	0.99	0.9821	0.9464	0.96
WID3_CID258, high	0.98	1.0	0.97	0.98	0.99	1.0	0.98	0.99	0.99	0.99	0.99	1.0	0.98	0.9831	0.9831	0.95	
WID3_CID258, low	1.0	0.98	0.99	0.99	0.99	1.0	0.98	0.99	0.98	0.96	0.98	1.0	1.0	0.99	0.971	0.971	0.94
WID3_CID259, default	1.0	0.97	1.0	1.0	0.99	1.0	0.99	0.98	0.99	1.0	0.99	0.97	0.96	0.99	0.9818	0.9818	0.95
WID3_CID259, high	0.99	1.0	0.99	0.98	1.0	0.99	0.99	0.98	0.98	0.97	1.0	0.98	0.97	0.98	0.9683	0.9524	0.96
WID3_CID259, low	0.99	0.97	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.97	1.0	0.99	1.0	0.9836	0.9672	0.96
WID3_CID260, default	1.0	0.99	0.99	1.0	0.98	0.99	0.96	0.99	0.98	0.99	1.0	0.99	1.0	0.98	0.9815	0.963	0.96
WID3_CID260, high	1.0	0.98	1.0	0.99	1.0	1.0	0.99	0.97	0.99	0.99	0.99	0.98	1.0	0.99	0.9559	0.9559	0.96
WID3_CID260, low	0.99	0.97	1.0	0.99	1.0	1.0	0.98	0.98	0.99	0.99	0.99	1.0	0.99	0.9857	0.9714	0.96	
WID3_CID261, default	0.98	0.99	0.98	0.99	1.0	0.95	1.0	0.99	0.99	0.98	0.98	1.0	1.0	0.99	0.9677	0.9677	0.96
WID3_CID261, high	1.0	0.99	1.0	1.0	0.98	0.99	0.97	0.96	0.98	1.0	0.98	0.98	0.97	0.99	0.9683	0.9841	0.94
WID3_CID261, low	1.0	0.98	1.0	1.0	0.99	0.98	0.98	0.98	0.99	0.98	0.98	0.97	1.0	0.98	0.9667	0.9667	0.96
WID3_CID316, default	0.98	1.0	0.98	0.99	1.0	1.0	1.0	1.0	0.99	1.0	0.99	1.0	1.0	0.9836	0.9672	0.96	
WID3_CID316, high	0.98	0.98	0.99	0.99	0.98	1.0	0.99	0.97	0.99	1.0	0.98	0.98	0.99	1.0	0.9718	0.9718	0.95
WID3_CID316, low	1.0	0.99	1.0	1.0	1.0	0.98	1.0	0.99	0.98	0.99	0.99	1.0	1.0	0.9683	0.9683	0.95	
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 7: TRNG0 -40 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -25.0 (°C), TRNG0, WID3
Fail ratio 2.18%, ~ 1.84% expected

WID3_CID246, default	0.99	0.98	0.99	0.99	0.99	0.97	1.0	0.97	0.98	0.99	0.98	1.0	0.98	1.0	0.9434	0.9434	0.96
WID3_CID246, high	1.0	0.99	1.0	1.0	1.0	1.0	1.0	0.99	0.98	0.99	0.98	0.96	1.0	0.99	0.9815	0.9815	0.96
WID3_CID246, low	0.99	0.98	1.0	0.99	0.99	1.0	1.0	0.99	1.0	1.0	0.99	1.0	0.99	1.0	0.9848	0.9697	0.95
WID3_CID253, default	1.0	1.0	1.0	1.0	0.97	1.0	0.98	0.99	0.98	0.98	1.0	0.99	0.99	1.0	0.9841	0.9841	0.95
WID3_CID253, high	0.99	1.0	0.98	0.99	0.99	0.99	1.0	1.0	0.99	0.99	1.0	0.99	0.99	1.0	0.9841	0.9524	0.96
WID3_CID253, low	0.98	0.99	0.98	0.99	1.0	0.99	1.0	1.0	1.0	0.98	1.0	1.0	0.99	0.98	0.9672	0.9508	0.96
WID3_CID254, default	1.0	1.0	1.0	1.0	0.99	1.0	0.98	0.99	1.0	0.99	0.98	1.0	0.99	1.0	0.9701	0.9701	0.95
WID3_CID254, high	1.0	0.97	1.0	1.0	0.99	0.98	0.97	0.98	0.98	0.98	1.0	1.0	1.0	1.0	0.9841	0.9683	0.97
WID3_CID254, low	0.99	1.0	0.98	0.99	0.98	1.0	1.0	0.98	0.99	0.97	1.0	1.0	0.99	1.0	0.9692	0.9692	0.94
WID3_CID255, default	1.0	1.0	1.0	1.0	0.99	0.99	1.0	0.99	1.0	0.99	0.97	0.99	0.98	1.0	0.9589	0.9726	0.95
WID3_CID255, high	1.0	1.0	0.99	1.0	0.99	1.0	0.99	0.97	1.0	0.98	0.99	0.99	1.0	1.0	0.9692	0.9385	0.96
WID3_CID255, low	1.0	0.99	0.99	1.0	0.99	1.0	1.0	0.99	1.0	0.99	1.0	0.99	0.99	0.99	0.9661	0.9831	0.96
WID3_CID256, default	0.99	0.99	0.99	0.99	0.99	0.99	0.99	1.0	1.0	1.0	0.98	0.99	0.97	1.0	0.9821	0.9821	0.95
WID3_CID256, high	0.99	0.96	0.99	1.0	0.97	0.96	0.98	0.97	0.98	0.98	0.98	0.99	1.0	0.96	0.9821	0.9821	0.96
WID3_CID256, low	0.98	0.99	0.97	0.98	0.99	0.98	1.0	0.99	0.99	0.98	1.0	0.98	0.99	0.99	0.9815	0.9815	0.96
WID3_CID258, default	1.0	1.0	1.0	1.0	0.99	0.98	0.98	0.97	0.99	0.99	1.0	1.0	1.0	0.98	0.9836	0.9672	0.96
WID3_CID258, high	0.99	0.99	0.99	0.99	0.98	1.0	0.97	1.0	0.99	0.99	0.98	1.0	1.0	0.99	0.9672	0.9672	0.96
WID3_CID258, low	0.99	0.99	0.99	0.99	1.0	1.0	0.98	0.99	0.97	0.99	1.0	0.99	0.99	0.99	0.9649	0.9649	0.95
WID3_CID259, default	1.0	0.99	1.0	1.0	1.0	1.0	0.99	1.0	0.99	1.0	1.0	0.97	0.97	0.96	0.9683	0.9683	0.95
WID3_CID259, high	0.98	0.98	0.99	0.99	0.99	1.0	0.99	0.99	0.99	0.97	1.0	0.99	0.98	0.99	0.9434	0.9623	0.95
WID3_CID259, low	0.99	0.98	0.99	0.98	0.98	1.0	0.96	1.0	0.99	0.98	1.0	0.98	1.0	0.99	0.9841	0.9841	0.96
WID3_CID260, default	1.0	0.99	1.0	0.99	0.98	0.98	0.98	0.99	1.0	1.0	0.99	1.0	0.98	0.99	0.9649	0.9825	0.95
WID3_CID260, high	1.0	0.98	1.0	1.0	1.0	1.0	0.97	0.98	1.0	0.99	1.0	0.97	0.96	0.98	0.9688	0.9688	0.94
WID3_CID260, low	0.99	1.0	0.98	0.99	1.0	1.0	0.99	0.99	0.99	0.99	0.98	0.99	0.97	0.98	0.9839	0.9516	0.95
WID3_CID261, default	0.98	1.0	0.99	0.98	0.99	0.98	0.99	1.0	0.99	0.99	0.99	0.98	0.97	0.99	0.9672	0.9836	0.95
WID3_CID261, high	0.99	1.0	0.99	0.99	0.99	0.99	1.0	0.99	0.97	1.0	0.98	0.99	1.0	0.99	0.9559	0.9706	0.95
WID3_CID261, low	0.99	1.0	0.99	0.99	0.99	0.99	0.98	1.0	0.98	1.0	0.99	1.0	1.0	1.0	0.9833	0.95	0.96
WID3_CID316, default	0.99	0.99	0.99	0.99	1.0	1.0	0.98	0.97	0.99	0.98	1.0	0.99	1.0	1.0	0.9833	0.9667	0.94
WID3_CID316, high	0.98	1.0	0.98	0.97	0.99	0.99	0.99	1.0	0.99	0.99	1.0	0.96	0.99	0.98	0.9649	0.9649	0.95
WID3_CID316, low	1.0	0.98	1.0	1.0	1.0	0.97	0.99	0.97	0.98	0.99	0.98	1.0	1.0	1.0	0.9706	0.9853	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 8: TRNG0 -25 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 25.0 (°C), TRNG0, WID3
Fail ratio 1.87%, ~ 1.84% expected

WID3_CID246, default	1.0	0.98	0.99	1.0	1.0	1.0	1.0	1.0	0.98	0.99	0.97	0.99	1.0	0.98	0.9848	0.9848	0.95
WID3_CID246, high	0.99	0.99	0.99	1.0	1.0	0.97	1.0	0.99	0.98	1.0	0.99	0.99	0.99	1.0	0.9636	0.9636	0.96
WID3_CID246, low	0.99	0.99	0.99	0.99	0.98	0.99	0.99	0.99	0.99	0.99	1.0	1.0	1.0	1.0	0.9672	0.9508	0.95
WID3_CID253, default	0.98	1.0	0.99	0.98	1.0	0.98	1.0	0.99	0.96	0.99	0.99	0.99	0.98	0.99	0.9677	0.9516	0.95
WID3_CID253, high	0.98	0.99	0.98	0.98	1.0	0.98	0.98	0.99	1.0	1.0	0.99	0.99	0.99	0.99	0.9365	0.9841	0.96
WID3_CID253, low	0.97	0.99	0.97	0.98	0.98	0.95	0.99	1.0	1.0	0.97	0.98	0.98	0.98	1.0	0.963	0.963	0.95
WID3_CID254, default	0.98	1.0	0.98	0.98	1.0	0.97	0.97	0.98	0.98	0.98	1.0	0.99	1.0	0.98	0.9538	0.9692	0.93
WID3_CID254, high	1.0	0.99	1.0	1.0	1.0	0.99	0.98	0.98	0.98	0.98	0.99	1.0	1.0	0.99	0.9683	0.9683	0.95
WID3_CID254, low	0.98	0.99	0.98	0.98	0.99	0.99	1.0	1.0	0.98	0.98	1.0	1.0	1.0	1.0	0.9825	0.9649	0.96
WID3_CID255, default	1.0	0.99	1.0	1.0	1.0	1.0	0.95	0.99	0.99	1.0	0.98	1.0	0.98	1.0	0.9818	0.9455	0.96
WID3_CID255, high	0.99	0.99	0.99	0.99	1.0	0.99	0.99	1.0	0.99	0.99	0.99	0.99	0.99	1.0	0.9857	0.9571	0.96
WID3_CID255, low	0.99	0.98	0.99	1.0	0.98	0.99	0.99	1.0	0.99	0.99	1.0	0.99	0.99	0.99	0.9492	0.9661	0.96
WID3_CID256, default	0.97	1.0	0.97	0.97	1.0	0.98	0.98	1.0	0.99	0.96	1.0	0.98	0.99	0.98	0.9726	0.9726	0.96
WID3_CID256, high	0.97	0.98	0.97	0.97	1.0	1.0	0.98	0.97	0.98	0.99	0.97	1.0	1.0	1.0	1.0	0.9811	0.96
WID3_CID256, low	0.99	0.97	0.99	0.99	1.0	1.0	1.0	1.0	0.98	0.97	1.0	1.0	0.99	0.99	0.9841	0.9841	0.96
WID3_CID258, default	1.0	0.99	1.0	1.0	1.0	0.99	1.0	0.99	0.98	1.0	0.99	0.98	0.98	1.0	0.9672	0.9836	0.95
WID3_CID258, high	0.99	0.99	0.99	0.99	1.0	0.97	0.98	0.98	1.0	0.99	1.0	1.0	1.0	0.96	0.9697	0.9394	0.95
WID3_CID258, low	1.0	0.99	1.0	0.98	0.99	0.99	1.0	0.98	0.99	1.0	0.99	1.0	1.0	0.97	0.9818	0.9636	0.95
WID3_CID259, default	0.99	1.0	0.98	0.99	0.98	1.0	1.0	0.99	0.99	0.98	1.0	0.99	0.99	1.0	0.9714	0.9857	0.96
WID3_CID259, high	0.99	1.0	0.99	0.99	0.99	1.0	1.0	0.98	0.98	0.98	1.0	1.0	1.0	0.98	0.9375	0.9531	0.96
WID3_CID259, low	0.99	0.99	0.98	0.98	0.99	0.99	0.99	0.98	1.0	0.98	0.99	0.98	0.98	0.99	0.9846	0.9846	0.96
WID3_CID260, default	1.0	0.99	1.0	0.99	1.0	0.99	0.98	0.99	0.98	0.97	1.0	0.99	0.98	0.97	0.9844	0.9844	0.96
WID3_CID260, high	0.99	0.99	0.98	0.99	1.0	1.0	0.99	0.99	1.0	0.97	1.0	1.0	0.99	1.0	0.9808	0.9615	0.96
WID3_CID260, low	0.98	0.99	0.99	0.98	1.0	0.99	1.0	0.97	0.99	1.0	1.0	1.0	0.99	1.0	0.9688	0.9844	0.96
WID3_CID261, default	0.98	1.0	0.99	0.98	0.98	0.98	1.0	0.98	0.99	1.0	0.99	1.0	0.99	0.97	0.9677	0.9839	0.96
WID3_CID261, high	1.0	0.99	0.99	1.0	0.97	0.98	0.99	1.0	0.98	0.97	0.99	1.0	0.98	0.98	0.9649	0.9649	0.95
WID3_CID261, low	0.99	0.98	0.99	0.99	1.0	1.0	0.98	0.98	1.0	0.99	0.98	1.0	0.99	0.98	0.9531	0.9688	0.96
WID3_CID316, default	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.97	0.99	0.99	0.99	0.99	1.0	0.95	0.9846	0.9846	0.96
WID3_CID316, high	0.99	0.99	0.99	0.99	1.0	0.99	1.0	0.99	1.0	1.0	0.99	0.99	0.97	0.99	0.9855	0.9855	0.97
WID3_CID316, low	1.0	0.99	1.0	1.0	0.98	0.99	0.98	1.0	0.98	0.98	0.98	1.0	1.0	0.99	0.9701	0.9851	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 9: TRNG0 25 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 105.0 (°C), TRNG0, WID3
Fail ratio 1.89%, ~ 1.84% expected

WID3_CID246, default	0.98	0.99	0.98	0.98	0.96	0.98	0.99	0.99	0.99	1.0	1.0	0.98	1.0	1.0	0.9828	0.9655	0.96
WID3_CID246, high	1.0	1.0	1.0	1.0	1.0	0.99	0.96	0.99	0.97	0.98	0.99	0.97	0.99	0.97	0.9692	0.9692	0.94
WID3_CID246, low	0.97	0.99	0.97	0.97	0.98	0.99	0.99	0.98	0.99	1.0	0.99	1.0	0.99	1.0	0.971	0.971	0.96
WID3_CID253, default	0.99	1.0	0.99	0.99	0.99	0.98	1.0	0.97	0.99	0.98	0.99	1.0	0.99	1.0	0.9706	0.9706	0.95
WID3_CID253, high	0.99	1.0	0.99	0.99	1.0	0.98	1.0	0.99	0.99	0.97	0.99	0.99	1.0	1.0	0.9692	0.9846	0.96
WID3_CID253, low	0.97	0.99	0.98	0.97	0.98	0.99	1.0	0.98	1.0	1.0	0.99	1.0	1.0	0.99	0.9697	0.9848	0.95
WID3_CID254, default	0.98	1.0	1.0	0.99	0.99	0.99	1.0	0.98	1.0	1.0	0.99	0.99	0.98	0.99	0.9697	0.9848	0.96
WID3_CID254, high	1.0	1.0	1.0	1.0	0.97	0.98	0.99	0.99	0.99	1.0	0.99	0.94	0.98	1.0	0.9851	0.9701	0.95
WID3_CID254, low	1.0	0.99	1.0	0.99	0.99	0.99	0.98	0.99	0.99	1.0	0.99	1.0	0.98	0.99	0.9538	0.9692	0.95
WID3_CID255, default	0.99	0.99	1.0	0.99	0.99	1.0	1.0	1.0	0.97	1.0	0.99	0.99	0.98	0.99	0.9538	0.9692	0.95
WID3_CID255, high	0.99	0.99	0.99	0.98	1.0	1.0	0.99	0.98	0.98	0.99	1.0	1.0	1.0	0.97	0.9683	0.9683	0.95
WID3_CID255, low	0.98	0.99	0.98	0.99	0.99	1.0	1.0	0.99	0.98	0.99	0.99	0.98	1.0	0.98	0.9545	0.9394	0.96
WID3_CID256, default	0.99	0.99	0.99	0.99	0.97	1.0	0.99	0.97	0.99	0.98	0.99	0.98	0.99	0.99	0.9688	0.9219	0.96
WID3_CID256, high	1.0	0.98	0.99	1.0	0.99	1.0	0.99	0.98	0.99	1.0	1.0	0.98	1.0	0.99	0.9403	0.9701	0.96
WID3_CID256, low	0.99	1.0	0.99	0.99	0.99	0.99	0.97	0.99	1.0	1.0	0.99	0.98	1.0	0.99	0.9697	0.9697	0.96
WID3_CID258, default	0.99	0.98	0.99	1.0	0.99	0.98	1.0	0.97	0.97	0.98	0.97	1.0	1.0	0.98	0.9855	0.9855	0.96
WID3_CID258, high	0.99	1.0	0.99	0.99	1.0	1.0	1.0	0.98	1.0	0.99	1.0	0.99	1.0	0.99	0.9636	0.9818	0.94
WID3_CID258, low	1.0	0.98	1.0	0.99	1.0	0.99	0.98	0.98	1.0	0.98	1.0	0.98	0.99	1.0	0.9792	0.9583	0.94
WID3_CID259, default	0.99	1.0	0.99	0.99	0.98	0.96	1.0	1.0	0.99	1.0	0.99	0.99	0.99	1.0	0.963	0.9815	0.97
WID3_CID259, high	0.97	0.97	0.97	0.97	0.98	1.0	0.99	1.0	0.98	0.98	0.99	0.99	0.99	0.99	0.9697	0.9848	0.96
WID3_CID259, low	1.0	1.0	1.0	1.0	0.99	0.99	1.0	0.98	0.99	0.97	0.98	0.99	1.0	0.99	0.9683	0.9683	0.95
WID3_CID260, default	1.0	1.0	1.0	1.0	0.99	0.98	0.99	0.97	0.98	0.99	0.98	0.97	1.0	1.0	1.0	0.9636	0.95
WID3_CID260, high	1.0	0.99	0.99	1.0	0.99	0.99	0.98	0.99	0.98	0.95	0.98	0.98	0.99	1.0	0.9706	0.9559	0.96
WID3_CID260, low	0.98	1.0	0.98	0.98	0.99	0.97	1.0	0.99	1.0	0.99	1.0	0.99	0.99	0.99	0.9508	0.9836	0.95
WID3_CID261, default	0.98	0.99	0.97	0.98	1.0	0.99	1.0	0.99	0.98	0.99	0.99	1.0	1.0	0.99	0.9828	0.931	0.93
WID3_CID261, high	1.0	0.99	1.0	1.0	0.99	0.98	0.98	0.99	0.98	0.97	1.0	0.98	1.0	0.99	0.9701	0.9701	0.96
WID3_CID261, low	1.0	0.98	0.99	1.0	0.97	1.0	0.98	0.99	0.95	0.99	1.0	0.99	0.98	1.0	0.9552	0.9552	0.96
WID3_CID316, default	0.99	0.99	0.99	1.0	0.98	1.0	0.98	0.99	0.99	1.0	0.99	1.0	1.0	0.97	0.9531	0.9531	0.95
WID3_CID316, high	1.0	0.99	1.0	1.0	1.0	0.99	0.99	0.98	0.97	0.98	0.99	0.99	0.99	1.0	0.9677	0.9355	0.95
WID3_CID316, low	1.0	1.0	1.0	1.0	0.98	0.99	1.0	0.99	0.99	0.98	0.98	1.0	0.98	0.97	0.9545	0.9697	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 10: TRNG0 105 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 125.0 (°C), TRNG0, WID3
Fail ratio 1.89%, ~ 1.84% expected

WID3_CID246, default	0.99	0.99	0.99	0.98	0.98	0.99	0.99	0.99	0.99	0.99	0.99	0.98	0.99	1.0	0.9643	0.9821	0.96
WID3_CID246, high	0.99	0.98	0.98	0.98	0.97	1.0	0.98	1.0	0.98	0.99	0.99	1.0	1.0	0.99	0.9322	0.9322	0.96
WID3_CID246, low	0.98	1.0	0.96	0.98	1.0	0.99	1.0	0.98	0.98	1.0	0.99	0.99	0.99	0.99	0.9508	0.9836	0.95
WID3_CID253, default	0.99	0.99	0.99	0.98	0.98	0.99	1.0	1.0	0.98	1.0	1.0	0.99	0.98	0.99	0.9672	0.9508	0.96
WID3_CID253, high	0.99	0.99	0.99	0.98	0.9899	1.0	1.0	1.0	1.0	0.99	0.99	1.0	1.0	0.99	0.9571	0.9714	0.95
WID3_CID253, low	0.99	1.0	0.99	0.98	1.0	0.97	1.0	1.0	1.0	1.0	0.98	0.99	0.99	0.99	0.9841	0.9841	0.93
WID3_CID254, default	0.97	0.99	0.97	0.98	1.0	1.0	0.98	0.97	0.99	0.99	0.99	0.99	0.97	0.99	0.9841	0.9683	0.95
WID3_CID254, high	0.98	0.97	0.98	0.98	1.0	1.0	0.97	1.0	0.97	0.98	0.99	1.0	1.0	0.99	1.0	0.9836	0.96
WID3_CID254, low	1.0	0.98	1.0	1.0	1.0	0.97	0.98	0.99	0.98	0.97	0.99	0.98	0.98	1.0	0.9649	0.9825	0.95
WID3_CID255, default	1.0	1.0	1.0	1.0	0.97	0.96	0.99	0.98	1.0	0.97	0.98	0.98	0.98	0.98	0.9661	0.9661	0.96
WID3_CID255, high	1.0	0.96	1.0	1.0	0.98	1.0	0.98	1.0	0.98	0.99	0.99	0.98	1.0	0.99	0.9649	0.9825	0.97
WID3_CID255, low	1.0	1.0	1.0	1.0	1.0	0.98	1.0	0.99	0.98	0.99	0.98	0.98	0.99	0.99	0.9853	0.9412	0.94
WID3_CID256, default	1.0	1.0	1.0	1.0	1.0	0.97	1.0	0.95	0.99	0.97	0.98	0.99	1.0	0.99	0.9565	0.971	0.95
WID3_CID256, high	0.99	0.99	0.99	0.99	0.99	0.97	1.0	0.99	0.99	0.99	0.98	0.98	0.98	1.0	0.9692	0.9538	0.96
WID3_CID256, low	1.0	0.98	1.0	0.99	0.99	0.99	1.0	1.0	1.0	1.0	1.0	0.98	1.0	1.0	0.9701	0.9851	0.96
WID3_CID258, default	0.99	0.98	0.99	0.99	1.0	1.0	1.0	0.99	0.99	0.99	1.0	0.99	0.99	1.0	0.9818	0.9636	0.96
WID3_CID258, high	0.99	0.99	1.0	1.0	0.99	0.99	0.99	0.99	1.0	1.0	0.99	1.0	0.99	0.99	0.95	0.9167	0.96
WID3_CID258, low	1.0	0.99	1.0	1.0	1.0	0.98	1.0	0.97	0.98	0.99	1.0	0.99	0.99	0.99	0.9846	1.0	0.96
WID3_CID259, default	1.0	0.98	1.0	1.0	0.99	1.0	0.98	0.99	1.0	1.0	0.99	0.95	0.99	0.99	0.9636	0.9818	0.96
WID3_CID259, high	0.99	0.99	1.0	0.99	0.98	0.98	0.99	0.98	0.98	0.99	0.99	1.0	0.97	0.99	0.9833	0.9833	0.96
WID3_CID259, low	0.99	1.0	0.98	0.97	1.0	0.98	1.0	0.99	0.99	0.99	0.99	0.98	0.99	1.0	0.9649	0.9649	0.95
WID3_CID260, default	0.98	0.99	0.98	0.97	0.98	0.99	0.99	0.99	1.0	0.98	1.0	1.0	0.98	0.99	0.9833	0.9667	0.95
WID3_CID260, high	0.99	0.99	1.0	0.99	0.99	0.99	0.98	0.96	0.96	0.97	1.0	1.0	0.99	1.0	0.9649	0.9649	0.93
WID3_CID260, low	1.0	0.99	1.0	1.0	1.0	0.99	0.99	0.99	1.0	1.0	1.0	1.0	1.0	1.0	0.9839	0.9677	0.96
WID3_CID261, default	0.99	0.97	0.99	0.99	0.98	0.99	1.0	0.99	0.99	0.97	0.98	1.0	0.99	1.0	0.9508	0.9836	0.96
WID3_CID261, high	0.99	0.98	0.99	0.98	0.96	1.0	0.99	0.98	0.99	0.98	0.99	1.0	1.0	1.0	0.9459	0.973	0.96
WID3_CID261, low	0.98	1.0	0.98	0.96	0.99	0.99	0.99	0.99	0.99	1.0	0.99	0.98	0.98	0.99	0.9344	0.9344	0.96
WID3_CID316, default	0.99	0.98	0.99	0.99	0.99	1.0	1.0	0.97	0.97	0.97	0.99	0.99	0.98	1.0	0.9615	0.9615	0.95
WID3_CID316, high	1.0	0.99	1.0	1.0	0.97	1.0	0.99	1.0	1.0	0.98	0.99	0.98	0.99	0.99	0.9706	0.9559	0.96
WID3_CID316, low	1.0	0.99	1.0	1.0	0.98	0.97	0.99	0.98	0.99	1.0	0.98	0.98	0.99	1.0	0.9571	0.9571	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 11: TRNG0 125 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -40.0 (°C), TRNG1, WID3
Fail ratio 1.60%, ~ 1.84% expected

WID3_CID246, default	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.98	0.99	0.98	1.0	0.99	1.0	0.9863	0.9726	0.96
WID3_CID246, high	1.0	0.96	1.0	0.99	1.0	0.99	0.99	0.98	0.99	0.99	0.99	1.0	1.0	1.0	0.9701	0.9403	0.95
WID3_CID246, low	0.99	1.0	0.99	0.99	0.99	0.98	0.99	0.99	1.0	1.0	0.98	0.99	1.0	1.0	0.9839	0.9677	0.96
WID3_CID253, default	1.0	0.99	1.0	0.99	0.98	0.98	1.0	1.0	1.0	1.0	1.0	0.97	1.0	0.99	0.9538	0.9692	0.96
WID3_CID253, high	0.99	0.99	0.98	0.99	0.99	0.98	1.0	0.97	0.98	0.99	1.0	0.99	1.0	0.99	0.9697	0.9697	0.96
WID3_CID253, low	0.99	0.99	0.99	1.0	1.0	0.96	0.98	1.0	0.98	0.99	0.99	1.0	1.0	0.99	0.9559	0.9853	0.94
WID3_CID254, default	1.0	1.0	0.99	1.0	1.0	1.0	0.98	0.99	0.99	0.99	1.0	1.0	1.0	0.99	0.9667	0.9667	0.96
WID3_CID254, high	0.99	0.99	0.99	1.0	0.99	0.99	0.99	0.99	1.0	0.99	1.0	1.0	0.97	0.99	0.9531	0.9688	0.96
WID3_CID254, low	0.98	0.97	0.98	0.99	0.99	1.0	1.0	0.99	0.97	1.0	0.99	0.98	1.0	0.97	0.9833	0.9667	0.96
WID3_CID255, default	1.0	1.0	1.0	1.0	0.99	0.98	0.99	1.0	1.0	0.99	0.99	0.99	0.96	0.99	0.9667	0.9667	0.96
WID3_CID255, high	0.99	1.0	0.99	0.99	1.0	0.99	0.98	0.99	0.98	1.0	1.0	1.0	0.99	0.98	0.9677	0.9839	0.94
WID3_CID255, low	0.99	1.0	0.98	0.97	1.0	0.99	0.98	0.97	0.99	0.99	0.99	1.0	0.99	1.0	0.9697	0.9848	0.96
WID3_CID256, default	0.98	1.0	0.98	0.98	0.99	0.98	0.99	0.98	0.99	0.98	1.0	1.0	0.98	0.98	0.9464	0.9821	0.96
WID3_CID256, high	0.97	0.98	0.97	0.98	0.99	0.99	0.99	0.98	0.98	1.0	0.99	0.99	0.98	0.99	1.0	0.9661	0.96
WID3_CID256, low	0.97	1.0	0.96	0.97	0.99	0.99	1.0	0.98	1.0	0.99	1.0	0.99	0.99	1.0	0.9836	0.9672	0.95
WID3_CID258, default	0.99	0.98	1.0	0.99	0.98	1.0	0.98	1.0	0.98	0.99	0.98	1.0	1.0	0.98	0.9552	0.9851	0.97
WID3_CID258, high	0.99	0.99	0.99	0.99	0.98	0.98	1.0	0.98	1.0	1.0	1.0	1.0	0.97	0.97	0.9655	0.9828	0.96
WID3_CID258, low	1.0	0.99	1.0	0.99	0.97	1.0	1.0	1.0	0.99	1.0	0.99	1.0	0.99	0.99	0.9714	0.9857	0.95
WID3_CID259, default	1.0	1.0	1.0	1.0	0.98	1.0	0.99	1.0	0.99	0.98	1.0	1.0	1.0	0.99	0.9818	0.9636	0.96
WID3_CID259, high	1.0	1.0	1.0	0.99	0.99	0.99	0.98	0.99	0.98	0.99	0.99	1.0	0.99	1.0	0.9672	0.9672	0.96
WID3_CID259, low	0.99	0.98	0.98	0.98	0.99	1.0	0.99	0.98	0.96	0.99	0.98	1.0	1.0	0.98	0.9853	0.9559	0.95
WID3_CID260, default	1.0	1.0	0.99	0.99	0.97	1.0	0.98	0.97	0.99	1.0	1.0	0.99	0.99	1.0	0.9655	0.9828	0.96
WID3_CID260, high	1.0	0.99	0.99	1.0	1.0	0.99	0.99	0.98	0.98	0.96	1.0	0.97	0.97	0.99	0.9688	0.9688	0.95
WID3_CID260, low	1.0	0.99	1.0	1.0	1.0	0.97	1.0	0.98	0.96	0.98	0.99	1.0	0.99	1.0	0.9688	0.9688	0.96
WID3_CID261, default	0.99	1.0	0.98	0.99	0.99	1.0	0.98	0.97	0.99	0.99	1.0	0.98	1.0	0.99	0.9649	0.9649	0.97
WID3_CID261, high	1.0	1.0	1.0	1.0	0.98	0.97	0.97	1.0	1.0	0.97	0.98	1.0	0.98	1.0	0.971	0.971	0.95
WID3_CID261, low	0.99	1.0	0.99	0.99	0.99	0.99	0.98	1.0	1.0	0.97	0.99	0.98	0.99	0.97	0.9677	0.9516	0.96
WID3_CID316, default	0.98	0.99	0.99	0.99	0.98	1.0	1.0	0.98	0.99	0.98	0.98	1.0	1.0	1.0	0.9701	0.9552	0.96
WID3_CID316, high	1.0	0.99	0.99	1.0	1.0	1.0	0.97	0.94	1.0	0.98	1.0	0.99	0.98	0.99	0.9524	0.9524	0.96
WID3_CID316, low	0.97	0.98	0.99	0.98	1.0	1.0	1.0	0.99	1.0	0.99	0.99	0.99	1.0	0.99	0.95	0.95	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 12: TRNG1 -40 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -25.0 (°C), TRNG1, WID3
Fail ratio 2.16%, ~ 1.84% expected

WID3_CID246, default	0.97	0.99	0.99	0.98	0.98	0.98	0.98	0.98	0.99	0.98	1.0	0.99	0.98	1.0	0.9667	0.9667	0.95
WID3_CID246, high	1.0	0.96	1.0	1.0	0.99	0.99	1.0	0.99	0.98	0.99	0.99	0.99	1.0	1.0	0.9706	0.9559	0.96
WID3_CID246, low	0.98	0.98	0.97	0.98	1.0	0.99	1.0	0.99	1.0	0.97	0.98	1.0	1.0	1.0	0.963	0.963	0.95
WID3_CID253, default	0.95	0.97	0.96	0.95	1.0	0.99	1.0	1.0	0.99	1.0	0.96	0.99	1.0	0.99	0.9683	0.9841	0.96
WID3_CID253, high	0.99	0.99	0.99	0.99	0.99	1.0	1.0	1.0	0.98	1.0	1.0	0.98	0.98	1.0	0.9667	0.95	0.96
WID3_CID253, low	0.99	0.99	0.99	0.99	0.99	1.0	0.99	0.99	0.99	0.99	0.99	1.0	1.0	0.98	0.9844	0.9844	0.96
WID3_CID254, default	1.0	0.99	0.99	0.99	1.0	1.0	1.0	0.99	0.98	1.0	1.0	1.0	1.0	0.98	0.9643	0.9643	0.94
WID3_CID254, high	0.99	0.99	0.99	0.99	0.96	1.0	1.0	0.96	0.99	0.98	0.99	0.99	0.99	0.99	0.9688	0.9531	0.95
WID3_CID254, low	0.99	1.0	0.99	0.98	0.99	1.0	0.98	0.99	0.99	1.0	0.99	1.0	0.99	0.97	0.9841	0.9841	0.96
WID3_CID255, default	0.97	0.98	0.97	0.96	0.99	0.99	1.0	1.0	1.0	0.99	0.99	0.99	1.0	0.99	0.9846	0.9692	0.97
WID3_CID255, high	0.98	0.99	0.99	0.99	0.99	0.99	1.0	1.0	0.96	0.99	0.98	0.97	1.0	1.0	0.9636	0.9636	0.96
WID3_CID255, low	0.99	0.99	1.0	1.0	1.0	1.0	1.0	0.99	1.0	1.0	0.98	0.99	1.0	0.99	0.9841	0.9524	0.96
WID3_CID256, default	1.0	0.99	1.0	1.0	0.98	1.0	1.0	1.0	1.0	0.97	0.99	0.99	0.98	1.0	0.9677	0.9516	0.94
WID3_CID256, high	1.0	0.99	1.0	1.0	0.97	1.0	0.99	0.99	0.99	0.99	0.99	1.0	0.98	0.99	0.9552	0.9701	0.95
WID3_CID256, low	1.0	0.99	1.0	1.0	0.99	0.99	1.0	0.99	1.0	0.97	1.0	1.0	0.99	1.0	0.9841	0.9683	0.96
WID3_CID258, default	1.0	1.0	1.0	1.0	0.99	0.99	1.0	0.99	0.99	1.0	0.99	0.99	1.0	0.98	0.9375	0.9844	0.95
WID3_CID258, high	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.99	0.98	0.99	0.99	0.99	0.98	0.9828	0.9828	0.96
WID3_CID258, low	0.99	0.97	0.99	0.99	0.99	1.0	1.0	0.98	0.98	0.98	1.0	0.99	0.98	0.98	0.9818	0.9636	0.96
WID3_CID259, default	0.99	0.99	1.0	1.0	1.0	0.98	1.0	0.98	1.0	1.0	1.0	0.99	0.98	1.0	0.9524	0.9683	0.95
WID3_CID259, high	1.0	0.99	1.0	0.99	0.97	0.99	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.9655	0.9828	0.97
WID3_CID259, low	0.99	0.97	0.98	0.99	0.99	0.99	0.95	1.0	0.99	0.99	0.98	1.0	0.99	0.99	0.963	0.9444	0.96
WID3_CID260, default	0.96	1.0	0.97	0.96	1.0	0.97	0.99	1.0	0.99	0.98	0.97	0.99	0.99	1.0	0.9677	0.9677	0.95
WID3_CID260, high	1.0	0.98	1.0	0.99	1.0	0.99	1.0	0.99	0.95	0.98	0.99	0.99	0.98	0.96	0.9857	0.9857	0.96
WID3_CID260, low	1.0	0.99	1.0	0.99	1.0	0.98	0.99	0.97	0.99	0.97	0.98	0.98	0.96	1.0	0.9688	0.9688	0.96
WID3_CID261, default	0.97	0.99	0.97	0.97	0.98	1.0	0.98	0.98	1.0	1.0	0.98	0.98	1.0	0.98	0.9667	0.9667	0.96
WID3_CID261, high	0.97	0.99	0.96	0.97	0.99	1.0	1.0	1.0	0.99	0.98	0.99	0.99	1.0	0.97	0.9677	0.9677	0.96
WID3_CID261, low	0.98	0.98	0.98	0.98	0.99	1.0	0.98	1.0	1.0	0.98	0.99	0.98	0.99	0.98	0.9683	0.9524	0.96
WID3_CID316, default	0.98	1.0	0.97	0.98	0.97	0.98	0.97	0.98	1.0	0.99	1.0	0.99	0.99	0.99	0.9429	0.9714	0.96
WID3_CID316, high	0.99	0.99	0.99	0.99	0.99	0.98	0.98	0.98	0.98	0.99	0.98	0.99	0.99	0.98	0.9677	0.9839	0.96
WID3_CID316, low	1.0	0.98	1.0	0.99	1.0	0.99	1.0	1.0	0.99	0.99	1.0	0.98	0.98	1.0	0.9706	0.9706	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 13: TRNG1 -25 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 25.0 (°C), TRNG1, WID3
Fail ratio 2.16%, ~ 1.84% expected

WID3_CID246, default	0.97	0.99	0.99	0.98	0.99	0.99	1.0	1.0	1.0	0.99	1.0	0.98	0.99	0.99	0.9833	0.9667	0.96
WID3_CID246, high	1.0	0.99	1.0	1.0	1.0	1.0	0.99	1.0	1.0	0.99	1.0	1.0	0.99	1.0	0.9524	0.9683	0.96
WID3_CID246, low	0.99	0.99	0.99	0.99	1.0	0.99	0.98	0.97	0.99	0.97	0.99	0.98	0.96	0.99	0.9688	0.9688	0.95
WID3_CID253, default	0.99	0.99	1.0	1.0	1.0	1.0	0.96	0.99	1.0	1.0	0.98	1.0	1.0	0.99	0.9692	0.9846	0.96
WID3_CID253, high	0.97	0.99	0.98	0.97	1.0	0.99	0.99	0.98	0.99	1.0	0.99	1.0	1.0	0.99	0.9661	0.9492	0.96
WID3_CID253, low	0.99	0.98	1.0	0.99	0.99	1.0	0.99	0.99	0.99	0.99	0.99	0.99	1.0	0.98	0.9853	0.9706	0.95
WID3_CID254, default	1.0	1.0	1.0	1.0	0.99	0.98	1.0	0.98	0.99	0.99	0.99	0.99	0.98	0.99	0.9508	0.9836	0.96
WID3_CID254, high	1.0	1.0	1.0	1.0	0.99	0.99	1.0	0.97	1.0	0.99	0.98	0.99	0.99	0.99	0.9828	1.0	0.96
WID3_CID254, low	0.98	1.0	0.97	0.98	0.99	0.99	1.0	0.98	0.99	0.98	1.0	1.0	0.99	0.98	0.9833	0.9667	0.96
WID3_CID255, default	1.0	0.99	1.0	1.0	0.98	1.0	1.0	0.99	0.98	0.99	0.99	1.0	0.99	1.0	0.963	0.9815	0.96
WID3_CID255, high	1.0	0.99	0.99	1.0	1.0	0.95	1.0	0.99	0.98	1.0	0.98	0.99	0.99	1.0	0.9841	0.9841	0.96
WID3_CID255, low	1.0	0.99	1.0	1.0	0.99	1.0	0.99	0.99	0.98	0.99	0.97	0.99	1.0	0.99	0.9667	0.9833	0.96
WID3_CID256, default	0.99	0.96	1.0	0.99	0.99	1.0	1.0	1.0	0.99	1.0	0.99	0.99	1.0	0.99	0.9825	0.9825	0.95
WID3_CID256, high	1.0	0.98	1.0	1.0	1.0	0.96	0.99	1.0	0.99	0.99	0.98	0.97	1.0	0.99	0.9677	0.9355	0.97
WID3_CID256, low	0.99	0.99	1.0	1.0	0.99	0.99	0.99	0.99	0.96	1.0	0.99	0.98	0.98	0.99	0.9855	0.9855	0.95
WID3_CID258, default	0.99	0.99	0.99	0.99	0.99	0.98	0.97	0.98	0.98	0.99	0.99	1.0	0.99	1.0	0.9623	0.9623	0.96
WID3_CID258, high	0.99	0.98	0.99	0.99	0.97	0.96	0.98	1.0	0.99	1.0	0.99	1.0	0.99	0.98	0.973	0.9595	0.95
WID3_CID258, low	0.98	0.98	1.0	0.98	0.99	0.97	1.0	0.99	0.99	0.98	1.0	1.0	1.0	0.99	0.9688	0.9844	0.95
WID3_CID259, default	1.0	0.99	1.0	1.0	1.0	0.99	1.0	0.97	0.98	1.0	0.98	0.99	0.99	0.98	0.9559	0.9706	0.96
WID3_CID259, high	0.99	1.0	1.0	0.99	0.99	0.99	1.0	0.99	0.99	1.0	1.0	0.97	1.0	0.99	0.9815	0.9815	0.96
WID3_CID259, low	1.0	1.0	1.0	1.0	0.99	0.99	0.98	0.99	0.97	1.0	0.99	1.0	1.0	1.0	0.9531	0.9375	0.96
WID3_CID260, default	0.99	0.99	0.97	0.99	0.99	0.98	1.0	0.97	1.0	0.96	0.99	1.0	1.0	1.0	0.9636	0.9636	0.96
WID3_CID260, high	0.98	1.0	0.97	0.97	1.0	0.98	0.99	1.0	1.0	0.96	1.0	1.0	0.99	0.99	0.9833	0.9833	0.95
WID3_CID260, low	0.99	1.0	0.99	0.99	1.0	1.0	0.98	1.0	1.0	1.0	0.97	0.97	0.98	0.98	0.9851	0.9851	0.96
WID3_CID261, default	0.98	1.0	0.98	0.99	0.99	1.0	0.99	1.0	0.99	0.99	1.0	1.0	1.0	1.0	0.9853	0.9853	0.95
WID3_CID261, high	0.97	1.0	0.97	0.98	0.99	1.0	1.0	0.98	1.0	0.99	0.98	0.99	1.0	1.0	0.9821	0.9464	0.96
WID3_CID261, low	0.98	0.99	0.98	0.96	0.99	1.0	1.0	0.98	0.99	1.0	0.98	0.97	0.99	0.98	0.9667	0.95	0.95
WID3_CID316, default	0.98	1.0	0.98	0.97	0.99	0.98	0.98	0.98	0.97	0.98	1.0	1.0	1.0	0.99	0.8909	0.9636	0.96
WID3_CID316, high	1.0	0.99	1.0	1.0	0.97	0.96	0.99	0.98	0.99	1.0	0.99	0.98	0.98	1.0	0.9655	0.9828	0.96
WID3_CID316, low	0.99	0.99	0.99	0.99	0.99	1.0	1.0	1.0	0.98	0.99	0.97	0.99	0.99	0.99	0.9444	0.9815	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 14: TRNG1 25 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 105.0 (°C), TRNG1, WID3
Fail ratio 1.83%, ~ 1.84% expected

WID3_CID246, default	1.0	0.99	1.0	1.0	0.98	0.98	1.0	0.98	1.0	0.99	0.99	1.0	1.0	1.0	0.9565	0.9565	0.96
WID3_CID246, high	1.0	0.98	1.0	0.99	0.97	0.99	0.99	0.99	0.99	0.98	0.99	1.0	0.99	0.99	0.95	0.9667	0.95
WID3_CID246, low	1.0	0.98	0.99	1.0	1.0	1.0	1.0	0.99	0.96	0.98	0.98	0.96	0.98	0.96	0.9841	0.9683	0.96
WID3_CID253, default	0.98	0.98	0.99	0.99	0.99	1.0	1.0	0.99	0.99	1.0	1.0	1.0	0.98	0.98	0.9538	0.9692	0.95
WID3_CID253, high	0.99	0.98	0.99	0.99	1.0	1.0	0.99	0.99	1.0	1.0	0.99	0.99	0.97	0.99	0.9636	0.9818	0.96
WID3_CID253, low	1.0	0.98	0.99	1.0	0.98	0.99	1.0	1.0	1.0	0.98	0.99	0.99	0.96	1.0	0.9815	0.9815	0.96
WID3_CID254, default	1.0	0.97	0.99	0.99	0.99	1.0	0.99	0.99	0.99	1.0	0.99	1.0	1.0	0.97	0.9636	0.9636	0.94
WID3_CID254, high	0.99	1.0	0.99	0.99	0.99	1.0	0.99	0.95	0.99	0.98	1.0	0.99	0.99	0.99	0.9643	0.9464	0.95
WID3_CID254, low	0.97	0.98	0.96	0.97	1.0	1.0	0.99	0.99	0.99	0.99	1.0	1.0	0.99	0.99	0.9825	0.9825	0.95
WID3_CID255, default	1.0	1.0	1.0	1.0	0.97	1.0	1.0	1.0	0.97	0.99	0.98	0.99	0.98	0.99	0.9844	0.9688	0.96
WID3_CID255, high	0.98	0.99	0.99	0.98	0.99	0.99	0.98	0.99	0.99	0.97	1.0	0.99	1.0	1.0	1.0	0.9831	0.96
WID3_CID255, low	0.98	0.98	0.99	0.99	0.99	0.98	1.0	1.0	1.0	0.99	1.0	0.96	0.99	0.99	0.9344	0.9836	0.95
WID3_CID256, default	0.99	0.99	0.99	0.99	0.99	0.98	1.0	0.98	1.0	0.99	0.98	1.0	0.99	0.99	0.9848	0.9697	0.96
WID3_CID256, high	1.0	0.99	0.99	0.99	1.0	0.97	0.98	1.0	0.99	0.97	0.99	0.98	0.97	0.99	0.9571	0.9714	0.96
WID3_CID256, low	0.98	1.0	0.98	0.99	1.0	1.0	0.97	0.98	0.98	0.98	1.0	0.99	0.99	0.96	0.9474	0.9649	0.96
WID3_CID258, default	1.0	0.99	1.0	1.0	0.99	0.99	1.0	0.99	0.98	0.99	0.98	1.0	1.0	1.0	0.9545	0.9697	0.95
WID3_CID258, high	0.99	0.98	0.99	0.99	1.0	0.99	0.97	0.99	1.0	0.96	1.0	1.0	0.99	1.0	0.971	0.971	0.95
WID3_CID258, low	0.99	0.99	0.99	0.99	0.99	0.99	0.98	1.0	1.0	0.97	1.0	0.99	0.99	0.99	0.9672	0.9672	0.94
WID3_CID259, default	0.99	0.98	1.0	0.99	1.0	1.0	0.97	0.96	1.0	1.0	1.0	1.0	1.0	0.98	0.9464	0.9286	0.97
WID3_CID259, high	0.98	0.98	0.97	0.98	0.99	1.0	0.99	1.0	0.97	0.97	1.0	0.99	0.99	1.0	0.9836	0.9672	0.97
WID3_CID259, low	1.0	0.97	1.0	0.99	0.99	1.0	0.99	0.99	1.0	1.0	0.99	1.0	1.0	1.0	0.9483	0.9828	0.96
WID3_CID260, default	1.0	0.97	1.0	1.0	1.0	1.0	0.99	1.0	0.98	0.99	0.99	0.99	1.0	1.0	0.9672	0.9672	0.96
WID3_CID260, high	1.0	1.0	1.0	1.0	1.0	1.0	0.98	0.99	0.99	0.98	1.0	0.99	1.0	1.0	0.9688	0.9688	0.96
WID3_CID260, low	0.99	1.0	1.0	0.99	1.0	1.0	0.98	1.0	1.0	0.99	0.98	1.0	1.0	1.0	0.9524	0.9841	0.95
WID3_CID261, default	0.99	1.0	0.97	0.99	1.0	0.99	0.99	0.98	0.99	0.98	0.99	0.98	0.99	0.98	0.9508	0.9836	0.96
WID3_CID261, high	1.0	1.0	0.99	0.99	0.97	0.99	1.0	0.99	0.99	1.0	1.0	1.0	1.0	0.98	0.9455	0.9636	0.96
WID3_CID261, low	1.0	0.97	0.99	1.0	0.96	0.99	0.98	0.99	0.99	0.99	0.96	0.99	1.0	0.97	0.9677	0.9516	0.96
WID3_CID316, default	1.0	0.98	1.0	0.99	0.98	0.98	0.96	1.0	0.99	1.0	1.0	0.99	0.97	0.99	0.9833	0.9833	0.97
WID3_CID316, high	0.97	0.99	0.99	1.0	0.99	0.98	1.0	0.98	0.97	0.99	1.0	0.97	0.96	1.0	0.95	0.95	0.96
WID3_CID316, low	0.98	0.98	0.98	0.99	0.97	1.0	0.98	0.99	0.99	0.98	0.99	1.0	0.99	0.99	0.9483	0.9655	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 15: TRNG1 105 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 125.0 (°C), TRNG1, WID3
Fail ratio 2.16%, ~ 1.84% expected

WID3_CID246, default	0.97	0.98	0.96	0.98	1.0	0.97	0.98	0.97	0.99	0.97	0.99	0.99	0.99	0.99	0.94	0.98	0.96
WID3_CID246, high	1.0	0.99	0.99	1.0	0.99	0.98	1.0	0.99	0.98	0.98	1.0	1.0	0.99	0.96	0.9851	0.9701	0.96
WID3_CID246, low	1.0	1.0	1.0	1.0	1.0	1.0	0.98	1.0	0.98	0.99	0.99	1.0	1.0	0.99	0.9661	0.9661	0.96
WID3_CID253, default	1.0	0.99	0.99	1.0	0.99	1.0	0.98	0.99	0.96	0.98	0.98	0.98	0.97	0.99	0.9683	0.9683	0.95
WID3_CID253, high	0.98	0.99	0.99	0.99	0.97	0.98	1.0	1.0	0.98	0.99	0.99	1.0	0.99	0.97	0.9855	0.971	0.96
WID3_CID253, low	1.0	1.0	1.0	1.0	1.0	0.99	0.99	0.99	0.99	0.99	0.99	1.0	0.98	0.99	0.9692	0.9846	0.96
WID3_CID254, default	1.0	1.0	1.0	0.99	0.97	0.97	1.0	0.97	1.0	0.98	0.98	0.99	1.0	0.98	0.95	0.9833	0.96
WID3_CID254, high	0.99	0.99	0.98	0.98	0.99	1.0	1.0	0.97	0.98	0.99	0.99	1.0	0.99	0.99	0.9714	0.9714	0.95
WID3_CID254, low	0.99	0.99	0.99	1.0	0.99	1.0	1.0	1.0	0.96	0.97	0.97	0.99	1.0	0.99	0.9688	0.9375	0.95
WID3_CID255, default	0.99	0.97	1.0	0.99	1.0	0.99	0.98	0.99	0.99	0.99	0.99	0.98	1.0	0.97	0.9818	0.9636	0.96
WID3_CID255, high	1.0	0.99	1.0	1.0	0.99	0.99	0.98	0.98	0.98	0.98	0.99	0.98	1.0	0.98	0.9692	0.9692	0.96
WID3_CID255, low	0.97	1.0	0.97	0.98	0.98	0.99	0.98	0.99	0.98	0.99	0.99	0.99	0.99	0.99	0.9677	0.9839	0.92
WID3_CID256, default	0.99	0.99	0.99	1.0	0.99	0.99	1.0	0.99	0.98	0.98	0.97	0.98	0.99	1.0	0.9706	0.9706	0.95
WID3_CID256, high	0.97	0.99	0.96	0.98	0.99	0.99	0.99	0.99	0.99	1.0	0.98	0.99	0.99	0.99	0.9677	0.9677	0.95
WID3_CID256, low	0.99	0.99	0.98	0.98	1.0	0.99	1.0	1.0	0.99	1.0	1.0	0.97	0.99	0.98	0.9818	0.9818	0.94
WID3_CID258, default	1.0	0.99	1.0	1.0	0.98	0.99	1.0	1.0	0.99	0.97	1.0	0.99	0.99	0.99	0.9672	0.9672	0.96
WID3_CID258, high	0.99	1.0	0.99	0.99	1.0	0.97	0.99	1.0	1.0	1.0	0.99	0.99	0.98	1.0	0.9677	0.9677	0.96
WID3_CID258, low	1.0	1.0	1.0	1.0	1.0	0.98	0.98	1.0	0.99	1.0	0.99	0.98	1.0	0.99	0.9688	0.9844	0.97
WID3_CID259, default	1.0	1.0	1.0	1.0	1.0	1.0	0.99	0.98	0.99	0.99	0.99	1.0	0.98	0.99	0.9853	0.9412	0.96
WID3_CID259, high	0.97	0.98	0.97	0.97	1.0	0.99	0.99	1.0	1.0	0.97	1.0	0.99	0.96	0.98	0.9552	0.9851	0.96
WID3_CID259, low	0.98	0.97	0.97	0.98	0.99	0.97	1.0	0.99	1.0	0.99	0.98	0.99	0.98	1.0	0.9516	0.9355	0.95
WID3_CID260, default	1.0	0.99	1.0	1.0	0.99	0.99	0.98	0.97	1.0	1.0	1.0	0.97	0.99	1.0	0.9508	0.9672	0.96
WID3_CID260, high	1.0	0.99	1.0	1.0	1.0	1.0	0.98	1.0	0.99	0.99	0.97	0.99	1.0	1.0	0.9649	0.9649	0.96
WID3_CID260, low	0.98	0.97	0.99	0.99	1.0	0.98	1.0	0.99	0.99	0.98	0.99	0.99	1.0	0.99	0.9672	0.9508	0.96
WID3_CID261, default	0.99	0.96	0.99	0.99	0.99	0.99	0.99	0.99	0.97	0.99	0.98	0.99	0.98	1.0	0.9701	0.9701	0.95
WID3_CID261, high	0.99	0.97	1.0	1.0	1.0	1.0	0.99	1.0	0.99	0.99	1.0	1.0	0.99	1.0	0.9851	0.9701	0.96
WID3_CID261, low	0.99	1.0	0.99	0.98	1.0	0.98	1.0	0.98	0.98	0.98	0.97	1.0	1.0	1.0	0.9836	0.9836	0.96
WID3_CID316, default	1.0	0.99	1.0	1.0	1.0	1.0	1.0	0.98	0.99	0.98	0.99	0.99	1.0	0.99	0.9688	0.9688	0.96
WID3_CID316, high	1.0	0.98	1.0	1.0	1.0	0.99	0.98	0.98	0.99	0.97	1.0	1.0	0.99	1.0	0.9683	0.9683	0.96
WID3_CID316, low	0.98	0.98	0.98	0.99	0.99	0.97	1.0	0.99	1.0	1.0	1.0	0.99	1.0	1.0	0.9677	0.9677	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 16: TRNG1 125 °C WID3 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -40.0 (°C), TRNG0, WID8
Fail ratio 2.10%, ~ 1.84% expected

WID8_CID31, default	0.99	0.99	0.98	0.99	1.0	0.99	0.99	0.98	1.0	0.99	0.99	1.0	0.99	0.98	0.9683	0.9683	0.95
WID8_CID31, high	0.99	1.0	0.99	0.99	0.99	1.0	1.0	0.99	0.98	0.99	0.99	1.0	0.99	0.99	0.9833	0.9833	0.94
WID8_CID31, low	1.0	1.0	1.0	1.0	0.99	0.99	1.0	1.0	0.98	1.0	0.99	0.98	1.0	0.97	0.9841	0.9524	0.96
WID8_CID32, default	0.99	1.0	0.99	1.0	0.99	1.0	0.99	0.99	0.98	0.99	0.99	0.99	0.99	0.99	1.0	0.9828	0.96
WID8_CID32, high	0.99	0.98	0.99	0.99	0.99	1.0	0.98	1.0	0.99	0.98	0.98	0.97	0.98	0.99	0.9846	0.9692	0.96
WID8_CID32, low	0.99	1.0	1.0	0.99	1.0	0.99	0.99	1.0	1.0	0.98	0.99	1.0	1.0	1.0	0.9839	0.9677	0.96
WID8_CID72, default	1.0	1.0	1.0	1.0	1.0	0.98	1.0	1.0	0.98	0.99	0.99	0.99	0.99	1.0	0.9661	0.9661	0.96
WID8_CID72, high	1.0	0.98	1.0	1.0	1.0	1.0	1.0	0.99	0.98	0.98	0.99	0.97	1.0	1.0	0.9857	0.9571	0.96
WID8_CID72, low	0.95	0.97	0.95	0.95	0.98	0.99	1.0	0.98	1.0	1.0	1.0	1.0	1.0	0.98	0.9649	0.9474	0.96
WID8_CID73, default	0.99	0.99	0.99	1.0	0.99	0.98	0.99	0.99	0.98	0.99	1.0	1.0	1.0	1.0	0.9538	0.9692	0.95
WID8_CID73, high	0.99	0.98	0.99	0.99	1.0	0.98	0.99	1.0	1.0	1.0	0.98	1.0	1.0	0.99	0.95	0.9667	0.95
WID8_CID73, low	1.0	0.99	1.0	1.0	0.99	1.0	0.99	0.98	1.0	0.99	1.0	1.0	1.0	1.0	0.9683	0.9683	0.94
WID8_CID74, default	0.98	1.0	0.98	0.97	0.99	0.99	0.99	0.99	0.99	0.99	1.0	0.99	0.98	0.96	0.9655	0.9828	0.95
WID8_CID74, high	1.0	0.99	1.0	1.0	1.0	0.98	0.99	0.99	0.98	0.96	1.0	1.0	0.99	0.97	0.9661	0.9661	0.96
WID8_CID74, low	1.0	1.0	0.99	1.0	0.99	0.99	1.0	0.98	0.97	0.99	0.99	0.98	0.99	0.99	0.9683	0.9683	0.96
WID8_CID75, default	1.0	0.99	1.0	1.0	1.0	0.99	1.0	0.97	1.0	0.97	1.0	0.99	0.97	0.99	0.9577	0.9718	0.94
WID8_CID75, high	1.0	0.99	1.0	1.0	0.98	0.98	1.0	0.99	1.0	0.98	0.99	0.98	0.99	0.99	0.971	0.9565	0.95
WID8_CID75, low	0.98	0.99	0.99	0.98	1.0	1.0	0.99	0.99	1.0	0.99	1.0	1.0	0.99	0.98	0.9833	0.9667	0.96
WID8_CID76, default	1.0	0.98	1.0	1.0	1.0	0.99	0.99	0.99	0.98	0.95	1.0	0.97	0.98	0.99	0.9538	0.9846	0.96
WID8_CID76, high	0.97	0.97	0.96	0.97	0.98	0.99	0.99	1.0	1.0	0.99	0.99	0.99	0.98	0.97	0.9355	0.9516	0.96
WID8_CID76, low	1.0	0.98	0.99	0.99	1.0	0.99	0.98	0.98	0.97	0.98	0.98	1.0	1.0	0.98	0.9688	0.9844	0.95
WID8_CID77, default	1.0	0.99	1.0	1.0	0.98	1.0	0.99	0.98	0.99	0.99	0.98	0.98	0.99	0.99	0.9636	0.9818	0.94
WID8_CID77, high	0.99	0.99	0.99	0.99	0.97	0.96	0.99	0.99	0.99	1.0	0.99	1.0	1.0	1.0	0.9322	0.9492	0.94
WID8_CID77, low	0.99	0.98	0.99	0.99	1.0	1.0	0.97	0.98	0.99	0.99	0.98	0.98	0.98	0.99	0.9474	0.9474	0.97
WID8_CID78, default	0.97	0.97	0.98	0.98	1.0	0.99	0.99	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.9677	0.9677	0.93
WID8_CID78, high	0.99	1.0	0.99	0.99	0.99	0.97	0.99	0.98	1.0	1.0	1.0	0.99	1.0	0.99	0.9688	0.9688	0.96
WID8_CID78, low	0.99	1.0	0.99	0.99	0.99	0.99	0.98	0.99	0.98	1.0	1.0	0.97	1.0	0.98	0.9636	0.9636	0.96
WID8_CID79, default	0.98	1.0	0.98	0.98	1.0	1.0	0.99	0.99	0.98	0.99	0.99	1.0	1.0	0.99	0.9123	0.9649	0.95
WID8_CID79, high	0.98	1.0	0.97	0.98	0.99	1.0	0.99	0.99	0.98	0.96	1.0	0.99	0.98	0.99	0.9649	0.9825	0.96
WID8_CID79, low	0.99	0.99	1.0	0.99	1.0	0.98	0.96	0.97	0.98	0.99	0.97	0.98	0.99	0.98	0.9649	0.9825	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 17: TRNG0 -40 °C WID8 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -25.0 (°C), TRNG0, WID8
Fail ratio 1.89%, ~ 1.84% expected

WID8_CID31, default	1.0	1.0	1.0	1.0	1.0	1.0	0.99	0.99	0.99	0.99	0.94	0.99	0.98	0.99	0.9672	0.9672	0.97
WID8_CID31, high	0.99	1.0	0.99	0.99	1.0	0.99	0.99	0.98	0.98	1.0	0.99	0.98	0.99	0.99	0.9531	0.9844	0.95
WID8_CID31, low	0.99	1.0	1.0	0.99	0.97	0.99	0.99	0.99	0.99	1.0	0.98	0.98	0.99	0.99	0.95	0.9667	0.96
WID8_CID32, default	1.0	0.98	1.0	1.0	0.99	1.0	1.0	0.99	1.0	0.96	0.99	0.99	0.99	1.0	0.9692	0.9692	0.97
WID8_CID32, high	0.99	0.98	0.99	0.99	1.0	1.0	0.97	0.98	1.0	0.98	0.98	1.0	1.0	0.99	1.0	0.9375	0.96
WID8_CID32, low	0.99	0.99	1.0	0.99	1.0	0.98	0.99	0.99	1.0	1.0	0.98	0.99	0.99	1.0	0.9859	0.9859	0.96
WID8_CID72, default	0.98	1.0	0.98	0.99	0.99	0.99	0.99	0.99	1.0	0.98	1.0	0.99	0.99	0.99	0.9661	0.9661	0.96
WID8_CID72, high	1.0	0.99	1.0	1.0	0.97	1.0	1.0	0.99	0.97	1.0	0.99	0.98	0.98	1.0	0.9538	0.9692	0.96
WID8_CID72, low	0.99	1.0	0.99	0.99	1.0	0.99	0.98	0.97	0.99	0.99	0.99	1.0	0.99	0.97	0.9841	0.9841	0.96
WID8_CID73, default	0.99	0.99	0.98	0.98	0.99	0.99	1.0	0.97	1.0	1.0	1.0	0.97	0.97	1.0	0.9833	0.95	0.96
WID8_CID73, high	1.0	0.99	1.0	0.99	0.97	0.99	0.98	0.98	0.98	1.0	0.99	1.0	0.97	0.98	0.9677	0.9516	0.95
WID8_CID73, low	1.0	1.0	0.99	1.0	1.0	1.0	0.99	1.0	0.99	0.96	1.0	0.99	0.99	0.99	0.9524	0.9841	0.96
WID8_CID74, default	0.98	0.99	0.99	0.97	0.99	1.0	0.99	1.0	1.0	0.98	0.99	0.99	1.0	0.99	0.9677	0.9677	0.96
WID8_CID74, high	0.98	1.0	0.98	0.98	0.99	0.99	0.99	1.0	0.99	1.0	1.0	0.96	0.97	1.0	0.9844	0.9688	0.96
WID8_CID74, low	0.99	1.0	0.99	0.99	0.99	0.99	0.99	0.99	0.98	0.98	0.98	1.0	0.99	0.99	0.9524	0.9683	0.96
WID8_CID75, default	0.98	1.0	0.99	0.99	0.97	0.99	0.99	0.99	0.96	0.99	0.99	1.0	0.99	1.0	0.9692	0.9692	0.95
WID8_CID75, high	0.98	0.99	0.98	0.98	0.99	0.99	0.98	1.0	0.99	0.98	1.0	0.99	0.99	0.99	0.9661	0.9831	0.94
WID8_CID75, low	1.0	1.0	1.0	1.0	0.99	0.98	0.98	1.0	0.98	0.98	0.98	0.99	0.99	0.99	0.9836	0.9672	0.95
WID8_CID76, default	0.98	0.97	0.98	0.98	0.99	0.98	1.0	0.98	0.99	0.99	0.98	1.0	0.99	0.98	0.9836	0.9836	0.96
WID8_CID76, high	0.99	1.0	1.0	0.99	1.0	0.99	1.0	1.0	0.97	1.0	0.98	1.0	1.0	0.98	0.9697	0.9848	0.94
WID8_CID76, low	0.98	0.99	0.99	0.98	1.0	0.99	0.98	1.0	0.98	1.0	0.99	0.99	0.99	1.0	0.9508	0.9836	0.95
WID8_CID77, default	1.0	0.98	1.0	1.0	0.99	0.99	1.0	1.0	0.98	0.96	0.99	0.99	1.0	0.99	0.9661	0.9661	0.96
WID8_CID77, high	1.0	1.0	0.99	1.0	0.98	0.99	0.98	1.0	1.0	0.98	0.98	1.0	0.99	1.0	0.9492	0.9661	0.96
WID8_CID77, low	0.99	0.97	0.99	0.99	0.98	1.0	0.99	0.97	1.0	0.99	1.0	1.0	1.0	1.0	0.9697	0.9697	0.96
WID8_CID78, default	0.96	0.99	0.96	0.97	0.99	1.0	0.99	0.99	1.0	0.97	0.97	0.99	1.0	0.99	0.9636	0.9818	0.96
WID8_CID78, high	0.99	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.99	0.99	0.99	1.0	0.99	0.9855	0.9855	0.95
WID8_CID78, low	0.99	0.98	0.98	0.99	1.0	0.99	0.98	0.99	1.0	0.99	1.0	1.0	0.99	0.98	0.9643	0.9643	0.95
WID8_CID79, default	0.99	0.98	0.98	0.98	1.0	0.98	0.99	0.97	0.98	0.99	1.0	1.0	1.0	0.97	0.971	0.9565	0.94
WID8_CID79, high	0.99	1.0	1.0	1.0	1.0	0.99	1.0	1.0	0.98	0.97	1.0	0.98	0.99	0.99	0.9608	0.9608	0.95
WID8_CID79, low	1.0	0.98	0.99	1.0	0.96	0.99	0.99	0.97	0.99	1.0	0.98	0.99	0.98	0.98	0.9688	0.9844	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 18: TRNG0 -25 °C WID8 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 25.0 (°C), TRNG0, WID8
Fail ratio 2.02%, ~ 1.84% expected

WID8_CID31, default	1.0	1.0	0.99	1.0	1.0	1.0	0.98	0.99	1.0	0.99	0.98	0.97	0.99	0.97	0.9677	0.9839	0.95
WID8_CID31, high	0.99	0.98	0.99	0.99	0.99	0.99	1.0	1.0	0.97	0.98	0.96	0.98	1.0	0.95	0.9643	0.9464	0.96
WID8_CID31, low	0.99	0.99	0.98	0.96	0.98	0.98	0.98	1.0	1.0	0.98	0.97	0.99	0.99	0.96	0.9855	0.9855	0.96
WID8_CID32, default	1.0	0.99	1.0	0.98	0.99	0.99	0.99	0.99	0.98	0.97	1.0	1.0	1.0	0.99	0.9825	0.9649	0.96
WID8_CID32, high	0.99	0.98	0.98	0.98	1.0	0.97	1.0	0.97	1.0	0.99	1.0	0.98	0.98	0.97	0.9825	0.9649	0.94
WID8_CID32, low	0.96	0.97	0.97	0.96	1.0	1.0	0.99	0.99	0.98	0.97	0.97	0.99	0.98	0.98	0.9649	0.9649	0.95
WID8_CID72, default	1.0	0.97	1.0	1.0	0.98	0.98	0.98	1.0	1.0	0.98	0.98	0.96	0.98	0.98	0.9833	0.9667	0.95
WID8_CID72, high	1.0	1.0	1.0	0.99	0.99	0.99	1.0	1.0	0.99	0.99	1.0	1.0	1.0	0.99	0.9508	0.9672	0.96
WID8_CID72, low	1.0	0.99	1.0	1.0	0.96	1.0	0.99	0.99	1.0	0.98	1.0	1.0	1.0	1.0	0.9524	0.9365	0.95
WID8_CID73, default	1.0	1.0	1.0	1.0	0.98	1.0	0.99	1.0	0.98	0.98	0.97	0.98	0.98	1.0	0.9677	0.9677	0.96
WID8_CID73, high	1.0	0.99	0.99	1.0	1.0	0.98	0.98	0.99	0.98	0.98	0.98	1.0	0.98	0.98	1.0	0.9672	0.95
WID8_CID73, low	1.0	0.99	1.0	1.0	0.98	0.98	1.0	0.99	0.98	0.99	0.99	1.0	0.99	0.99	0.9861	0.9722	0.96
WID8_CID74, default	1.0	1.0	1.0	1.0	0.99	1.0	0.96	0.96	0.99	0.99	1.0	1.0	0.99	1.0	0.9524	0.9524	0.95
WID8_CID74, high	1.0	0.99	0.99	1.0	0.99	1.0	0.99	1.0	0.98	0.99	0.98	0.99	0.97	1.0	0.9697	0.9545	0.94
WID8_CID74, low	0.99	0.97	0.99	0.99	0.98	1.0	1.0	0.99	0.98	1.0	1.0	0.99	0.99	1.0	0.9661	0.9661	0.96
WID8_CID75, default	1.0	0.97	0.99	1.0	0.98	0.99	0.99	0.99	0.99	0.98	0.98	0.99	0.98	0.98	0.9697	0.9697	0.95
WID8_CID75, high	1.0	1.0	1.0	1.0	0.98	0.99	1.0	0.99	0.98	0.98	1.0	0.98	0.99	0.98	0.9848	0.9848	0.95
WID8_CID75, low	1.0	0.99	1.0	1.0	0.99	0.99	0.99	0.97	0.99	1.0	0.99	1.0	1.0	0.98	0.9853	0.9853	0.96
WID8_CID76, default	1.0	1.0	1.0	1.0	1.0	0.99	0.99	0.99	0.99	0.99	0.99	0.98	0.99	0.99	0.9524	0.9683	0.94
WID8_CID76, high	1.0	0.99	1.0	1.0	0.99	0.99	0.99	0.98	1.0	1.0	0.99	0.99	0.98	1.0	0.9636	0.9818	0.95
WID8_CID76, low	1.0	0.99	0.99	1.0	0.98	1.0	0.99	0.96	1.0	0.99	1.0	0.99	0.99	0.99	0.9672	0.9672	0.95
WID8_CID77, default	0.99	0.99	0.99	0.97	0.98	0.99	0.97	0.99	1.0	0.98	0.99	0.99	1.0	1.0	0.9688	0.9844	0.93
WID8_CID77, high	0.99	1.0	0.99	0.99	0.97	0.99	1.0	0.96	0.98	1.0	0.98	0.99	1.0	1.0	0.9825	0.9474	0.96
WID8_CID77, low	1.0	0.99	1.0	1.0	0.99	1.0	0.98	0.98	0.99	0.98	0.99	0.99	1.0	1.0	0.9701	0.9851	0.96
WID8_CID78, default	1.0	0.98	0.99	0.99	0.98	0.99	1.0	0.99	1.0	0.99	1.0	1.0	0.98	0.99	0.9841	0.9683	0.96
WID8_CID78, high	0.99	0.99	1.0	1.0	0.99	0.99	0.99	0.99	0.99	0.98	1.0	1.0	0.99	0.99	0.9516	0.9677	0.96
WID8_CID78, low	0.98	1.0	0.98	0.99	0.99	0.99	0.99	0.97	1.0	0.99	0.97	0.99	0.99	0.98	0.9683	0.9683	0.97
WID8_CID79, default	1.0	1.0	1.0	1.0	1.0	0.98	0.97	1.0	1.0	0.97	0.99	0.98	0.97	0.99	0.9275	0.971	0.96
WID8_CID79, high	0.98	0.98	0.98	0.99	1.0	0.99	0.99	1.0	0.97	0.98	1.0	1.0	0.98	1.0	0.9846	0.9692	0.96
WID8_CID79, low	0.97	1.0	0.98	0.97	0.99	1.0	1.0	0.98	0.99	1.0	0.99	1.0	1.0	0.99	0.9615	0.9615	0.94
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 19: TRNG0 25 °C WID8 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 105.0 (°C), TRNG0, WID8
Fail ratio 2.53%, ~ 1.84% expected

WID8_CID31, default	1.0	0.99	1.0	1.0	0.99	0.99	1.0	1.0	0.99	0.95	0.99	1.0	0.98	1.0	0.9412	0.9706	0.96
WID8_CID31, high	0.97	0.98	0.97	0.97	1.0	1.0	0.97	0.99	0.98	1.0	0.99	1.0	0.99	0.99	0.9524	0.9841	0.96
WID8_CID31, low	0.99	0.98	0.99	0.99	1.0	1.0	0.99	0.99	0.99	0.98	1.0	1.0	0.99	1.0	0.9841	0.9841	0.95
WID8_CID32, default	0.98	0.99	0.98	0.99	0.98	0.98	1.0	0.98	0.99	1.0	0.99	1.0	0.99	0.98	0.9254	0.9701	0.96
WID8_CID32, high	1.0	0.98	1.0	0.98	0.99	1.0	1.0	1.0	1.0	0.98	0.99	0.99	1.0	0.99	0.9538	0.9692	0.94
WID8_CID32, low	0.99	0.98	0.99	0.99	0.99	1.0	0.99	1.0	0.99	0.99	0.98	0.99	1.0	1.0	0.9726	0.9726	0.96
WID8_CID72, default	0.99	0.98	0.99	0.98	0.99	1.0	1.0	1.0	0.97	1.0	0.99	1.0	0.99	1.0	0.9828	0.9828	0.95
WID8_CID72, high	0.98	0.99	0.97	0.98	0.99	0.98	0.98	0.97	1.0	1.0	1.0	0.97	0.96	0.99	0.95	0.9667	0.95
WID8_CID72, low	1.0	0.99	1.0	0.99	0.99	0.99	0.99	0.97	1.0	0.98	0.99	0.98	0.98	1.0	0.9821	0.9643	0.96
WID8_CID73, default	0.99	1.0	0.99	0.99	0.97	1.0	1.0	0.99	0.96	0.99	1.0	0.98	1.0	1.0	0.9825	0.9649	0.96
WID8_CID73, high	0.96	0.98	0.96	0.96	1.0	0.99	0.99	0.99	0.99	0.98	0.99	1.0	1.0	1.0	0.9836	0.9508	0.96
WID8_CID73, low	0.99	1.0	0.99	0.99	1.0	1.0	0.96	0.99	0.99	0.99	1.0	1.0	1.0	1.0	0.9688	0.9688	0.94
WID8_CID74, default	0.96	0.99	0.96	0.96	0.98	0.99	0.99	0.98	1.0	0.98	0.99	1.0	0.98	1.0	0.9552	0.9552	0.95
WID8_CID74, high	0.99	0.99	1.0	0.98	0.97	1.0	1.0	0.99	1.0	0.98	0.98	1.0	1.0	0.99	0.9853	0.9853	0.95
WID8_CID74, low	1.0	0.99	0.99	1.0	1.0	0.97	0.98	0.99	1.0	0.99	1.0	1.0	0.98	0.99	0.9667	0.9667	0.96
WID8_CID75, default	0.98	1.0	0.98	0.98	1.0	0.99	0.99	0.99	0.99	1.0	1.0	0.99	1.0	0.99	0.9273	0.9636	0.96
WID8_CID75, high	0.97	0.98	0.98	0.97	0.98	1.0	0.96	0.99	0.98	0.99	1.0	0.99	0.99	0.98	0.9861	0.9722	0.94
WID8_CID75, low	0.99	1.0	0.99	0.99	0.99	1.0	0.99	1.0	0.99	0.99	0.98	0.97	1.0	0.97	0.9508	0.9672	0.94
WID8_CID76, default	1.0	0.99	1.0	1.0	0.99	0.98	1.0	0.97	0.96	1.0	1.0	0.98	0.99	0.98	0.9821	0.9464	0.96
WID8_CID76, high	0.97	0.99	0.98	1.0	0.99	0.99	1.0	0.99	0.98	0.99	1.0	0.99	0.99	0.96	0.9851	0.9701	0.96
WID8_CID76, low	1.0	1.0	1.0	0.97	0.98	1.0	0.99	0.96	0.98	0.98	0.99	0.99	1.0	0.97	0.9831	1.0	0.95
WID8_CID77, default	0.98	0.98	0.99	0.98	1.0	0.99	1.0	1.0	0.99	1.0	0.99	1.0	1.0	0.98	0.9831	0.9831	0.96
WID8_CID77, high	0.99	0.99	0.97	0.99	0.9899	1.0	1.0	0.99	0.99	1.0	0.99	0.99	0.99	1.0	0.9661	0.9661	0.96
WID8_CID77, low	0.96	0.99	0.96	0.96	0.99	0.95	0.97	0.98	0.98	1.0	1.0	0.97	0.99	1.0	0.9552	0.9851	0.95
WID8_CID78, default	0.99	0.99	0.99	0.98	0.99	1.0	1.0	1.0	0.99	1.0	0.99	0.99	1.0	1.0	0.9538	0.9692	0.96
WID8_CID78, high	1.0	0.98	1.0	1.0	1.0	1.0	0.96	0.99	0.99	1.0	1.0	1.0	0.99	0.98	0.9531	0.9531	0.95
WID8_CID78, low	0.98	0.98	0.98	0.99	0.98	1.0	0.99	1.0	0.98	0.96	1.0	1.0	1.0	0.97	0.9718	0.9718	0.96
WID8_CID79, default	1.0	0.98	1.0	1.0	0.99	0.98	1.0	1.0	1.0	0.99	1.0	0.99	0.99	0.99	0.9655	0.9483	0.96
WID8_CID79, high	1.0	1.0	1.0	1.0	1.0	0.99	1.0	1.0	1.0	0.97	0.99	1.0	1.0	0.99	0.9853	0.9706	0.96
WID8_CID79, low	0.97	0.99	0.97	1.0	0.99	0.99	1.0	0.99	1.0	0.98	0.97	1.0	0.99	0.99	0.9688	0.9844	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 20: TRNG0 105 °C WID8 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 125.0 (°C), TRNG0, WID8
Fail ratio 1.85%, ~ 1.84% expected

WID8_CID31, default	0.99	1.0	0.99	0.99	1.0	0.99	0.99	1.0	0.99	0.99	0.99	0.97	0.99	1.0	0.9714	0.9714	0.96
WID8_CID31, high	0.99	0.99	0.99	0.99	0.99	1.0	0.97	1.0	0.99	0.99	0.99	0.99	0.99	0.99	0.9429	0.9714	0.95
WID8_CID31, low	1.0	1.0	1.0	0.99	0.99	0.99	0.98	0.98	1.0	0.98	0.99	0.99	0.99	0.98	0.9701	0.9701	0.95
WID8_CID32, default	0.98	1.0	0.99	0.98	0.99	0.99	1.0	0.99	0.99	1.0	0.99	1.0	1.0	0.99	0.9667	0.9667	0.96
WID8_CID32, high	0.99	0.98	0.99	0.99	0.99	0.99	0.99	0.99	0.98	1.0	0.99	0.98	0.97	0.99	0.9655	0.9655	0.96
WID8_CID32, low	1.0	1.0	1.0	0.99	1.0	0.98	0.98	0.97	1.0	0.97	0.99	1.0	1.0	0.99	0.971	0.971	0.96
WID8_CID72, default	1.0	0.99	1.0	1.0	0.98	0.98	0.98	0.99	1.0	1.0	1.0	0.98	1.0	1.0	0.9538	0.9385	0.96
WID8_CID72, high	0.99	0.99	0.99	0.99	0.99	1.0	0.99	1.0	0.99	0.99	0.99	0.99	0.99	0.98	0.9851	0.9552	0.97
WID8_CID72, low	1.0	1.0	1.0	1.0	0.99	0.98	1.0	0.97	0.99	0.99	1.0	0.99	0.98	0.99	0.9643	0.9464	0.95
WID8_CID73, default	1.0	1.0	1.0	0.99	0.99	0.97	1.0	1.0	1.0	0.97	0.97	0.99	0.99	0.98	0.9474	0.9825	0.96
WID8_CID73, high	0.98	0.98	0.98	0.98	0.99	1.0	0.99	0.99	0.99	1.0	0.98	1.0	1.0	0.98	0.9848	0.9545	0.95
WID8_CID73, low	0.98	0.99	0.98	0.98	1.0	1.0	1.0	1.0	0.99	0.96	0.97	1.0	1.0	1.0	0.9643	0.9643	0.95
WID8_CID74, default	0.99	0.98	0.99	0.99	0.99	0.99	1.0	0.99	0.98	0.96	0.99	0.99	0.99	0.99	0.9857	0.9714	0.96
WID8_CID74, high	0.98	0.98	0.98	0.98	0.99	1.0	1.0	0.99	1.0	1.0	0.98	0.99	0.97	1.0	0.9821	0.9643	0.95
WID8_CID74, low	0.98	1.0	0.97	0.98	1.0	1.0	0.98	1.0	0.98	0.99	1.0	0.98	0.99	1.0	0.9667	0.9667	0.95
WID8_CID75, default	0.98	0.99	0.97	0.96	0.99	0.98	1.0	1.0	0.97	1.0	1.0	0.99	0.99	0.99	0.9706	0.9706	0.95
WID8_CID75, high	1.0	0.98	1.0	1.0	0.99	1.0	1.0	0.99	1.0	0.99	0.97	0.97	0.99	0.98	0.9697	0.9697	0.96
WID8_CID75, low	1.0	0.98	1.0	1.0	0.96	1.0	0.99	1.0	0.99	0.98	0.99	0.99	0.96	0.99	0.95	0.9667	0.96
WID8_CID76, default	1.0	0.98	0.99	1.0	1.0	0.96	0.97	0.98	0.99	0.99	0.98	0.98	1.0	0.98	0.9839	0.9677	0.94
WID8_CID76, high	0.99	0.99	1.0	1.0	0.99	0.98	1.0	0.98	0.99	1.0	1.0	1.0	1.0	1.0	0.95	0.9833	0.96
WID8_CID76, low	0.99	0.99	0.99	0.99	0.99	0.96	0.99	0.98	0.99	0.98	0.98	0.98	0.99	0.99	0.9615	0.9808	0.96
WID8_CID77, default	0.99	0.99	0.99	0.99	0.99	0.99	0.98	0.99	1.0	0.97	0.99	1.0	0.99	0.99	0.9677	0.9516	0.96
WID8_CID77, high	0.97	0.99	0.98	0.97	0.98	0.99	0.97	0.96	1.0	0.98	0.99	0.99	0.99	1.0	0.9683	0.9841	0.96
WID8_CID77, low	0.99	1.0	0.99	0.99	0.98	0.99	1.0	1.0	0.99	0.98	0.97	0.97	0.99	0.96	0.9701	0.9701	0.97
WID8_CID78, default	0.97	1.0	0.97	0.97	0.97	0.98	0.99	1.0	0.97	0.98	1.0	1.0	0.98	1.0	0.9841	0.9683	0.96
WID8_CID78, high	0.99	1.0	0.99	0.99	1.0	0.99	0.98	1.0	1.0	0.99	0.99	0.98	1.0	1.0	0.9298	0.9649	0.96
WID8_CID78, low	0.99	1.0	0.98	1.0	0.98	0.98	1.0	0.98	1.0	1.0	0.99	1.0	0.98	0.98	0.9608	0.9412	0.96
WID8_CID79, default	1.0	1.0	0.99	0.99	1.0	1.0	0.96	0.99	0.98	0.97	0.99	1.0	0.98	0.96	0.9655	0.931	0.97
WID8_CID79, high	0.95	0.99	0.96	0.97	1.0	0.99	1.0	0.99	0.99	1.0	0.98	1.0	1.0	1.0	0.9831	0.9661	0.96
WID8_CID79, low	0.98	0.98	0.98	0.97	1.0	1.0	0.99	1.0	1.0	0.98	0.98	0.99	0.98	1.0	0.971	0.971	0.97
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 21: TRNG0 125 °C WID8 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -40.0 (°C), TRNG1, WID8
Fail ratio 1.44%, ~ 1.84% expected

WID8_CID31, default	1.0	0.98	1.0	1.0	0.97	1.0	1.0	1.0	0.98	0.98	1.0	0.98	0.99	1.0	0.9667	0.9667	0.97
WID8_CID31, high	1.0	0.98	1.0	0.99	0.97	0.98	1.0	0.97	0.99	1.0	0.99	0.98	1.0	1.0	0.9697	0.9697	0.94
WID8_CID31, low	1.0	0.99	1.0	0.99	0.99	0.99	1.0	1.0	1.0	0.98	1.0	1.0	1.0	0.96	0.9677	0.9516	0.96
WID8_CID32, default	0.98	1.0	0.98	0.98	0.99	1.0	0.99	0.99	0.99	1.0	0.99	0.98	0.99	0.99	0.9636	0.9818	0.96
WID8_CID32, high	0.99	0.98	0.99	0.99	0.99	0.98	0.98	0.98	0.99	0.98	0.99	0.99	0.95	1.0	0.9726	0.9589	0.95
WID8_CID32, low	1.0	0.99	1.0	1.0	1.0	0.98	0.99	1.0	0.98	1.0	1.0	0.99	0.98	0.99	0.9692	0.9538	0.96
WID8_CID72, default	1.0	1.0	1.0	1.0	0.99	1.0	0.99	0.99	1.0	0.97	0.99	1.0	0.99	0.96	0.9818	0.9636	0.95
WID8_CID72, high	0.99	1.0	0.99	0.99	0.99	1.0	1.0	0.96	0.99	1.0	0.99	1.0	0.98	1.0	0.9636	0.9455	0.96
WID8_CID72, low	0.98	0.98	0.99	0.98	0.99	0.99	0.99	1.0	1.0	0.99	1.0	0.99	0.98	0.98	0.9683	0.9841	0.96
WID8_CID73, default	0.99	1.0	1.0	0.98	0.99	1.0	1.0	1.0	0.99	0.97	1.0	0.97	0.99	0.99	0.9692	0.9538	0.96
WID8_CID73, high	1.0	1.0	1.0	1.0	1.0	0.99	0.99	1.0	1.0	0.97	1.0	0.97	0.98	1.0	0.9153	0.9492	0.96
WID8_CID73, low	0.97	0.98	0.98	0.97	0.99	0.96	0.99	1.0	0.99	1.0	0.98	0.99	0.97	0.99	0.9508	0.9672	0.96
WID8_CID74, default	0.99	0.99	1.0	1.0	1.0	1.0	0.97	1.0	0.96	1.0	1.0	0.99	1.0	0.99	0.973	0.9865	0.95
WID8_CID74, high	0.99	1.0	1.0	1.0	0.98	1.0	1.0	0.98	1.0	0.99	1.0	1.0	1.0	0.98	0.9623	0.9623	0.96
WID8_CID74, low	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0.99	0.98	0.99	1.0	1.0	0.99	0.99	0.9655	0.9828	0.97
WID8_CID75, default	1.0	0.99	1.0	0.99	0.99	0.99	1.0	0.98	0.99	1.0	1.0	1.0	1.0	1.0	0.9677	0.9677	0.96
WID8_CID75, high	0.97	1.0	0.97	0.97	0.99	0.99	0.98	0.99	0.99	1.0	1.0	0.99	1.0	0.99	0.9552	0.9701	0.96
WID8_CID75, low	1.0	1.0	1.0	1.0	1.0	1.0	0.99	1.0	0.97	0.95	1.0	1.0	0.99	1.0	0.9508	0.9836	0.96
WID8_CID76, default	0.99	0.99	0.99	0.99	1.0	1.0	0.99	1.0	0.97	0.98	0.98	0.98	0.97	0.99	0.9552	0.9701	0.95
WID8_CID76, high	0.99	0.98	0.99	1.0	1.0	1.0	0.99	1.0	0.97	1.0	1.0	1.0	1.0	0.98	0.9365	0.9524	0.95
WID8_CID76, low	1.0	1.0	0.99	1.0	0.99	1.0	1.0	0.99	0.99	0.99	0.99	0.99	0.99	1.0	0.9804	0.9608	0.95
WID8_CID77, default	0.99	0.99	0.99	0.99	1.0	1.0	1.0	0.98	0.99	0.96	1.0	0.99	1.0	0.97	0.9636	0.9636	0.96
WID8_CID77, high	0.99	0.98	0.98	0.98	0.98	0.97	0.99	1.0	0.99	0.99	0.98	0.98	1.0	0.99	0.9661	0.9831	0.96
WID8_CID77, low	1.0	1.0	1.0	1.0	0.99	0.97	0.98	0.99	1.0	0.98	1.0	0.96	1.0	0.99	0.9483	0.9655	0.96
WID8_CID78, default	0.98	0.97	0.99	0.98	0.99	0.99	0.99	1.0	1.0	0.99	0.99	0.98	0.98	1.0	0.9531	0.9844	0.96
WID8_CID78, high	0.97	0.99	0.97	0.98	0.98	0.99	1.0	0.99	1.0	1.0	0.99	1.0	0.99	0.99	0.9672	0.9508	0.93
WID8_CID78, low	0.99	0.98	0.99	1.0	1.0	1.0	1.0	1.0	1.0	0.96	1.0	1.0	0.99	0.99	0.9851	0.9701	0.96
WID8_CID79, default	1.0	1.0	0.99	0.99	0.97	1.0	1.0	0.99	0.98	0.99	0.94	1.0	0.99	0.98	0.9677	0.9677	0.97
WID8_CID79, high	0.98	1.0	0.96	0.97	0.99	1.0	1.0	0.98	1.0	0.98	0.99	0.99	0.99	0.99	0.9655	0.9655	0.97
WID8_CID79, low	1.0	0.99	1.0	1.0	0.99	1.0	1.0	1.0	0.99	0.96	1.0	0.98	1.0	0.98	0.9538	0.9538	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 22: TRNG1 -40 °C WID8 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -25.0 (°C), TRNG1, WID8
Fail ratio 1.48%, ~ 1.84% expected

WID8_CID31, default	0.99	0.99	0.99	0.98	1.0	0.99	0.99	0.99	0.99	0.99	0.98	1.0	0.99	0.98	0.9836	0.9672	0.97
WID8_CID31, high	1.0	0.97	1.0	1.0	1.0	0.99	1.0	0.99	0.98	1.0	0.99	0.99	1.0	1.0	0.9649	0.9825	0.96
WID8_CID31, low	1.0	1.0	1.0	1.0	1.0	0.98	0.98	0.99	0.98	0.98	1.0	1.0	0.99	0.99	0.9655	0.9828	0.96
WID8_CID32, default	1.0	1.0	0.99	0.99	0.98	0.98	1.0	0.99	0.99	0.98	1.0	1.0	0.97	0.9643	0.9643	0.95	
WID8_CID32, high	1.0	1.0	0.99	0.99	0.98	1.0	1.0	0.99	0.99	0.98	1.0	0.99	0.99	1.0	0.9855	0.971	0.96
WID8_CID32, low	0.99	0.99	1.0	0.99	0.99	0.98	0.96	0.97	0.99	0.99	0.99	0.99	1.0	0.99	0.9583	0.9583	0.96
WID8_CID72, default	0.98	1.0	0.99	0.99	0.99	0.99	0.99	1.0	0.98	0.99	0.98	1.0	0.99	1.0	0.9577	0.9718	0.96
WID8_CID72, high	0.99	1.0	0.98	0.98	1.0	1.0	0.98	0.98	0.99	0.99	0.99	1.0	1.0	0.99	0.9667	0.9833	0.96
WID8_CID72, low	0.97	0.98	0.98	0.98	1.0	0.98	0.99	0.99	0.99	1.0	1.0	0.99	1.0	0.98	0.9608	0.9608	0.96
WID8_CID73, default	0.98	1.0	0.98	0.98	0.98	0.98	0.98	1.0	0.99	1.0	1.0	0.99	1.0	0.99	0.9508	0.9508	0.96
WID8_CID73, high	0.99	1.0	0.99	0.99	1.0	0.98	0.98	1.0	0.98	0.99	0.99	0.97	0.99	0.99	0.9863	0.9863	0.96
WID8_CID73, low	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.98	1.0	0.98	0.99	0.98	0.99	0.99	0.9839	0.9839	0.96
WID8_CID74, default	0.99	0.99	1.0	0.99	1.0	0.99	0.98	0.99	0.98	0.98	1.0	1.0	1.0	1.0	0.96	0.96	0.96
WID8_CID74, high	1.0	0.99	1.0	0.98	1.0	1.0	1.0	0.98	0.99	1.0	0.99	0.99	0.98	0.97	0.9833	0.95	0.96
WID8_CID74, low	1.0	0.99	1.0	0.99	0.99	0.98	0.99	0.98	0.97	0.99	1.0	0.98	1.0	1.0	0.9833	0.9667	0.96
WID8_CID75, default	0.98	0.99	0.98	0.98	0.98	1.0	0.98	0.99	1.0	0.99	1.0	1.0	1.0	1.0	0.9483	0.9483	0.96
WID8_CID75, high	0.99	0.99	1.0	1.0	0.99	0.99	0.99	0.98	0.99	0.97	1.0	0.98	0.98	1.0	0.9655	0.9655	0.95
WID8_CID75, low	0.99	0.99	0.99	0.98	0.99	0.97	1.0	0.99	0.98	1.0	0.97	0.98	1.0	0.98	0.9672	0.9836	0.97
WID8_CID76, default	0.99	0.99	0.99	0.99	0.99	1.0	1.0	0.95	1.0	0.99	0.97	0.99	0.98	0.97	0.9841	0.9524	0.95
WID8_CID76, high	1.0	1.0	1.0	0.99	0.99	0.99	0.99	0.97	0.97	1.0	1.0	0.99	0.99	1.0	0.9839	0.9677	0.96
WID8_CID76, low	1.0	1.0	0.99	1.0	0.98	1.0	0.99	1.0	0.98	1.0	1.0	0.99	0.99	1.0	0.9836	0.9508	0.96
WID8_CID77, default	1.0	1.0	1.0	1.0	1.0	0.97	0.98	0.99	0.99	0.99	0.96	0.99	0.99	1.0	0.9815	0.9815	0.95
WID8_CID77, high	1.0	0.99	1.0	1.0	0.99	1.0	1.0	0.99	0.99	0.99	1.0	0.98	0.99	1.0	0.9643	0.9643	0.95
WID8_CID77, low	0.99	1.0	1.0	0.99	0.99	1.0	0.99	0.98	1.0	1.0	0.98	0.99	0.98	1.0	0.9833	0.9833	0.96
WID8_CID78, default	1.0	1.0	1.0	1.0	0.99	1.0	0.99	0.99	1.0	0.97	0.97	0.99	0.98	0.98	0.9677	0.9839	0.96
WID8_CID78, high	0.98	1.0	0.99	0.99	1.0	0.99	1.0	0.99	0.98	0.99	0.97	0.98	0.99	0.97	0.9697	0.9697	0.96
WID8_CID78, low	1.0	0.98	1.0	1.0	0.99	0.97	0.99	0.99	0.99	1.0	1.0	0.99	1.0	1.0	0.9667	0.9833	0.95
WID8_CID79, default	0.99	0.98	0.98	0.99	1.0	1.0	0.99	1.0	1.0	0.99	0.98	1.0	1.0	1.0	0.9559	0.9853	0.97
WID8_CID79, high	0.99	0.98	0.99	0.99	1.0	0.97	0.99	0.98	0.99	1.0	0.99	0.99	1.0	0.99	0.9655	0.9828	0.97
WID8_CID79, low	0.99	0.98	0.99	1.0	0.97	0.98	1.0	1.0	1.0	0.99	0.99	1.0	1.0	1.0	0.9841	0.9683	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 23: TRNG1 -25 °C WID8 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 25.0 (°C), TRNG1, WID8
Fail ratio 2.10%, ~ 1.84% expected

WID8_CID31, default	0.99	0.98	0.99	1.0	0.97	0.99	1.0	0.97	0.99	0.99	0.98	1.0	0.98	0.99	0.9661	0.9831	0.96
WID8_CID31, high	0.99	0.97	0.98	0.99	1.0	0.98	1.0	0.99	0.99	1.0	0.99	1.0	0.98	1.0	0.9483	0.9483	0.96
WID8_CID31, low	1.0	1.0	0.99	1.0	0.99	0.98	0.98	0.99	0.99	0.98	0.97	0.98	0.97	0.99	0.9672	0.9672	0.96
WID8_CID32, default	0.98	0.98	0.97	0.97	1.0	1.0	0.97	0.99	0.99	0.96	0.97	0.98	0.99	1.0	0.9859	0.9577	0.97
WID8_CID32, high	1.0	1.0	1.0	1.0	0.99	0.99	0.99	0.99	1.0	0.99	1.0	1.0	1.0	0.99	0.9516	0.9839	0.96
WID8_CID32, low	0.99	0.98	0.99	0.99	0.98	1.0	0.98	0.97	0.99	0.99	0.99	0.99	1.0	1.0	0.9828	0.9655	0.96
WID8_CID72, default	1.0	0.99	1.0	1.0	0.99	1.0	1.0	1.0	0.99	0.98	1.0	1.0	0.99	0.98	0.971	0.971	0.95
WID8_CID72, high	0.99	0.98	0.99	0.97	1.0	0.99	1.0	0.96	0.98	0.99	0.99	1.0	0.99	0.99	0.9667	0.9667	0.96
WID8_CID72, low	0.99	0.97	1.0	1.0	0.99	0.99	1.0	0.98	0.99	1.0	0.99	0.99	1.0	1.0	0.9464	0.9821	0.97
WID8_CID73, default	1.0	0.98	1.0	1.0	0.99	0.99	1.0	0.99	0.99	1.0	0.99	1.0	1.0	1.0	0.9706	0.9559	0.95
WID8_CID73, high	1.0	0.98	1.0	0.99	0.99	0.98	0.99	0.99	0.99	0.98	1.0	0.98	1.0	0.97	0.973	0.9865	0.96
WID8_CID73, low	0.99	0.99	1.0	1.0	1.0	0.99	1.0	0.99	0.96	0.99	0.98	0.98	0.98	0.98	0.9565	0.9565	0.96
WID8_CID74, default	0.97	1.0	0.97	0.97	0.99	0.99	1.0	0.99	1.0	0.99	1.0	1.0	1.0	0.98	0.9636	0.9818	0.96
WID8_CID74, high	1.0	0.98	1.0	1.0	1.0	0.99	0.99	1.0	0.99	0.99	1.0	0.98	0.99	0.99	0.9808	0.9808	0.95
WID8_CID74, low	1.0	0.99	1.0	1.0	0.98	0.98	0.97	1.0	0.99	0.99	0.97	1.0	0.99	0.99	0.9692	0.9538	0.96
WID8_CID75, default	0.96	0.99	0.97	0.94	0.98	0.98	1.0	0.99	1.0	0.98	0.99	0.99	0.99	1.0	0.9615	0.9615	0.96
WID8_CID75, high	0.99	0.99	0.99	0.99	1.0	0.99	1.0	1.0	1.0	0.98	0.97	1.0	1.0	1.0	0.9661	0.9661	0.95
WID8_CID75, low	1.0	0.99	1.0	1.0	0.98	0.98	1.0	0.98	0.98	0.99	0.99	0.99	0.99	0.98	0.9344	0.9836	0.96
WID8_CID76, default	0.99	0.98	1.0	1.0	0.99	0.99	0.98	1.0	0.99	1.0	0.99	0.99	0.97	0.99	0.9836	0.9672	0.94
WID8_CID76, high	1.0	1.0	0.99	0.99	0.98	1.0	0.98	0.97	0.97	0.97	0.98	1.0	1.0	0.98	0.9464	0.9821	0.95
WID8_CID76, low	0.98	0.99	0.99	1.0	1.0	1.0	0.96	0.97	1.0	0.98	0.98	1.0	0.99	0.99	0.9677	0.9355	0.95
WID8_CID77, default	0.97	1.0	0.99	0.97	0.99	0.98	0.99	1.0	0.99	0.98	1.0	0.97	0.98	1.0	0.9545	0.9697	0.93
WID8_CID77, high	0.99	1.0	0.99	0.99	0.99	1.0	0.97	1.0	0.98	0.99	1.0	1.0	1.0	0.98	0.9839	0.9839	0.96
WID8_CID77, low	0.99	0.99	0.99	1.0	0.98	0.96	0.99	1.0	0.96	1.0	0.99	0.99	1.0	1.0	0.9851	0.9851	0.96
WID8_CID78, default	1.0	0.98	1.0	1.0	1.0	0.99	1.0	0.99	0.99	0.97	0.98	1.0	0.98	0.97	0.9649	0.9825	0.95
WID8_CID78, high	0.98	0.99	0.99	0.98	0.99	0.98	0.98	0.98	1.0	0.97	0.96	0.98	0.97	1.0	0.9649	0.9825	0.96
WID8_CID78, low	0.99	0.98	0.99	0.99	1.0	0.99	1.0	0.98	0.99	1.0	0.99	1.0	1.0	0.99	0.9677	0.9677	0.96
WID8_CID79, default	0.99	0.98	0.99	0.99	1.0	0.97	1.0	1.0	0.97	1.0	0.99	1.0	1.0	0.96	0.9692	0.9846	0.96
WID8_CID79, high	1.0	0.98	1.0	1.0	0.98	0.98	0.99	0.97	0.99	1.0	0.99	0.99	0.99	0.99	0.9804	0.9804	0.96
WID8_CID79, low	0.99	1.0	1.0	1.0	0.98	0.98	0.98	1.0	0.99	0.98	1.0	0.98	1.0	0.98	0.9636	0.9818	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 24: TRNG1 25 °C WID8 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 105.0 (°C), TRNG1, WID8
Fail ratio 2.06%, ~ 1.84% expected

WID8_CID31, default	0.98	1.0	0.98	0.98	0.99	1.0	0.99	0.97	0.99	0.98	1.0	0.98	0.97	1.0	0.9661	0.9831	0.96
WID8_CID31, high	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	1.0	0.99	0.99	0.98	0.99	0.98	0.9836	0.9672	0.96
WID8_CID31, low	1.0	1.0	1.0	1.0	0.98	1.0	0.98	0.96	0.99	0.97	0.99	0.99	1.0	1.0	0.9836	0.9836	0.96
WID8_CID32, default	1.0	1.0	1.0	1.0	1.0	0.98	1.0	1.0	1.0	0.99	0.99	0.99	1.0	1.0	0.9692	1.0	0.94
WID8_CID32, high	0.96	1.0	0.96	0.97	0.98	0.99	1.0	1.0	0.96	0.99	1.0	1.0	1.0	0.98	0.9677	0.9677	0.96
WID8_CID32, low	1.0	1.0	1.0	1.0	1.0	0.98	1.0	0.99	1.0	1.0	0.99	0.99	0.99	1.0	0.9429	0.9714	0.96
WID8_CID72, default	1.0	1.0	0.98	1.0	1.0	1.0	1.0	1.0	0.99	0.98	0.99	0.99	0.99	0.99	0.9531	0.9844	0.95
WID8_CID72, high	0.99	0.99	0.99	0.99	0.97	1.0	1.0	1.0	1.0	0.99	0.99	0.97	0.98	0.99	0.9649	0.9825	0.96
WID8_CID72, low	1.0	1.0	1.0	1.0	0.99	0.99	0.97	0.97	0.96	0.99	0.99	1.0	1.0	0.97	0.9677	0.9839	0.96
WID8_CID73, default	0.99	1.0	0.99	0.99	0.99	0.99	1.0	0.98	0.96	1.0	1.0	1.0	1.0	1.0	0.9825	0.9649	0.96
WID8_CID73, high	0.99	1.0	1.0	0.99	1.0	0.98	1.0	0.99	0.98	0.98	1.0	1.0	1.0	0.99	0.9464	0.9643	0.96
WID8_CID73, low	0.98	0.98	0.98	0.98	0.98	0.99	0.97	0.98	0.97	0.98	0.99	1.0	0.98	0.98	0.9831	0.9831	0.94
WID8_CID74, default	1.0	1.0	1.0	1.0	0.99	0.99	1.0	0.99	0.98	0.99	0.99	0.97	0.97	1.0	0.9701	0.9403	0.95
WID8_CID74, high	0.98	1.0	0.98	0.98	0.99	1.0	0.98	1.0	0.99	1.0	1.0	0.99	0.98	0.96	0.9683	0.9683	0.94
WID8_CID74, low	0.99	0.99	1.0	0.99	0.98	0.99	0.99	1.0	0.99	0.99	0.99	0.99	1.0	1.0	0.9833	0.9833	0.94
WID8_CID75, default	1.0	0.99	1.0	1.0	0.98	0.99	1.0	1.0	0.99	0.99	1.0	1.0	1.0	0.98	0.9672	0.9836	0.95
WID8_CID75, high	1.0	0.98	1.0	0.99	0.99	1.0	0.97	0.97	1.0	1.0	0.99	0.99	0.99	0.99	0.9412	0.9706	0.93
WID8_CID75, low	0.99	0.98	0.99	1.0	1.0	0.98	1.0	0.96	1.0	0.99	1.0	0.98	0.98	0.99	0.9868	0.9868	0.96
WID8_CID76, default	1.0	1.0	0.99	1.0	0.98	0.98	0.99	1.0	0.99	0.99	1.0	0.99	1.0	1.0	0.9545	0.9697	0.95
WID8_CID76, high	0.98	1.0	0.98	0.97	1.0	0.98	0.98	0.98	1.0	0.98	0.99	1.0	0.99	0.99	0.9846	0.9538	0.96
WID8_CID76, low	0.99	0.99	0.99	0.99	1.0	0.99	0.98	1.0	1.0	0.98	1.0	1.0	0.99	0.99	0.9538	0.9846	0.94
WID8_CID77, default	1.0	0.99	1.0	0.99	1.0	0.96	0.99	0.98	0.99	0.98	0.98	0.99	0.98	0.97	0.9636	0.9818	0.95
WID8_CID77, high	1.0	0.98	1.0	1.0	0.96	0.98	1.0	0.97	0.98	0.99	0.99	0.99	1.0	0.99	0.9811	0.9811	0.97
WID8_CID77, low	1.0	1.0	1.0	1.0	0.98	0.99	0.98	1.0	0.98	1.0	1.0	0.98	0.97	1.0	0.9559	0.9559	0.96
WID8_CID78, default	0.99	0.94	0.99	0.98	1.0	0.97	1.0	0.99	0.99	0.98	0.99	1.0	1.0	0.99	0.9672	0.9836	0.95
WID8_CID78, high	1.0	1.0	1.0	0.99	0.97	1.0	0.98	0.98	1.0	0.99	0.97	1.0	1.0	0.94	0.9524	0.9524	0.97
WID8_CID78, low	0.99	1.0	0.99	0.99	1.0	1.0	0.99	0.98	0.99	1.0	0.98	0.99	1.0	0.99	0.9545	0.9848	0.96
WID8_CID79, default	1.0	0.99	1.0	1.0	1.0	0.99	0.98	0.99	0.98	0.98	0.99	0.98	0.97	1.0	0.9608	0.9608	0.94
WID8_CID79, high	0.99	0.99	0.99	1.0	1.0	0.99	1.0	1.0	0.98	1.0	0.98	0.99	0.98	1.0	0.94	0.96	0.95
WID8_CID79, low	0.99	1.0	0.98	0.99	0.99	0.99	0.96	0.99	0.98	0.98	0.99	1.0	1.0	1.0	0.9683	0.9365	0.97
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 25: TRNG1 105 °C WID8 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 125.0 (°C), TRNG1, WID8
Fail ratio 1.98%, ~ 1.84% expected

WID8_CID31, default	0.97	1.0	0.96	0.97	1.0	1.0	1.0	1.0	0.97	0.98	0.99	1.0	1.0	1.0	0.9661	0.9661	0.95
WID8_CID31, high	1.0	0.99	0.99	1.0	0.98	0.98	0.99	0.98	0.99	1.0	1.0	0.97	0.98	0.98	0.9701	0.9552	0.96
WID8_CID31, low	0.97	0.98	0.98	0.98	0.97	0.99	0.99	1.0	1.0	0.99	1.0	0.99	0.97	0.98	0.9636	0.9636	0.95
WID8_CID32, default	0.97	0.97	0.98	0.96	1.0	0.99	0.98	0.98	0.98	0.98	0.99	0.98	0.99	1.0	0.9649	0.9474	0.96
WID8_CID32, high	0.99	1.0	0.99	1.0	0.99	0.99	1.0	1.0	0.99	0.97	0.98	0.99	0.99	1.0	0.9672	0.9836	0.95
WID8_CID32, low	1.0	1.0	1.0	1.0	0.98	0.99	0.99	0.97	0.99	1.0	1.0	0.99	0.99	0.99	0.9538	0.9846	0.96
WID8_CID72, default	0.97	0.98	0.98	0.97	0.98	1.0	0.98	0.96	0.99	0.99	0.99	0.99	1.0	1.0	0.9545	0.9848	0.96
WID8_CID72, high	1.0	0.99	1.0	1.0	0.99	0.99	1.0	0.99	1.0	1.0	0.99	1.0	0.99	0.98	0.9649	0.9825	0.96
WID8_CID72, low	0.97	1.0	0.97	0.98	0.96	0.97	0.99	0.98	1.0	0.99	0.98	0.99	0.96	0.99	0.9688	0.9844	0.96
WID8_CID73, default	1.0	0.98	0.99	1.0	0.99	1.0	0.98	1.0	1.0	0.98	0.99	0.97	0.99	0.98	0.9661	0.9831	0.96
WID8_CID73, high	0.97	0.99	0.96	0.97	1.0	1.0	1.0	0.98	0.99	0.97	1.0	0.98	0.99	0.99	0.9667	0.9667	0.95
WID8_CID73, low	0.99	0.97	0.98	0.98	0.98	1.0	0.97	0.99	1.0	0.99	1.0	0.98	0.98	1.0	0.9839	0.9839	0.94
WID8_CID74, default	0.99	1.0	0.99	0.99	0.97	1.0	0.99	1.0	0.99	0.99	1.0	0.99	0.97	0.99	0.9672	0.9672	0.95
WID8_CID74, high	0.99	0.95	0.99	1.0	1.0	0.96	0.99	0.99	0.99	1.0	1.0	1.0	0.99	0.97	0.9508	0.9836	0.96
WID8_CID74, low	1.0	0.98	0.99	1.0	0.99	0.99	0.99	0.98	0.98	0.96	1.0	0.99	1.0	0.98	0.9851	0.9701	0.96
WID8_CID75, default	1.0	1.0	1.0	1.0	1.0	0.99	1.0	1.0	0.98	1.0	0.98	1.0	1.0	0.99	0.9825	0.9825	0.95
WID8_CID75, high	0.99	1.0	0.99	0.99	1.0	0.99	1.0	1.0	0.97	0.97	0.99	0.99	1.0	1.0	0.9672	0.9672	0.96
WID8_CID75, low	1.0	0.99	1.0	1.0	0.97	0.98	0.99	0.99	1.0	1.0	0.98	0.98	0.99	1.0	0.9643	0.9643	0.95
WID8_CID76, default	1.0	0.98	1.0	0.99	1.0	0.98	0.98	0.98	1.0	1.0	0.99	0.99	0.99	0.99	0.9667	0.9833	0.95
WID8_CID76, high	1.0	1.0	1.0	0.99	0.98	0.99	0.96	1.0	0.99	0.99	0.99	0.98	1.0	1.0	0.9841	0.9683	0.96
WID8_CID76, low	0.98	0.99	0.99	0.98	0.98	0.99	1.0	0.99	0.98	1.0	0.99	0.98	0.97	0.99	0.9655	0.9655	0.96
WID8_CID77, default	0.99	1.0	0.99	0.99	0.98	1.0	0.99	0.99	0.99	0.99	1.0	0.98	0.98	0.98	0.9697	0.9848	0.95
WID8_CID77, high	1.0	0.99	1.0	1.0	1.0	0.97	0.99	0.98	1.0	0.99	0.98	0.99	0.99	1.0	0.9643	0.9821	0.94
WID8_CID77, low	0.97	1.0	0.97	0.99	0.99	0.98	0.97	0.96	0.99	1.0	0.99	0.99	0.99	0.99	0.9821	0.9821	0.96
WID8_CID78, default	1.0	0.99	1.0	1.0	0.96	1.0	0.99	0.99	0.98	0.99	0.97	0.98	0.98	0.99	0.9821	0.9643	0.96
WID8_CID78, high	0.99	0.98	0.99	0.99	0.99	0.99	1.0	0.99	0.94	0.97	0.99	0.97	0.99	1.0	0.9661	0.9492	0.95
WID8_CID78, low	0.97	0.98	0.97	0.97	0.97	0.99	0.99	0.96	1.0	0.97	0.99	1.0	0.99	0.99	0.9677	0.9677	0.96
WID8_CID79, default	0.99	0.99	0.99	1.0	0.98	0.99	0.96	1.0	1.0	1.0	0.99	1.0	1.0	0.99	0.9846	0.9692	0.95
WID8_CID79, high	0.98	1.0	0.99	0.97	0.98	1.0	0.99	0.98	0.99	0.99	1.0	1.0	0.99	1.0	0.9677	0.9839	0.96
WID8_CID79, low	1.0	0.99	1.0	1.0	1.0	0.97	0.98	0.98	0.99	0.97	0.99	1.0	0.99	0.98	0.9846	0.9692	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 26: TRNG1 125 °C WID8 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -40.0 (°C), TRNG0, WID11
Fail ratio 1.63%, ~ 1.84% expected

WID11_CID101, default	0.99	0.99	0.99	0.99	1.0	0.98	1.0	0.97	0.98	1.0	0.98	1.0	0.99	0.99	0.9836	0.9836	0.96
WID11_CID101, high	1.0	0.99	1.0	1.0	0.99	0.99	0.99	0.97	0.99	0.99	1.0	1.0	0.98	0.99	0.9688	0.9688	0.96
WID11_CID101, low	0.98	0.99	0.99	0.98	0.97	0.99	1.0	0.99	0.99	0.99	0.99	0.99	1.0	0.99	0.9692	0.9692	0.95
WID11_CID103, default	0.99	0.99	0.99	0.99	0.98	0.99	1.0	0.97	0.97	0.97	0.99	0.99	1.0	0.99	0.971	0.971	0.97
WID11_CID103, high	1.0	1.0	1.0	1.0	0.99	1.0	1.0	0.98	0.99	0.99	0.98	0.97	1.0	1.0	0.9677	0.9677	0.93
WID11_CID103, low	0.99	0.99	0.98	0.99	0.99	0.99	1.0	1.0	0.99	1.0	0.99	0.99	0.99	1.0	0.9655	0.9828	0.94
WID11_CID104, default	0.99	0.98	0.99	0.99	1.0	1.0	1.0	0.99	0.99	0.99	0.99	1.0	1.0	0.99	0.9841	0.9683	0.96
WID11_CID104, high	0.99	0.99	0.99	0.99	0.99	0.98	0.99	0.98	0.99	0.98	0.99	0.99	0.99	0.99	0.9677	0.9677	0.96
WID11_CID104, low	0.97	0.99	0.97	0.97	0.98	1.0	1.0	0.99	1.0	0.99	1.0	0.99	0.98	0.99	0.9808	0.9423	0.96
WID11_CID107, default	0.99	0.98	0.99	0.98	1.0	0.98	0.97	0.98	0.99	1.0	1.0	0.99	0.98	0.99	0.9677	0.9677	0.96
WID11_CID107, high	1.0	0.98	1.0	1.0	1.0	0.99	1.0	1.0	0.98	1.0	0.99	0.99	0.98	1.0	0.9701	0.9851	0.97
WID11_CID107, low	1.0	0.99	1.0	1.0	0.98	0.99	0.99	1.0	0.99	1.0	0.96	0.98	0.99	0.99	0.9667	0.9667	0.95
WID11_CID120, default	0.99	0.99	1.0	0.99	0.99	0.99	0.99	1.0	0.99	0.99	0.99	1.0	0.99	0.99	0.95	0.9	0.96
WID11_CID120, high	0.99	0.98	0.98	1.0	1.0	0.98	1.0	1.0	0.98	0.99	0.98	0.99	0.99	0.99	0.9677	0.9839	0.94
WID11_CID120, low	1.0	0.99	1.0	1.0	0.98	1.0	1.0	0.99	0.99	1.0	0.99	0.98	0.99	1.0	0.9655	0.9828	0.95
WID11_CID35, default	1.0	1.0	0.99	1.0	0.99	0.99	1.0	1.0	0.99	1.0	0.99	1.0	0.97	1.0	0.9718	0.9718	0.96
WID11_CID35, high	0.98	1.0	0.98	0.98	0.98	1.0	1.0	0.97	1.0	0.99	1.0	0.99	1.0	1.0	0.9677	0.9516	0.96
WID11_CID35, low	0.99	0.98	0.99	0.99	1.0	0.99	1.0	1.0	0.99	0.99	0.99	0.99	0.99	1.0	0.9636	0.9636	0.95
WID11_CID60, default	0.99	0.99	0.99	0.99	0.98	0.98	0.99	0.99	1.0	0.99	1.0	0.99	1.0	1.0	0.9643	0.9821	0.96
WID11_CID60, high	0.99	0.99	0.99	0.99	1.0	0.99	0.98	0.98	1.0	1.0	0.99	0.99	1.0	1.0	0.9815	0.9815	0.95
WID11_CID60, low	0.97	0.99	0.98	0.98	0.99	1.0	0.99	1.0	0.97	0.98	0.98	1.0	0.98	1.0	0.9623	0.9623	0.95
WID11_CID97, default	1.0	0.97	1.0	1.0	0.99	1.0	0.99	0.97	0.97	0.97	1.0	1.0	1.0	0.98	0.9839	0.9839	0.96
WID11_CID97, high	0.99	0.99	0.99	1.0	1.0	0.99	0.99	0.99	0.99	0.98	0.97	0.99	1.0	1.0	0.9857	0.9571	0.94
WID11_CID97, low	0.97	0.99	0.98	0.97	0.98	0.99	1.0	1.0	1.0	0.97	1.0	0.98	1.0	0.98	0.9464	0.9643	0.95
WID11_CID98, default	0.98	1.0	0.99	0.99	0.99	0.98	0.97	0.99	0.99	0.99	1.0	0.99	1.0	0.99	0.9655	0.9655	0.94
WID11_CID98, high	0.99	1.0	0.98	0.99	0.99	0.98	0.98	0.99	1.0	0.96	1.0	1.0	0.99	1.0	0.9194	0.9839	0.96
WID11_CID98, low	1.0	1.0	0.99	1.0	0.98	0.99	1.0	0.99	0.98	1.0	0.99	1.0	1.0	0.98	0.9242	0.9697	0.95
WID11_CID99, default	0.99	0.98	1.0	0.98	0.98	0.99	0.98	0.98	1.0	0.99	0.98	0.97	0.99	1.0	0.9677	0.9677	0.96
WID11_CID99, high	0.97	1.0	0.99	0.97	0.9899	1.0	0.99	0.99	1.0	1.0	0.98	1.0	0.98	1.0	0.9848	0.9697	0.94
WID11_CID99, low	0.97	1.0	0.97	0.97	1.0	0.97	1.0	1.0	0.99	1.0	0.99	0.99	0.99	1.0	0.9844	0.9844	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 27: TRNG0 -40 °C WID11 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -25.0 (°C), TRNG0, WID11
Fail ratio 1.69%, ~ 1.84% expected

WID11_CID101, default	0.99	1.0	0.99	0.99	0.97	1.0	0.96	0.98	0.99	0.98	0.97	0.97	1.0	0.97	0.9643	0.9821	0.95
WID11_CID101, high	1.0	1.0	0.99	1.0	0.99	1.0	0.99	1.0	1.0	1.0	1.0	0.99	0.99	1.0	0.9583	0.9583	0.96
WID11_CID101, low	0.99	1.0	0.99	0.99	0.99	0.97	0.99	0.99	1.0	0.99	0.98	0.98	0.99	1.0	0.9828	0.9828	0.96
WID11_CID103, default	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.97	1.0	1.0	0.98	0.99	0.99	0.98	0.9836	0.9672	0.95
WID11_CID103, high	0.99	0.97	1.0	0.99	0.98	0.99	0.99	0.96	0.96	0.98	1.0	0.99	0.99	1.0	0.9655	0.9655	0.95
WID11_CID103, low	0.99	1.0	0.99	0.99	0.99	0.98	1.0	0.99	0.99	0.97	1.0	1.0	0.99	0.98	0.9833	0.9667	0.95
WID11_CID104, default	0.98	0.98	0.98	0.99	0.99	0.97	0.99	0.99	0.98	1.0	0.98	1.0	1.0	0.99	0.9836	0.9836	0.96
WID11_CID104, high	0.98	0.99	1.0	0.98	0.99	0.97	0.99	1.0	0.97	1.0	0.98	0.99	1.0	0.99	0.9524	0.9524	0.97
WID11_CID104, low	0.99	0.98	0.99	0.99	0.98	0.97	0.98	0.98	1.0	1.0	0.99	1.0	1.0	0.98	0.9508	0.9836	0.96
WID11_CID107, default	0.99	1.0	1.0	0.98	0.98	1.0	0.99	1.0	0.99	0.98	0.99	1.0	1.0	1.0	0.9655	0.9655	0.95
WID11_CID107, high	0.99	0.99	0.98	0.99	1.0	1.0	0.99	0.97	0.98	0.96	1.0	0.99	0.99	0.98	0.9825	0.9649	0.95
WID11_CID107, low	1.0	0.99	1.0	1.0	0.98	0.98	0.99	0.97	0.99	0.96	1.0	0.97	0.99	1.0	0.9531	0.9688	0.96
WID11_CID120, default	1.0	1.0	0.99	1.0	0.99	0.99	1.0	1.0	0.98	1.0	0.99	0.99	0.99	0.98	0.9848	0.9697	0.95
WID11_CID120, high	0.98	0.99	0.99	0.98	1.0	0.99	0.99	1.0	0.99	0.99	0.98	0.99	1.0	1.0	1.0	0.9683	0.96
WID11_CID120, low	0.98	0.98	0.98	0.98	0.99	1.0	0.97	0.99	0.99	1.0	0.99	0.99	1.0	0.99	0.9677	0.9677	0.95
WID11_CID35, default	1.0	0.98	1.0	1.0	1.0	0.98	0.98	0.98	0.98	1.0	1.0	0.97	0.98	0.99	0.9833	0.95	0.96
WID11_CID35, high	0.97	0.97	0.97	0.97	1.0	0.97	1.0	0.99	0.98	0.98	0.98	0.99	1.0	0.98	0.9434	0.9623	0.97
WID11_CID35, low	1.0	0.98	1.0	1.0	0.98	0.99	0.99	1.0	0.99	0.98	0.98	1.0	1.0	0.99	0.9701	0.9851	0.97
WID11_CID60, default	0.98	0.98	0.97	0.98	1.0	0.98	0.98	1.0	0.99	1.0	0.98	0.98	0.99	0.96	0.9661	0.9831	0.95
WID11_CID60, high	0.97	0.99	0.98	0.98	1.0	0.98	1.0	0.99	0.98	1.0	1.0	1.0	1.0	0.98	0.9697	0.9848	0.95
WID11_CID60, low	0.98	0.99	0.98	0.99	0.98	1.0	0.99	0.99	0.99	1.0	1.0	0.99	0.98	1.0	0.9672	1.0	0.95
WID11_CID97, default	0.98	1.0	0.99	0.98	1.0	1.0	0.97	0.99	0.99	0.99	0.99	0.98	0.98	0.99	0.9841	0.9683	0.96
WID11_CID97, high	1.0	1.0	1.0	1.0	0.99	0.98	0.99	0.98	0.97	0.98	0.98	0.99	0.99	1.0	0.9844	0.9688	0.95
WID11_CID97, low	0.99	1.0	0.99	0.99	0.99	1.0	1.0	0.98	0.99	0.98	1.0	0.98	0.97	0.98	0.9855	0.971	0.96
WID11_CID98, default	1.0	1.0	1.0	0.99	0.99	0.98	0.99	0.99	0.99	0.99	0.99	0.98	0.95	0.99	0.9846	0.9692	0.96
WID11_CID98, high	1.0	0.98	1.0	1.0	0.99	0.98	1.0	0.99	0.97	0.99	0.97	1.0	0.99	0.98	0.95	0.95	0.96
WID11_CID98, low	1.0	0.97	1.0	1.0	0.97	0.98	0.99	0.98	0.98	0.96	1.0	0.98	0.97	0.99	0.9841	0.9683	0.95
WID11_CID99, default	1.0	0.98	1.0	1.0	0.98	0.98	0.99	1.0	0.98	1.0	0.99	1.0	0.98	0.97	0.9853	0.9853	0.95
WID11_CID99, high	0.98	1.0	0.97	0.98	0.99	1.0	1.0	0.99	1.0	0.98	1.0	0.99	1.0	0.99	0.9531	0.9844	0.95
WID11_CID99, low	0.97	0.98	0.97	0.97	0.99	1.0	1.0	1.0	0.99	0.99	0.99	0.99	0.98	0.99	0.9683	0.9683	0.94
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 28: TRNG0 -25 °C WID11 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 25.0 (°C), TRNG0, WID11
Fail ratio 1.91%, ~ 1.84% expected

WID11_CID101, default	0.99	0.99	0.99	0.98	0.98	1.0	0.96	0.97	0.97	1.0	1.0	1.0	0.98	0.99	0.9531	0.9688	0.95
WID11_CID101, high	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.96	0.98	0.98	0.98	1.0	1.0	0.97	0.9559	0.9853	0.95
WID11_CID101, low	1.0	0.98	1.0	1.0	0.98	0.98	1.0	1.0	0.99	0.99	1.0	1.0	0.99	1.0	0.9643	0.9643	0.95
WID11_CID103, default	1.0	0.99	0.99	1.0	0.99	0.98	0.99	0.99	0.99	1.0	0.99	1.0	0.99	0.99	0.9848	0.9697	0.96
WID11_CID103, high	1.0	1.0	0.98	1.0	0.98	1.0	0.99	0.99	0.99	0.98	0.98	1.0	0.99	0.98	0.9857	1.0	0.94
WID11_CID103, low	0.99	0.98	0.99	0.98	1.0	0.97	0.99	0.98	0.96	0.99	0.99	0.99	0.99	1.0	0.9844	0.9688	0.95
WID11_CID104, default	1.0	1.0	1.0	1.0	0.99	0.99	0.97	0.96	0.99	0.99	0.99	0.99	1.0	1.0	0.9718	0.9718	0.96
WID11_CID104, high	1.0	1.0	1.0	1.0	0.98	1.0	0.98	0.99	0.99	1.0	0.99	1.0	0.99	1.0	0.9688	0.9688	0.97
WID11_CID104, low	1.0	1.0	1.0	1.0	0.99	0.99	0.99	1.0	0.99	0.98	0.98	0.96	0.98	1.0	0.9531	0.9688	0.96
WID11_CID107, default	1.0	0.98	0.99	0.99	1.0	0.98	1.0	1.0	1.0	1.0	1.0	0.99	0.99	0.98	0.9683	0.9841	0.95
WID11_CID107, high	0.98	1.0	0.98	0.96	1.0	0.99	0.98	0.99	0.97	0.98	0.98	0.98	1.0	1.0	0.9825	0.9825	0.95
WID11_CID107, low	0.99	0.99	0.99	0.99	0.98	0.97	0.99	0.99	0.99	1.0	0.99	1.0	1.0	0.98	0.9615	0.9808	0.96
WID11_CID120, default	0.99	0.99	1.0	0.99	1.0	0.99	0.97	0.95	0.99	0.98	1.0	0.99	0.96	0.99	0.9677	0.9516	0.95
WID11_CID120, high	1.0	1.0	1.0	1.0	0.97	0.99	0.95	0.97	0.98	0.98	0.96	0.98	0.99	0.99	0.9833	0.9833	0.96
WID11_CID120, low	1.0	1.0	1.0	0.99	1.0	0.98	0.96	0.98	0.99	0.99	0.99	0.99	1.0	0.99	0.9778	0.9778	0.96
WID11_CID35, default	0.97	0.99	0.97	0.98	0.98	1.0	0.99	0.99	0.99	0.99	1.0	0.98	0.99	0.98	0.9839	0.9677	0.95
WID11_CID35, high	0.99	0.98	0.99	0.99	1.0	0.99	0.99	0.98	0.99	0.98	0.99	1.0	0.95	0.98	0.9831	0.9831	0.96
WID11_CID35, low	0.99	0.98	0.99	0.99	0.99	1.0	0.99	0.99	0.99	0.98	1.0	0.99	0.99	0.97	0.9661	0.9322	0.96
WID11_CID60, default	1.0	0.99	0.99	1.0	0.98	0.99	0.99	0.98	1.0	1.0	0.98	0.99	1.0	1.0	0.9661	0.9661	0.96
WID11_CID60, high	0.99	1.0	0.99	0.99	1.0	0.98	0.98	1.0	1.0	0.99	0.97	0.98	1.0	0.99	0.9531	0.9531	0.96
WID11_CID60, low	0.99	1.0	1.0	1.0	1.0	0.98	0.98	0.98	0.97	0.99	1.0	0.99	0.98	1.0	0.9841	0.9683	0.95
WID11_CID97, default	0.98	0.99	0.98	0.99	0.99	0.99	0.99	1.0	0.98	0.99	0.99	1.0	1.0	0.98	0.9808	0.9808	0.96
WID11_CID97, high	0.99	1.0	0.99	0.99	0.99	0.99	1.0	0.99	0.99	0.99	0.98	0.99	0.99	1.0	0.9655	0.9828	0.96
WID11_CID97, low	1.0	0.99	1.0	1.0	1.0	0.99	1.0	1.0	0.97	0.97	0.99	0.99	0.98	1.0	0.9394	0.9848	0.96
WID11_CID98, default	1.0	0.98	0.99	1.0	0.98	0.98	0.99	0.99	0.99	1.0	0.98	0.98	0.99	1.0	0.9492	0.9492	0.96
WID11_CID98, high	0.97	0.98	0.97	0.99	0.99	1.0	0.99	0.98	1.0	1.0	1.0	0.99	0.99	1.0	0.9492	0.9831	0.95
WID11_CID98, low	0.97	0.99	0.99	0.99	0.99	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.98	0.97	0.9833	0.9667	0.95
WID11_CID99, default	0.99	0.98	0.99	0.99	1.0	0.98	1.0	0.98	1.0	0.97	1.0	0.98	1.0	0.99	0.9831	0.9661	0.95
WID11_CID99, high	0.99	1.0	0.99	0.97	0.99	0.98	1.0	0.99	0.99	0.98	0.99	0.99	1.0	1.0	0.9608	0.9608	0.95
WID11_CID99, low	0.98	1.0	0.98	0.97	0.98	0.99	1.0	1.0	1.0	1.0	0.97	1.0	1.0	1.0	0.971	0.9855	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 29: TRNG0 25 °C WID11 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 105.0 (°C), TRNG0, WID11
Fail ratio 2.02%, ~ 1.84% expected

WID11_CID101, default	0.99	1.0	0.99	0.99	0.99	1.0	0.98	0.98	1.0	1.0	0.99	0.98	1.0	0.97	0.9683	0.9524	0.95
WID11_CID101, high	1.0	0.99	1.0	0.98	1.0	0.98	0.97	0.98	0.96	1.0	1.0	1.0	0.99	0.99	0.9697	0.9848	0.95
WID11_CID101, low	0.99	0.98	0.99	0.99	0.99	1.0	1.0	0.99	1.0	0.98	0.97	0.98	1.0	1.0	0.9538	0.9692	0.95
WID11_CID103, default	0.99	0.99	0.99	0.98	0.98	0.99	1.0	0.96	1.0	0.99	0.99	0.96	0.97	1.0	0.9828	0.9828	0.96
WID11_CID103, high	1.0	0.98	1.0	0.99	0.99	1.0	1.0	0.99	0.98	0.99	0.99	1.0	1.0	1.0	0.9815	0.963	0.95
WID11_CID103, low	1.0	1.0	1.0	1.0	0.99	1.0	0.99	0.98	0.98	0.99	1.0	0.98	0.98	0.99	0.9851	0.9552	0.95
WID11_CID104, default	1.0	0.99	1.0	1.0	0.98	0.99	0.98	1.0	0.99	0.99	1.0	0.99	0.98	1.0	0.9839	0.9677	0.96
WID11_CID104, high	0.95	1.0	0.96	0.96	0.98	1.0	1.0	0.98	1.0	0.98	1.0	0.99	1.0	0.99	0.9714	0.9857	0.96
WID11_CID104, low	0.99	0.99	0.99	0.99	0.99	0.99	1.0	1.0	1.0	0.98	1.0	1.0	0.99	0.99	0.9677	0.9516	0.96
WID11_CID107, default	1.0	1.0	1.0	1.0	0.98	0.99	0.96	1.0	0.97	1.0	0.98	0.99	1.0	0.97	0.9672	0.9508	0.95
WID11_CID107, high	1.0	0.99	1.0	1.0	0.98	1.0	0.98	0.99	0.98	0.98	0.99	0.97	0.99	1.0	0.9661	0.9831	0.95
WID11_CID107, low	1.0	0.98	1.0	0.98	0.99	0.98	0.98	0.98	1.0	0.96	0.98	0.98	0.99	1.0	0.9831	0.9831	0.96
WID11_CID120, default	0.99	0.98	0.98	1.0	0.96	0.98	0.99	0.99	0.99	0.99	0.99	1.0	1.0	1.0	0.9718	0.9577	0.96
WID11_CID120, high	1.0	1.0	0.99	1.0	0.99	1.0	0.99	1.0	0.99	0.98	0.98	0.99	0.99	0.99	0.971	0.9565	0.95
WID11_CID120, low	1.0	1.0	1.0	1.0	1.0	0.99	0.99	1.0	1.0	0.98	1.0	0.99	1.0	0.98	0.9516	0.9677	0.95
WID11_CID35, default	1.0	1.0	1.0	1.0	0.99	1.0	0.99	0.98	0.99	1.0	1.0	0.99	0.97	0.99	0.9821	0.9643	0.96
WID11_CID35, high	0.99	0.99	0.99	0.99	0.99	1.0	0.98	1.0	0.98	1.0	1.0	0.99	0.98	0.99	0.9833	0.9667	0.96
WID11_CID35, low	1.0	0.99	0.98	1.0	1.0	0.98	1.0	1.0	0.97	0.99	0.98	1.0	1.0	1.0	0.9661	0.9661	0.96
WID11_CID60, default	0.99	1.0	0.99	1.0	0.99	1.0	1.0	1.0	1.0	0.98	1.0	0.99	1.0	1.0	0.9545	0.9394	0.95
WID11_CID60, high	0.99	0.97	0.99	0.99	1.0	1.0	0.98	0.99	0.98	0.98	0.97	0.98	0.98	0.98	0.9672	0.9344	0.96
WID11_CID60, low	0.98	0.99	0.97	0.96	1.0	1.0	0.98	1.0	1.0	0.98	0.99	1.0	1.0	0.97	0.9655	0.9828	0.96
WID11_CID97, default	1.0	0.99	1.0	1.0	0.99	1.0	1.0	0.98	0.99	0.98	0.99	0.99	1.0	1.0	0.9697	0.9394	0.95
WID11_CID97, high	1.0	1.0	1.0	1.0	0.98	0.99	0.99	0.95	1.0	0.98	0.99	0.99	0.99	0.97	0.9508	0.9672	0.93
WID11_CID97, low	1.0	0.98	0.99	1.0	1.0	0.99	0.99	0.98	0.98	0.99	1.0	1.0	1.0	0.97	0.9683	0.9683	0.96
WID11_CID98, default	0.99	0.98	0.99	0.97	1.0	0.97	1.0	0.99	1.0	0.99	1.0	1.0	1.0	0.98	0.9636	0.9818	0.96
WID11_CID98, high	1.0	0.98	1.0	1.0	0.99	0.97	1.0	0.99	0.99	0.99	1.0	0.99	0.99	0.99	0.9857	0.9714	0.96
WID11_CID98, low	0.98	0.99	0.99	0.99	1.0	0.99	1.0	0.99	0.99	0.98	1.0	0.99	0.98	0.98	0.9821	0.9643	0.95
WID11_CID99, default	0.99	0.97	0.99	1.0	1.0	1.0	0.99	0.99	0.99	0.99	0.99	0.99	0.97	0.98	0.9697	0.9848	0.95
WID11_CID99, high	0.98	0.99	1.0	0.99	0.99	0.98	1.0	0.95	1.0	1.0	1.0	1.0	1.0	1.0	0.9649	0.9825	0.96
WID11_CID99, low	0.99	1.0	0.99	0.99	1.0	0.99	0.98	0.99	0.99	0.98	0.99	1.0	1.0	1.0	0.9706	0.9559	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 30: TRNG0 105 °C WID11 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 125.0 (°C), TRNG0, WID11
Fail ratio 1.77%, ~ 1.84% expected

WID11_CID101, default	0.99	0.97	0.99	0.98	0.99	1.0	0.99	0.97	1.0	0.97	1.0	0.97	0.98	0.98	0.9692	0.9846	0.95
WID11_CID101, high	0.97	1.0	0.99	0.97	0.99	0.95	0.99	0.98	0.98	1.0	0.98	1.0	1.0	0.99	0.9821	1.0	0.96
WID11_CID101, low	1.0	0.98	0.99	1.0	0.98	0.99	0.99	1.0	1.0	1.0	0.98	0.97	0.97	1.0	0.9655	0.9655	0.95
WID11_CID103, default	0.99	0.98	0.99	0.99	1.0	1.0	0.96	1.0	1.0	0.99	0.99	0.98	0.98	1.0	0.9672	0.9672	0.95
WID11_CID103, high	0.99	1.0	0.99	0.99	0.99	1.0	0.98	0.97	0.97	1.0	1.0	0.98	1.0	0.99	0.9841	0.9683	0.95
WID11_CID103, low	0.99	0.99	1.0	0.99	0.97	0.96	0.99	0.97	0.98	0.98	0.98	0.99	1.0	0.98	0.9831	0.9831	0.96
WID11_CID104, default	1.0	1.0	1.0	1.0	0.99	1.0	0.98	0.99	1.0	1.0	1.0	0.98	0.98	1.0	0.9667	0.9833	0.95
WID11_CID104, high	0.99	1.0	0.99	1.0	1.0	0.98	0.99	1.0	0.97	1.0	1.0	0.99	0.98	0.99	0.9839	0.9677	0.96
WID11_CID104, low	0.98	0.99	0.98	0.98	1.0	0.99	0.99	0.97	1.0	0.99	1.0	0.98	0.99	1.0	0.9636	0.9636	0.95
WID11_CID107, default	0.99	0.98	0.99	0.99	0.99	0.97	0.98	0.98	0.99	0.98	1.0	0.98	0.99	1.0	0.9828	0.9655	0.95
WID11_CID107, high	0.99	1.0	1.0	0.98	0.99	1.0	0.97	0.99	0.95	0.99	1.0	1.0	0.98	0.99	0.9206	0.9683	0.95
WID11_CID107, low	1.0	1.0	1.0	1.0	1.0	1.0	0.98	0.98	0.98	0.99	1.0	1.0	1.0	1.0	0.9851	0.9851	0.96
WID11_CID120, default	1.0	0.99	1.0	0.99	1.0	0.99	1.0	0.99	1.0	0.97	1.0	1.0	1.0	1.0	0.9655	0.931	0.96
WID11_CID120, high	0.97	0.98	0.99	0.99	0.98	1.0	0.99	0.97	1.0	0.97	0.99	0.99	1.0	0.98	0.9808	0.9808	0.95
WID11_CID120, low	1.0	1.0	1.0	1.0	0.99	0.98	1.0	1.0	0.98	0.97	0.99	1.0	1.0	0.98	0.9697	0.9545	0.96
WID11_CID35, default	0.98	1.0	0.97	0.98	1.0	0.99	1.0	0.97	1.0	1.0	0.95	0.99	1.0	1.0	0.9531	0.9688	0.96
WID11_CID35, high	0.99	0.99	0.98	0.98	1.0	1.0	0.99	0.98	1.0	0.99	0.99	0.99	0.99	0.98	0.9808	0.9808	0.96
WID11_CID35, low	0.99	0.99	0.99	0.99	1.0	0.98	1.0	1.0	0.98	1.0	1.0	0.99	0.99	0.98	0.9841	0.9683	0.96
WID11_CID60, default	0.99	0.98	0.99	0.99	0.99	0.99	0.99	0.98	0.98	0.98	1.0	0.99	0.99	0.99	0.9688	0.9688	0.96
WID11_CID60, high	0.99	0.98	0.97	0.99	0.99	0.97	1.0	0.99	0.98	0.99	0.99	0.98	1.0	0.99	0.9649	0.9825	0.96
WID11_CID60, low	0.98	0.98	0.98	0.98	1.0	0.99	0.98	0.98	0.99	0.97	0.97	0.96	0.97	0.98	0.9636	0.9636	0.97
WID11_CID97, default	0.99	1.0	0.99	0.99	0.99	1.0	1.0	0.98	0.99	1.0	0.98	1.0	0.99	0.98	0.9655	0.9828	0.96
WID11_CID97, high	1.0	1.0	1.0	1.0	1.0	0.98	1.0	0.97	0.99	0.98	0.99	0.99	0.97	0.97	0.9483	0.9483	0.96
WID11_CID97, low	1.0	1.0	1.0	1.0	0.99	0.98	1.0	0.98	0.99	1.0	1.0	0.98	0.99	1.0	0.9848	0.9848	0.96
WID11_CID98, default	1.0	0.99	1.0	1.0	0.96	1.0	0.99	0.99	1.0	0.99	0.99	1.0	0.99	0.99	0.9701	0.9701	0.96
WID11_CID98, high	0.99	0.99	1.0	1.0	0.98	1.0	0.98	0.97	1.0	0.99	0.98	0.99	0.99	0.98	0.9516	0.9839	0.96
WID11_CID98, low	0.99	0.99	0.99	0.99	0.97	0.99	0.99	0.98	1.0	0.98	0.99	0.98	0.98	0.99	0.9828	0.9655	0.96
WID11_CID99, default	0.99	0.99	0.99	0.99	0.99	0.98	1.0	0.95	1.0	0.98	0.98	1.0	0.99	0.99	0.9492	0.9661	0.95
WID11_CID99, high	0.98	1.0	0.98	0.97	1.0	1.0	1.0	0.99	0.99	0.96	1.0	1.0	1.0	0.99	0.9667	0.9667	0.97
WID11_CID99, low	0.99	0.98	1.0	1.0	1.0	0.98	1.0	1.0	1.0	0.99	0.99	0.99	0.99	1.0	0.9692	0.9692	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 31: TRNG0 125 °C WID11 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -40.0 (°C), TRNG1, WID11
Fail ratio 1.87%, ~ 1.84% expected

WID11_CID101, default	0.98	0.97	0.98	0.98	0.99	0.97	1.0	0.99	0.99	0.98	0.99	0.98	0.98	0.99	0.9841	0.9683	0.95
WID11_CID101, high	0.98	1.0	0.99	0.99	0.99	0.99	0.99	1.0	1.0	0.96	0.99	0.98	1.0	1.0	0.9692	1.0	0.96
WID11_CID101, low	1.0	1.0	0.99	1.0	0.99	0.98	1.0	1.0	0.99	0.99	0.98	0.99	0.99	0.99	0.9848	0.9697	0.96
WID11_CID103, default	1.0	0.98	1.0	0.99	0.98	0.99	0.99	0.96	0.99	0.97	1.0	1.0	1.0	0.99	0.9831	0.9322	0.96
WID11_CID103, high	0.99	0.99	0.99	1.0	1.0	0.97	0.99	0.99	1.0	1.0	1.0	1.0	0.98	1.0	0.9655	0.9828	0.96
WID11_CID103, low	0.98	1.0	0.98	0.98	0.99	1.0	1.0	0.99	0.98	0.98	1.0	1.0	1.0	0.99	0.9846	0.9846	0.96
WID11_CID104, default	1.0	0.99	1.0	1.0	1.0	0.99	0.99	0.97	1.0	1.0	0.99	0.98	0.98	1.0	0.9697	0.9697	0.96
WID11_CID104, high	0.99	0.98	0.99	0.99	0.99	0.99	0.96	0.99	1.0	0.99	0.97	0.99	0.96	1.0	0.9423	0.9615	0.94
WID11_CID104, low	1.0	1.0	1.0	0.99	0.98	0.99	0.99	0.98	0.99	1.0	0.99	1.0	1.0	0.98	0.9672	0.9836	0.96
WID11_CID107, default	0.99	0.99	0.99	1.0	0.99	1.0	0.98	0.99	0.99	0.99	0.98	0.96	0.99	0.99	0.9571	0.9857	0.95
WID11_CID107, high	1.0	0.99	1.0	1.0	0.98	0.98	0.98	1.0	0.97	1.0	1.0	0.98	0.98	0.99	0.9474	0.9825	0.95
WID11_CID107, low	0.98	0.99	0.99	0.98	1.0	0.99	0.99	0.98	0.98	1.0	0.98	1.0	1.0	1.0	0.9661	0.9492	0.96
WID11_CID120, default	0.98	1.0	0.97	0.98	0.98	0.99	0.99	0.99	0.98	0.99	0.99	0.99	1.0	0.97	0.9516	0.9677	0.94
WID11_CID120, high	0.99	0.98	1.0	0.99	0.99	0.99	1.0	0.98	0.98	0.98	0.98	1.0	0.99	1.0	0.9375	0.9688	0.97
WID11_CID120, low	1.0	0.98	1.0	1.0	1.0	0.99	0.97	1.0	0.97	0.98	1.0	0.98	1.0	1.0	0.9661	0.9831	0.96
WID11_CID35, default	0.98	0.99	0.99	0.98	0.99	0.99	0.99	0.99	1.0	0.99	0.99	0.99	1.0	0.99	0.9474	0.9298	0.96
WID11_CID35, high	0.99	1.0	1.0	0.99	0.98	0.97	0.99	1.0	1.0	0.99	1.0	0.99	0.98	0.98	0.9692	0.9692	0.96
WID11_CID35, low	1.0	1.0	0.99	1.0	0.99	1.0	0.99	1.0	1.0	1.0	1.0	0.98	0.98	0.99	0.9692	0.9538	0.95
WID11_CID60, default	0.99	0.99	0.99	0.98	0.98	0.96	0.99	0.98	1.0	0.97	1.0	0.99	0.99	1.0	0.9841	0.9683	0.96
WID11_CID60, high	1.0	1.0	0.99	1.0	0.96	0.99	0.98	0.99	0.99	0.99	1.0	1.0	1.0	0.99	0.9677	0.9677	0.96
WID11_CID60, low	0.99	0.99	0.99	0.98	0.98	1.0	0.99	1.0	0.98	0.97	0.97	0.99	0.98	0.99	0.9839	0.9677	0.95
WID11_CID97, default	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.98	1.0	0.99	1.0	1.0	1.0	0.97	0.9677	0.9677	0.95
WID11_CID97, high	0.98	0.99	0.99	0.99	0.99	1.0	1.0	0.97	0.99	0.99	0.99	0.99	0.99	0.99	0.9545	0.9545	0.95
WID11_CID97, low	0.99	1.0	1.0	0.98	0.98	0.99	0.99	0.98	0.97	0.98	0.99	0.98	0.99	0.98	0.9811	0.9623	0.95
WID11_CID98, default	1.0	0.99	1.0	1.0	0.99	0.97	0.99	0.98	1.0	0.99	0.99	0.97	1.0	0.98	0.9841	0.9841	0.95
WID11_CID98, high	1.0	1.0	1.0	1.0	0.99	0.99	0.98	0.97	0.99	1.0	1.0	1.0	1.0	0.99	0.963	0.963	0.95
WID11_CID98, low	0.96	1.0	0.97	0.96	0.97	0.97	1.0	1.0	1.0	0.97	0.99	1.0	0.99	0.99	0.9818	0.9636	0.96
WID11_CID99, default	0.97	0.99	0.95	0.97	1.0	0.99	0.99	1.0	1.0	1.0	1.0	0.99	1.0	0.99	0.9841	0.9841	0.94
WID11_CID99, high	0.99	1.0	0.99	0.99	1.0	1.0	0.99	0.99	0.97	1.0	1.0	0.99	0.98	1.0	0.9362	0.9574	0.96
WID11_CID99, low	1.0	0.99	1.0	1.0	0.97	0.99	0.99	0.98	0.98	0.99	0.99	0.99	0.99	0.95	0.9545	0.9697	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 32: TRNG1 -40 °C WID11 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for -25.0 (°C), TRNG1, WID11
Fail ratio 1.93%, ~ 1.84% expected

WID11_CID101, default	1.0	0.99	1.0	0.99	0.97	1.0	0.99	1.0	0.98	0.99	1.0	0.99	1.0	0.98	0.9722	0.9583	0.95
WID11_CID101, high	0.99	1.0	1.0	0.99	1.0	0.95	1.0	0.97	0.99	0.99	0.99	0.99	0.99	0.99	0.9833	0.9833	0.96
WID11_CID101, low	1.0	0.99	1.0	1.0	1.0	1.0	1.0	0.98	0.99	1.0	0.99	1.0	1.0	0.99	0.9839	0.9516	0.96
WID11_CID103, default	0.99	0.99	0.99	0.99	0.99	0.98	0.97	1.0	0.97	0.98	0.99	0.99	1.0	1.0	0.963	0.963	0.96
WID11_CID103, high	0.99	0.99	0.99	1.0	0.97	0.99	0.99	1.0	1.0	1.0	1.0	1.0	0.99	1.0	0.9697	0.9697	0.96
WID11_CID103, low	1.0	0.99	0.99	1.0	0.99	1.0	0.99	0.99	0.99	1.0	0.99	0.99	0.98	1.0	0.9815	0.963	0.95
WID11_CID104, default	0.97	0.99	0.97	0.97	1.0	1.0	0.97	0.98	0.98	0.99	1.0	0.98	0.98	0.99	0.9804	0.9608	0.95
WID11_CID104, high	0.99	1.0	0.98	0.99	0.99	0.99	0.99	1.0	0.98	0.97	0.98	0.98	0.99	0.98	0.9655	0.9655	0.95
WID11_CID104, low	1.0	0.99	1.0	1.0	0.97	0.98	0.99	1.0	0.99	1.0	1.0	1.0	1.0	0.99	0.9552	0.9701	0.96
WID11_CID107, default	0.99	0.99	0.98	0.99	0.99	1.0	1.0	0.99	1.0	0.98	1.0	1.0	0.99	0.99	0.9667	0.9833	0.95
WID11_CID107, high	0.98	0.98	0.98	0.97	0.98	0.99	1.0	0.99	1.0	0.98	0.99	0.99	1.0	1.0	0.9649	0.9825	0.97
WID11_CID107, low	0.99	1.0	0.99	0.99	1.0	1.0	0.98	0.98	1.0	1.0	0.98	0.99	1.0	0.99	0.9688	0.9688	0.94
WID11_CID120, default	1.0	0.99	1.0	1.0	0.98	1.0	0.99	0.98	1.0	0.98	0.99	1.0	1.0	0.99	0.9692	0.9846	0.94
WID11_CID120, high	0.99	1.0	0.99	0.99	1.0	0.97	0.98	0.98	0.97	1.0	0.99	0.99	1.0	0.98	0.9697	0.9697	0.97
WID11_CID120, low	0.99	0.98	0.99	0.98	0.99	1.0	0.99	1.0	0.98	0.98	1.0	1.0	0.98	0.98	0.9677	0.9839	0.95
WID11_CID35, default	0.98	0.99	0.98	0.97	0.99	1.0	0.99	0.99	1.0	0.99	1.0	0.99	1.0	0.99	0.9701	0.9851	0.95
WID11_CID35, high	0.98	0.99	0.99	0.98	1.0	1.0	0.98	1.0	0.98	0.99	0.99	1.0	0.99	0.98	0.9375	0.9688	0.96
WID11_CID35, low	1.0	0.99	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.98	1.0	1.0	1.0	1.0	0.9516	0.9839	0.95
WID11_CID60, default	0.99	0.97	0.99	0.98	1.0	1.0	0.98	1.0	1.0	1.0	1.0	0.98	0.99	1.0	0.9672	0.9672	0.96
WID11_CID60, high	0.99	1.0	0.99	0.99	0.96	0.99	0.99	0.99	0.99	0.99	0.98	1.0	1.0	0.99	0.9672	0.9836	0.95
WID11_CID60, low	1.0	0.98	1.0	1.0	0.97	0.98	1.0	1.0	0.99	0.98	1.0	1.0	0.99	0.99	0.9831	0.9831	0.95
WID11_CID97, default	0.99	1.0	0.99	0.98	0.99	1.0	1.0	0.99	0.98	0.98	0.99	0.97	0.98	0.99	0.95	0.9667	0.96
WID11_CID97, high	0.99	1.0	1.0	0.99	1.0	0.99	0.98	0.99	0.99	0.98	1.0	0.99	0.99	0.99	0.973	0.973	0.96
WID11_CID97, low	1.0	0.98	0.99	1.0	1.0	0.99	0.98	0.99	1.0	0.99	1.0	0.96	1.0	0.99	0.9672	0.9836	0.96
WID11_CID98, default	0.99	0.99	0.99	0.99	0.97	0.99	0.98	0.99	1.0	0.99	0.99	0.99	0.99	1.0	0.9667	0.95	0.96
WID11_CID98, high	0.99	1.0	1.0	1.0	1.0	0.99	1.0	0.96	0.99	0.98	0.99	1.0	0.99	0.98	0.9825	0.9649	0.95
WID11_CID98, low	0.99	0.99	0.99	0.99	1.0	0.98	0.98	0.99	0.98	1.0	0.98	1.0	0.99	1.0	0.9474	0.9123	0.96
WID11_CID99, default	0.99	1.0	0.99	0.99	1.0	1.0	0.99	0.97	1.0	1.0	1.0	0.98	0.99	0.97	0.9483	0.9655	0.95
WID11_CID99, high	0.99	0.99	1.0	1.0	1.0	0.97	0.96	1.0	0.99	1.0	0.99	0.99	1.0	0.99	0.9655	0.9655	0.96
WID11_CID99, low	0.97	0.99	0.97	0.97	0.98	0.97	0.99	1.0	0.99	0.98	1.0	1.0	0.99	0.99	0.9672	0.9672	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 33: TRNG1 -25 °C WID11 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 25.0 (°C), TRNG1, WID11
Fail ratio 1.87%, ~ 1.84% expected

WID11_CID101, default	0.98	0.99	0.98	0.98	1.0	0.98	1.0	0.99	0.99	0.98	0.98	0.99	0.98	0.98	0.9655	0.9483	0.96
WID11_CID101, high	0.97	0.99	0.97	0.97	0.99	0.98	1.0	0.98	0.99	1.0	0.99	0.99	0.97	1.0	0.9683	0.9683	0.96
WID11_CID101, low	0.98	0.98	0.98	0.98	0.97	0.99	0.99	0.99	1.0	0.99	0.99	0.99	0.98	0.99	0.9184	0.9796	0.97
WID11_CID103, default	0.99	0.99	0.99	0.99	1.0	1.0	0.98	1.0	0.99	0.98	1.0	0.98	0.98	1.0	0.9825	0.9649	0.95
WID11_CID103, high	0.97	0.99	0.98	0.95	0.99	0.99	0.99	0.99	0.99	0.99	1.0	0.99	0.96	0.99	0.9844	0.9688	0.95
WID11_CID103, low	1.0	1.0	0.99	0.99	0.99	0.99	1.0	0.99	1.0	0.98	0.99	1.0	0.98	1.0	0.9688	0.9844	0.94
WID11_CID104, default	0.98	1.0	0.98	0.98	1.0	0.99	1.0	0.99	0.99	0.97	0.99	0.98	0.98	1.0	0.9643	0.9643	0.96
WID11_CID104, high	0.99	1.0	0.99	0.99	0.99	0.99	0.98	1.0	0.98	0.98	1.0	1.0	1.0	1.0	0.9492	0.9661	0.96
WID11_CID104, low	0.98	0.99	0.98	0.98	1.0	1.0	1.0	1.0	0.98	0.99	0.98	0.99	1.0	0.97	0.9733	0.9333	0.96
WID11_CID107, default	0.99	1.0	0.99	0.99	1.0	0.99	0.99	0.99	0.97	0.98	0.98	0.99	0.98	0.97	0.9559	0.9706	0.94
WID11_CID107, high	0.99	0.98	0.99	0.99	1.0	0.99	0.99	1.0	0.99	0.99	0.98	1.0	0.99	1.0	0.9701	0.9701	0.97
WID11_CID107, low	1.0	0.99	1.0	1.0	0.98	0.99	0.99	0.98	1.0	0.98	0.98	0.99	1.0	0.99	0.9688	0.9688	0.95
WID11_CID120, default	1.0	0.99	1.0	1.0	0.98	1.0	0.97	0.99	0.99	0.99	0.99	1.0	1.0	0.99	0.9844	0.9688	0.94
WID11_CID120, high	0.98	0.98	0.98	0.97	1.0	1.0	1.0	0.99	0.96	0.98	0.96	0.99	1.0	1.0	0.9545	0.9697	0.96
WID11_CID120, low	1.0	0.99	1.0	1.0	0.98	1.0	0.99	0.97	1.0	0.99	0.97	0.99	0.98	0.99	0.9836	0.9836	0.96
WID11_CID35, default	0.99	0.97	0.98	1.0	0.99	1.0	0.98	1.0	0.95	0.97	1.0	0.99	0.99	0.99	0.9692	0.9385	0.96
WID11_CID35, high	0.99	0.97	1.0	1.0	1.0	1.0	1.0	1.0	0.98	0.97	1.0	0.99	0.98	0.97	0.9692	0.9692	0.93
WID11_CID35, low	0.99	1.0	0.99	0.99	1.0	0.99	1.0	0.99	0.95	1.0	0.97	0.99	0.99	0.98	0.9661	0.9661	0.96
WID11_CID60, default	0.99	0.98	0.98	0.99	1.0	0.99	1.0	0.98	0.99	1.0	1.0	1.0	0.98	1.0	0.9672	0.9672	0.96
WID11_CID60, high	1.0	0.98	0.99	1.0	0.99	0.98	1.0	0.98	0.98	0.99	0.99	1.0	1.0	0.98	0.9667	0.9667	0.96
WID11_CID60, low	1.0	0.99	1.0	1.0	1.0	0.99	0.98	1.0	0.99	0.97	0.98	0.97	0.98	0.99	0.9683	0.9841	0.96
WID11_CID97, default	0.97	0.98	0.98	0.97	1.0	0.98	1.0	0.99	1.0	0.99	0.99	0.98	0.99	1.0	0.9718	0.9577	0.97
WID11_CID97, high	1.0	0.99	1.0	1.0	1.0	0.98	1.0	0.99	1.0	1.0	0.99	0.97	0.96	1.0	0.9701	0.9701	0.95
WID11_CID97, low	0.99	0.98	0.99	0.99	0.97	0.97	0.98	0.98	0.99	1.0	0.96	0.98	0.99	1.0	0.9821	0.9464	0.96
WID11_CID98, default	1.0	0.98	1.0	1.0	1.0	1.0	1.0	0.98	0.97	0.99	0.99	1.0	1.0	1.0	0.9836	0.9672	0.96
WID11_CID98, high	0.99	1.0	0.99	0.99	0.99	1.0	1.0	1.0	1.0	1.0	0.99	0.98	0.98	1.0	0.9688	0.9688	0.96
WID11_CID98, low	1.0	0.99	1.0	1.0	1.0	1.0	1.0	0.99	0.98	0.99	1.0	1.0	1.0	0.98	0.95	0.9833	0.93
WID11_CID99, default	1.0	0.99	1.0	1.0	1.0	0.98	0.98	1.0	1.0	0.98	1.0	1.0	0.96	0.99	0.9667	0.9667	0.96
WID11_CID99, high	0.99	0.98	0.99	0.99	1.0	0.99	1.0	1.0	0.99	0.98	0.97	1.0	0.99	1.0	0.9636	0.9636	0.95
WID11_CID99, low	1.0	1.0	1.0	1.0	0.96	0.99	1.0	1.0	0.99	0.99	0.99	1.0	1.0	1.0	1.0	0.9692	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 34: TRNG1 25 °C WID11 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 105.0 (°C), TRNG1, WID11
Fail ratio 1.89%, ~ 1.84% expected

WID11_CID101, default	1.0	0.99	1.0	1.0	0.99	0.99	0.99	0.98	1.0	0.99	0.99	1.0	0.97	1.0	0.9672	0.9672	0.96
WID11_CID101, high	0.98	1.0	0.99	0.98	0.98	0.99	1.0	0.99	0.98	0.97	0.97	0.98	0.97	1.0	0.9839	0.9677	0.95
WID11_CID101, low	1.0	0.99	1.0	1.0	1.0	0.98	0.99	0.98	0.98	1.0	0.95	1.0	1.0	1.0	0.9655	0.9655	0.97
WID11_CID103, default	0.99	1.0	1.0	0.99	1.0	0.99	0.99	0.99	1.0	1.0	1.0	1.0	1.0	0.98	0.9474	0.9474	0.97
WID11_CID103, high	1.0	0.99	1.0	0.99	0.98	0.99	0.97	1.0	0.98	1.0	1.0	1.0	0.98	0.97	0.9636	0.9455	0.96
WID11_CID103, low	1.0	0.99	1.0	1.0	0.98	0.99	0.99	1.0	0.96	0.99	1.0	0.98	1.0	0.98	0.9524	0.9841	0.95
WID11_CID104, default	0.98	0.97	0.98	0.98	1.0	1.0	1.0	1.0	0.96	1.0	0.99	0.97	0.99	0.99	0.9677	0.9516	0.96
WID11_CID104, high	1.0	1.0	1.0	0.99	0.99	1.0	1.0	0.98	1.0	1.0	1.0	0.99	0.99	0.99	0.9531	0.9531	0.96
WID11_CID104, low	1.0	0.98	0.99	0.99	0.97	0.98	0.98	1.0	1.0	0.97	0.99	1.0	0.98	0.99	0.9697	0.9848	0.96
WID11_CID107, default	1.0	1.0	1.0	1.0	0.99	0.97	0.99	0.98	1.0	0.98	1.0	0.99	0.99	0.99	0.9683	0.9841	0.96
WID11_CID107, high	0.98	0.99	0.98	0.99	1.0	0.99	1.0	0.99	0.98	1.0	0.99	0.97	0.99	0.99	0.9661	0.9661	0.96
WID11_CID107, low	1.0	0.98	0.99	1.0	0.98	1.0	1.0	0.98	0.98	0.99	1.0	0.99	1.0	0.99	0.95	0.9833	0.95
WID11_CID120, default	0.99	0.98	0.98	0.99	0.99	1.0	1.0	1.0	0.97	1.0	1.0	1.0	1.0	0.99	0.9655	0.9828	0.95
WID11_CID120, high	1.0	1.0	0.99	0.99	0.97	0.99	1.0	0.99	0.99	1.0	0.99	1.0	0.98	0.97	0.9697	0.9394	0.95
WID11_CID120, low	1.0	0.96	1.0	0.98	0.96	0.99	0.99	0.99	1.0	0.98	0.99	0.98	0.99	0.98	0.9683	0.9683	0.95
WID11_CID35, default	1.0	1.0	1.0	1.0	1.0	1.0	0.99	0.99	0.99	0.99	1.0	0.99	1.0	0.95	0.9661	0.9831	0.96
WID11_CID35, high	0.98	1.0	0.99	0.99	1.0	0.98	0.98	0.98	0.97	0.99	0.99	0.99	1.0	0.99	0.9545	0.9697	0.96
WID11_CID35, low	0.96	1.0	0.97	0.97	1.0	1.0	0.99	0.99	1.0	0.99	0.99	0.99	0.98	1.0	0.9688	0.9531	0.96
WID11_CID60, default	1.0	1.0	0.99	1.0	0.98	1.0	1.0	0.99	0.99	0.99	0.99	1.0	0.98	0.98	0.9545	0.9394	0.96
WID11_CID60, high	0.99	0.97	0.99	0.99	0.97	0.99	0.99	1.0	0.99	1.0	1.0	0.98	0.99	0.98	0.9697	0.9848	0.95
WID11_CID60, low	1.0	1.0	0.99	0.98	0.99	0.99	1.0	1.0	0.97	0.98	0.99	1.0	1.0	0.99	0.9706	0.9853	0.96
WID11_CID97, default	1.0	1.0	1.0	1.0	1.0	1.0	0.97	1.0	0.99	0.99	0.97	0.98	1.0	1.0	0.9649	0.9825	0.95
WID11_CID97, high	0.99	1.0	1.0	1.0	0.99	1.0	0.99	0.95	0.99	1.0	1.0	1.0	0.99	0.97	0.9831	0.9831	0.96
WID11_CID97, low	0.97	1.0	0.97	0.97	0.99	0.98	0.97	0.98	1.0	0.98	0.98	0.99	1.0	1.0	0.9375	0.9688	0.96
WID11_CID98, default	1.0	1.0	1.0	1.0	0.99	0.99	0.99	0.98	0.99	0.99	0.98	0.99	0.99	1.0	0.9683	0.9524	0.95
WID11_CID98, high	0.99	0.97	0.98	0.99	1.0	1.0	0.99	0.99	0.99	0.99	1.0	0.99	0.98	1.0	0.9828	0.9655	0.95
WID11_CID98, low	1.0	0.99	1.0	1.0	0.99	1.0	0.99	0.98	1.0	0.99	0.98	1.0	1.0	0.99	0.9828	0.9483	0.96
WID11_CID99, default	0.98	0.99	0.98	0.97	0.96	1.0	0.97	0.99	0.99	1.0	0.99	0.99	1.0	1.0	0.9828	0.9828	0.96
WID11_CID99, high	0.98	0.99	0.98	0.98	1.0	0.99	1.0	1.0	0.99	0.99	1.0	0.98	0.97	0.99	0.9683	0.9524	0.95
WID11_CID99, low	1.0	1.0	1.0	1.0	0.99	0.98	0.97	0.98	0.98	0.98	0.99	1.0	1.0	1.0	0.9444	0.963	0.96
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 35: TRNG1 105 °C WID11 statistical test results.



TROPIC01
ODR_TR01_PSR_004_TRNG_TropicSquare
TROPIC01 True Random Number Generator (TRNG)

NIST Test Suite Summary for 125.0 (°C), TRNG1, WID11
Fail ratio 1.83%, ~ 1.84% expected

WID11_CID101, default	1.0	0.99	1.0	1.0	0.98	0.99	1.0	0.99	0.96	0.99	0.97	0.99	0.99	0.99	0.9524	0.9841	0.95
WID11_CID101, high	1.0	0.99	1.0	1.0	1.0	0.99	0.99	1.0	0.99	0.99	0.99	1.0	0.98	0.99	0.9828	0.9828	0.95
WID11_CID101, low	0.99	0.99	0.99	0.99	0.98	0.99	0.98	1.0	1.0	0.97	0.99	0.99	1.0	1.0	0.9833	0.9667	0.96
WID11_CID103, default	1.0	1.0	1.0	0.99	0.99	1.0	1.0	0.98	1.0	0.97	0.99	1.0	0.99	0.99	0.9688	0.9688	0.95
WID11_CID103, high	0.99	1.0	0.99	0.99	0.99	0.98	1.0	0.99	0.99	1.0	1.0	1.0	0.99	0.99	0.9697	0.9848	0.96
WID11_CID103, low	0.98	0.98	0.97	0.97	0.99	1.0	0.98	1.0	0.99	0.99	1.0	0.96	1.0	0.99	0.9859	0.9577	0.95
WID11_CID104, default	1.0	1.0	0.99	0.99	0.99	1.0	0.98	0.98	0.98	0.99	0.98	0.99	1.0	0.99	0.9833	0.95	0.96
WID11_CID104, high	0.99	0.99	0.99	0.99	1.0	0.99	0.99	1.0	1.0	0.98	0.98	1.0	1.0	1.0	0.9683	0.9841	0.95
WID11_CID104, low	0.98	1.0	0.98	0.98	0.98	0.99	1.0	1.0	0.99	0.99	0.99	1.0	1.0	0.98	0.9846	0.9692	0.95
WID11_CID107, default	0.99	0.99	1.0	0.98	1.0	0.99	1.0	0.98	0.98	1.0	0.97	0.99	0.99	0.99	0.9846	0.9692	0.95
WID11_CID107, high	1.0	0.99	0.99	1.0	0.98	1.0	1.0	0.99	0.98	0.98	1.0	1.0	1.0	0.99	0.9365	0.9683	0.95
WID11_CID107, low	1.0	0.95	1.0	1.0	0.98	1.0	0.98	0.99	1.0	1.0	0.99	0.99	1.0	0.98	0.9524	0.9524	0.97
WID11_CID120, default	0.98	0.99	0.97	0.97	0.98	0.99	0.97	1.0	0.98	0.99	0.99	0.99	1.0	0.99	0.9846	0.9846	0.96
WID11_CID120, high	0.98	0.98	0.98	0.99	0.99	0.99	0.99	0.98	1.0	1.0	0.99	0.99	0.99	0.99	0.9844	0.9844	0.96
WID11_CID120, low	1.0	1.0	1.0	1.0	0.99	0.99	1.0	0.98	0.96	0.99	1.0	0.96	0.97	1.0	0.9697	0.9545	0.97
WID11_CID35, default	0.98	1.0	0.98	0.98	0.99	1.0	1.0	0.99	1.0	1.0	0.99	1.0	0.99	0.99	0.9661	0.9661	0.96
WID11_CID35, high	0.99	0.99	0.98	0.99	0.98	0.98	0.98	0.98	1.0	1.0	1.0	0.99	0.98	0.99	0.9559	0.9706	0.96
WID11_CID35, low	0.99	0.98	0.99	0.97	0.99	0.98	0.98	0.99	0.97	1.0	0.98	1.0	0.99	1.0	0.9531	0.9688	0.96
WID11_CID60, default	0.98	1.0	0.99	0.99	0.99	0.98	1.0	0.98	1.0	1.0	0.99	1.0	1.0	1.0	0.9851	0.9701	0.95
WID11_CID60, high	1.0	0.99	0.99	0.99	0.95	0.99	1.0	0.99	0.98	1.0	1.0	0.98	0.99	0.99	0.9623	0.9811	0.97
WID11_CID60, low	1.0	0.99	0.99	1.0	0.99	0.99	0.97	0.99	1.0	0.99	1.0	0.99	0.98	0.99	0.9688	0.9844	0.95
WID11_CID97, default	0.98	0.99	0.98	0.98	0.99	1.0	1.0	0.99	1.0	0.99	0.97	1.0	0.98	1.0	0.963	0.9815	0.96
WID11_CID97, high	1.0	0.99	1.0	1.0	1.0	1.0	0.98	0.98	0.99	0.96	0.99	0.99	0.97	0.99	0.9846	0.9692	0.97
WID11_CID97, low	1.0	0.99	1.0	1.0	1.0	1.0	0.99	0.99	1.0	0.99	0.99	0.99	0.99	0.99	0.9714	0.9857	0.95
WID11_CID98, default	1.0	0.99	1.0	1.0	0.99	1.0	0.99	0.99	0.99	0.99	1.0	0.97	0.98	0.96	0.9844	0.9688	0.96
WID11_CID98, high	0.99	0.99	0.99	0.99	0.98	0.99	0.99	1.0	0.98	1.0	1.0	0.99	0.97	1.0	0.9661	0.9661	0.96
WID11_CID98, low	0.99	0.99	0.98	0.99	0.99	0.99	1.0	0.99	0.99	1.0	0.98	0.97	0.98	1.0	0.9672	0.9836	0.95
WID11_CID99, default	1.0	0.99	1.0	0.99	0.97	0.99	0.98	1.0	0.98	0.99	0.98	1.0	1.0	0.99	0.9643	0.9643	0.96
WID11_CID99, high	0.99	0.98	0.99	0.99	0.99	0.99	0.98	0.99	0.96	1.0	0.98	0.98	1.0	0.99	0.9836	0.9672	0.96
WID11_CID99, low	0.98	1.0	0.98	0.98	0.98	1.0	0.98	0.99	0.99	0.99	0.99	1.0	0.99	1.0	0.9701	0.9552	0.95
	Frequency	BlockFrequency	CumulativeSums	CumulativeSums	Runs	LongestRun	Rank	FFT	OverlappingTemplate	Universal	ApproximateEntropy	Serial	Serial	LinearComplexity	*RandomExcursionsVariantMin	*RandomExcursionsMin	*NonOverlappingTemplateMin

Figure 36: TRNG1 125 °C WID11 statistical test results.



3.2 NIST SP 800-90B min entropy assessment

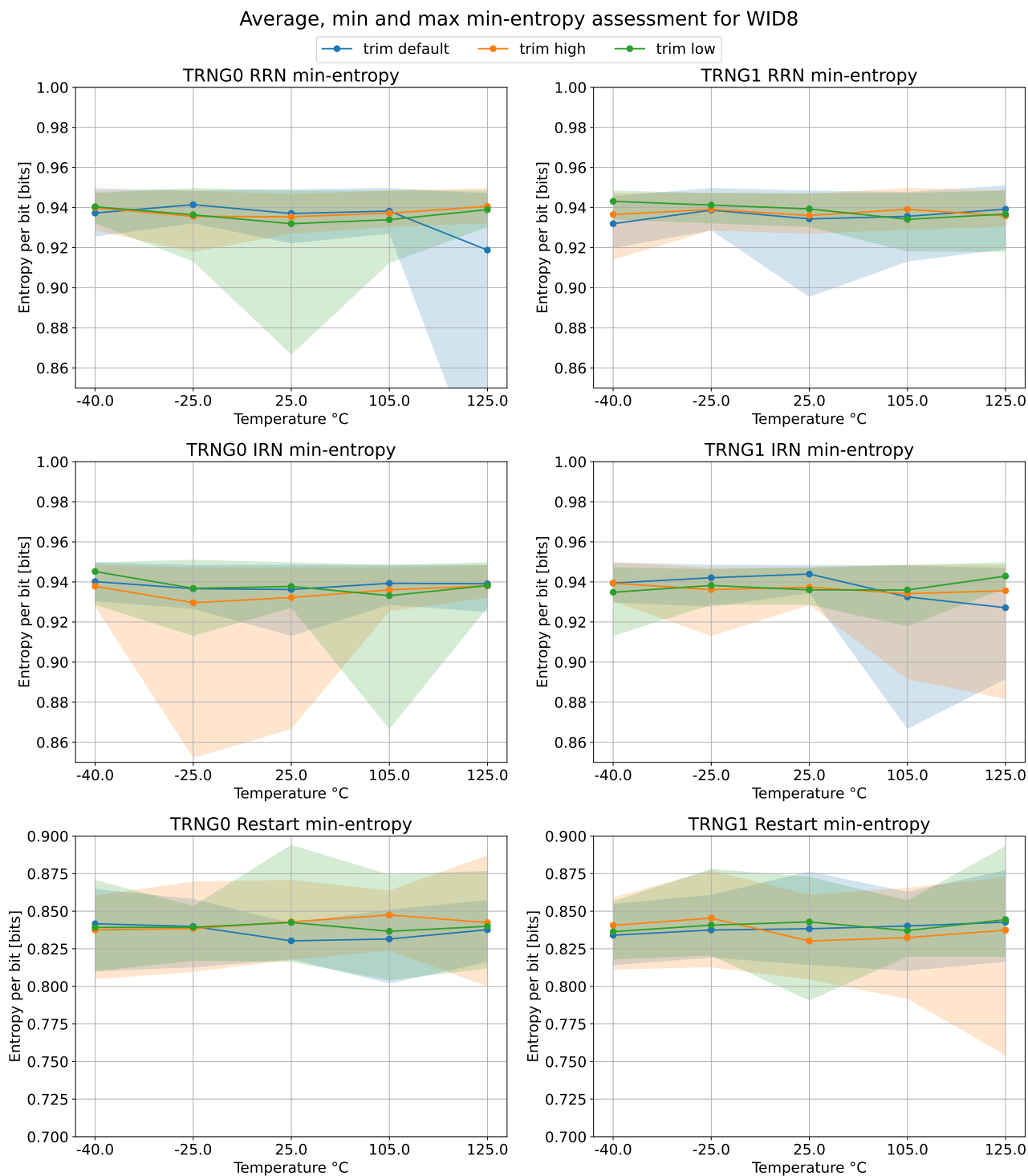


Figure 37: Combined entropy assessment for WID8.

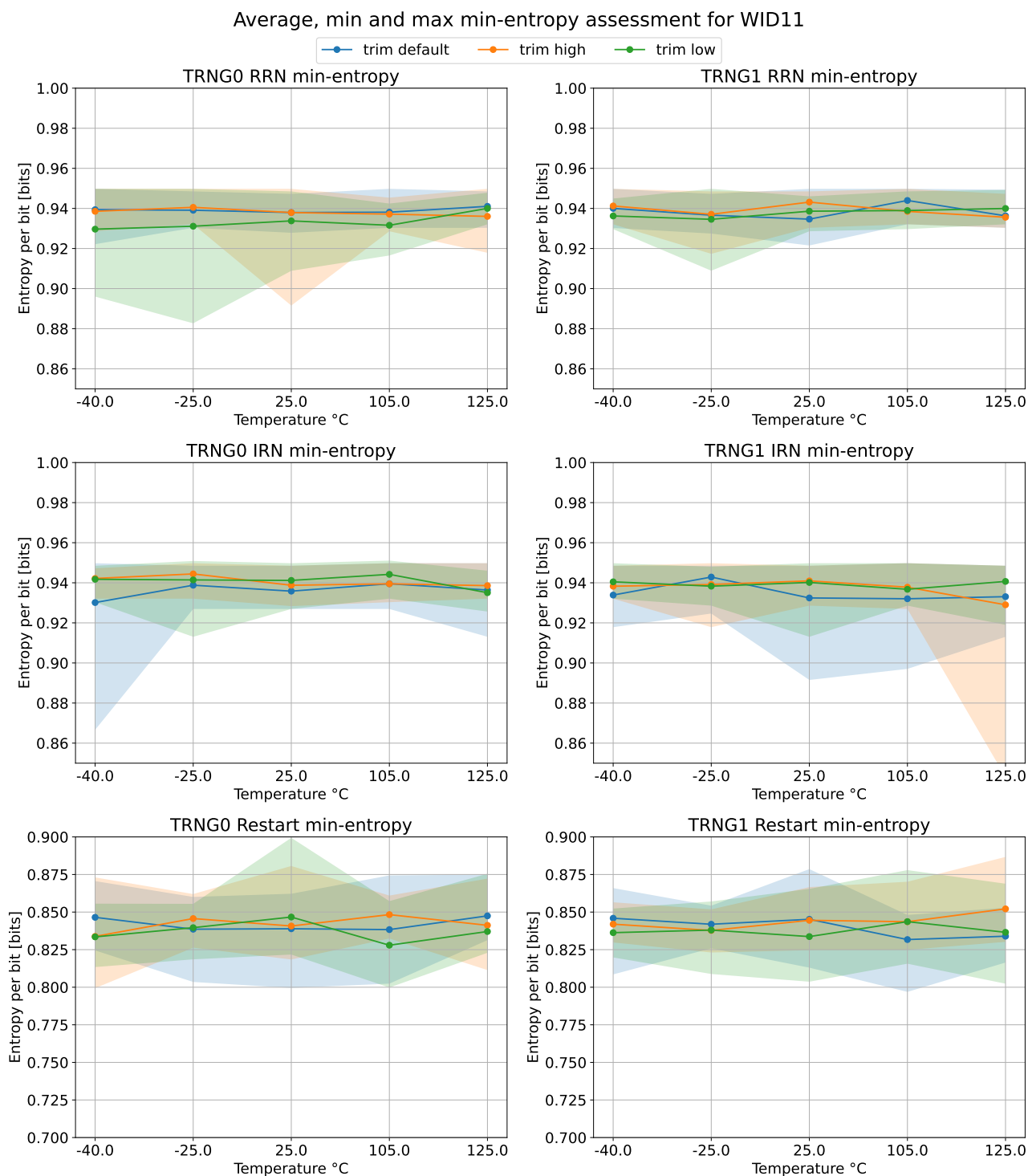


Figure 38: Combined entropy assessment for WID11.



Figure 39: TRNG0 entropy assessment at all conditions for WID8.



Figure 40: TRNG1 entropy assessment at all conditions for WID8.



Figure 41: TRNG0 entropy assessment at all conditions for WID11.

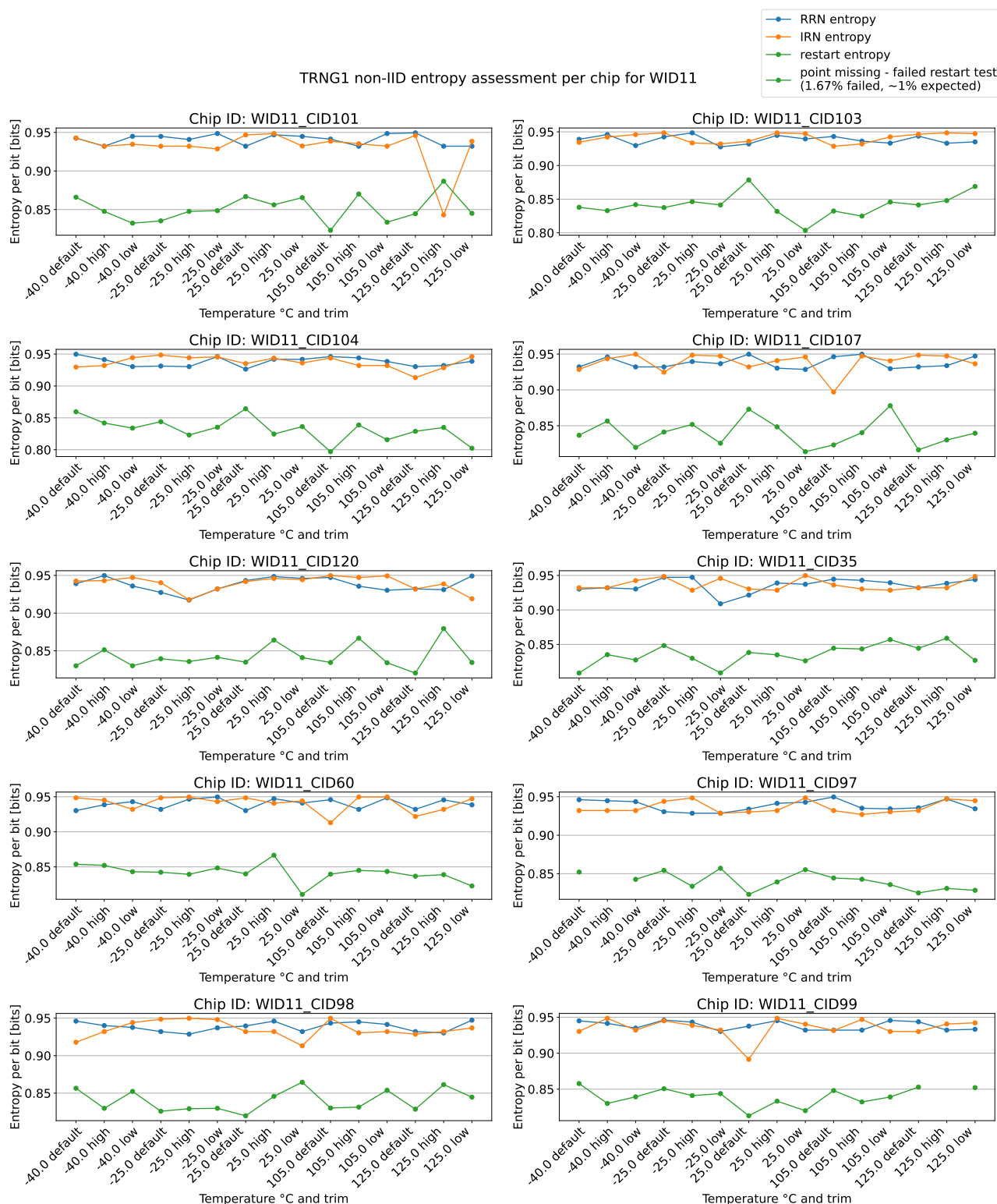


Figure 42: TRNG1 entropy assessment at all conditions for WID11.



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